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ProTrend User Guide

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About This Document

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

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GUSPro Installation Guide	GP05-150
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GUSPro Software Change Notice	GPro150-SCN
GUSPro Specification and Technical Data	GP03-110
Display Builder User's Guide	GU23
Display Scripting User's Guide	GU24
Display Authoring Tutorial	GU07

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.

Symbol Definitions

Contents

1. Introduction	17
1.1 Overview	17
1.2 ProTrend Object.....	17
Overview	17
Prerequisites	18
Interfaces	18
Scrolling	18
Zooming	18
Supported Data Sources	19
Data Aggregates	20
1.3 ProTrend Tool	23
Functions and Features	23
ProTrend Tool Architecture	26
1.4 ProTrend Templates	28
Group Trend.....	28
Overview Trend Templates	29
2. ProTrend Tool	31
2.1 Initial ProTrend Tool Configuration	31
Set Up the ProTrend Tool Database	31
Configure the Connection to the ProTrend Tool Database	39
2.2 ProTrend Tool Configuration Utility.....	45
Configure the ProTrend Tool General Settings	48
Configure the ProTrend Tool Default Settings.....	55
Configure the ProTrend Tool Time Spans.....	58
Configure the ProTrend Tool Data Sources	61
2.3 ProTrend Tool User Interface	63
Trend Pane Functions and Features	65
Trend Control Area.....	71
Review the Current Trend Trace Status	74
Set the Trace Details.....	76
Set the Trend Details	84
Set the Time Settings	90
Save Load or Delete the Trend Settings	91
Set the Data Source Settings	94

3.	ProTrend Templates Reference	97
3.1	General Trend Template Information	97
3.2	Group Trend.....	98
Purpose.....		98
Sub_Title_Group Trend (Trend Title)		101
Sub_Trend_Group Trend (Trend).....		102
Sub_TrendDetails_Group Trend (Trend Details).....		107
3.3	Overview Trend Template.....	110
Purpose.....		110
Sub_Title / subMT_Title(Trend Title).....		111
Trend Graphs: Sub_Trend_4-Point_PV_Small		112
Trend Details: Sub_TrendDetails_4-Point		115
4.	ProTrend Object Scripting.....	119
4.1	Programmatic Interface	119
The ProTrend Object Model		120
4.2	Data Source Types	121
5.	GProTrend Object	123
5.1	Introduction.....	123
5.2	GProTrend Properties.....	123
Name.....		123
Description		124
Title		125
Font.....		125
UpdatePeriod		126
VisibleTimeSpan		127
BufferTimeSpan		129
EndTime.....		131
BeforeNow		132
VisibleEarliestTime.....		133
VisibleLatestTime		134
SelectedTrace		134
AxesMode		135
HairlineCursor		137
HairlineTimeFormat.....		139
YScaleVisible		140
TimeScaleVisible.....		141
LegendPosition.....		142
LegendStyle		143

PlotAreaOnly	144
MinimumDataChangePeriod	145
LastUpdateTime	146
ColorPallet.....	147
Colors.....	148
GridColor	149
BackgroundColor	150
HairlineColor	150
BorderColor	151
LegendColor.....	151
AxesColor.....	152
TraceWidth.....	152
ZoomEnabled.....	153
5.3 GProTrend Methods	154
Traces	154
CollectHistory	155
InsertColor.....	156
DeleteColor	156
DateToTimeSpan	157
TimeSpanToDate	158
ZoomOut	159
ShiftHairlineLeft.....	159
ShiftHairlineRight	160
CenterHairline	161
ExportTrace.....	161
5.4 GProTrend Events	163
OnUpdateCollected	163
OnHistoryCollected	165
OnDataCollectError	166
OnHairlineCursor	167
OnScrollLeft	169
OnScrollRight	169
OnZoomIn	169
OnZoomOut	170
6. GProDataSources Object	173
6.1 Introduction	173
6.2 GProDataSources Properties	174
_NewEnum.....	174
Count	175
6.3 GProDataSources Methods	176
Item	176
Add.....	177

Contents

Remove.....	178
Trend.....	179
7. GProDataSource Object	179
7.1 Introduction.....	179
7.2 GProDataSource Properties.....	180
long Type	181
BSTR Name	181
BSTR Host	181
BSTR Username	182
BSTR Password	182
BSTR Server	182
BSTR HCIServerAliasName.....	182
DATE LastCollectionTime	182
double Timeout.....	182
long State	182
BSTR NativeTagPrefix	184
7.3 GProDataSource Methods.....	184
Activate()	184
Deactivate()	185
8. GProTag Object.....	187
8.1 Introduction.....	187
8.2 GProTag Properties	187
TagID	187
Type	188
InternalID.....	189
ValueDataSource	190
RangingDataSource	191
ReadFromCache	192
RequestedCacheUpdatePeriod.....	192
RetrievedCacheUpdatePeriod.....	193
RangeHigh	194
RangeLow	195
DescriptionTagID.....	196
Description	197
State.....	198
8.3 GProTag Methods	199
TagPair.....	199
Activate	199
Deactivate	200

9. GProTagPair Object.....	201
9.1 Introduction	201
9.2 GProTagPair Properties	201
PrimaryTag.....	201
Type	202
AggregateType.....	203
AggregateOffset	205
AggregateSampleInterval	206
AggregateInterval	207
9.3 GProTagPair Methods	208
ActiveTag	208
Trace	208
ActivatePrimaryTag	209
RemovePrimaryTag	209
CreatePrimaryTag	210
10. GProTraces Object	211
10.1Introduction	211
10.2GProTraces Properties.....	211
_NewEnum.....	211
Count	212
10.3GProTraces Methods.....	213
Item	213
Add.....	214
Remove.....	215
GetEldest	216
Trend.....	216
11. GProTrace Object	217
11.1Introduction	217
11.2GProTrace Properties.....	217
Name	217
Description	218
Color	219
Deletable	219
CanonicalDataType.....	220
Value	221
TimeStamp.....	222
hResult	222

Contents

OPCQuality	223
PercentConfidence	224
ConfidenceThreshold	224
CollectionUnit	225
hResultCollectionUnit	226
RequestUnit	227
HairlineDateTime	228
HairlineValue	228
HairlineDataValid	229
ScalingMode	230
ManualRangeHigh	231
ManualRangeLow	231
RangeHighFixed	232
RangeLowFixed	233
RetrievedRangeHigh	234
RetrievedRangeLow	234
Assigning RetrievedRangeHigh and RetrievedRangeLow Properties	235
DynamicRangeHigh	236
DynamicRangeLow	237
RetrievedDescription	238
hResultDescription	239
VisibleRangeHighLimit	240
VisibleRangeLowLimit	240
CollectedRangeHigh	241
CollectedRangeLow	241
LatestTimeCollected	242
EarliestTimeCollected	243
Visible	243
TraceWidth	244
LineStyle	244
11.3GProTrace Methods	245
HistoryTagPair	245
UpdateTagPair	246
Trend	246
Refresh	247
12. ProTrend Scripting Examples	249
12.1ProTrend Object Scripting Examples.....	249
Adding a Trace to a Trend	249
Removing a Trace	250
Setting Trend Properties on Startup	250
12.2ProTrend Tool Interface Examples.....	251
Adding a Tag to the ProTrend Tool	251
Loading A Saved Trend in the ProTrend Tool	251

13. Appendix253

13.1Color Registry Settings..... 253

13.2Group Trend Registry Settings 255

13.3Overview Trend Template Registry Settings 257

13.4Supported Variant Types 258

Contents

1. Introduction

1.1 Overview

ProTrend is a fully functional trend package that can display trend data from a range of data sources, including Honeywell's TotalPlant Network (TPN/LCN) and Plant History Database (PHD).

At the heart of ProTrend is the ProTrend Object. This trend object provides the trend display features (such as trend traces, gridlines, axes, and legends), as well as data collection and error handling functionality. The ProTrend Object can be embedded in GUS schematics.

In order for users to quickly take advantage of the functions and features available with the ProTrend Object, a standalone ProTrend Tool application and two ProTrend GUS graphic templates are provided with the ProTrend package. The ProTrend Tool and each ProTrend template include the ProTrend Object, to make it immediately accessible to the user without the need for extensive configuration.

The ProTrend Tool is a standalone, trend display application that can be used as is. It allows the operator to dynamically build trends for monitoring and analysis. It is set up with a central ProTrend Tool Server and associated database, which are used to store common trend display settings that apply to every instance of the ProTrend Tool in a given console area. The database is also used to store the detailed trend settings for identified groups of points. These trend displays are then available for future use and may be accessed from any station in the console.

The ProTrend templates may be modified to create custom GUS graphics that include the ProTrend Object or can be used as is. The ProTrend Object can also be embedded in any GUS display and scripted for fully customizable trend displays.

1.2 ProTrend Object

Overview

The ProTrend Object is an ActiveX control for trends that can be embedded in GUS schematics. It provides the trend display features (for example, trend traces, gridlines, axes, and legends), as well as data collection and error handling functionality.

The ProTrend Object functions as an ActiveX control that does the following:

- Responds directly to user input.

1. Introduction

1.2. ProTrend Object

- Exposes an automation interface for control by GUS Picture Builder schematics.

The ProTrend Object also provides the trending function for GPB displays and for the ProTrend Tool.

Prerequisites

PHD OLE Automation Server must be installed on each computer where the ProTrend Object is to be run. This includes computers where the ProTrend Tool is to be run, as well as computers where GUS graphics or custom applications that use the ProTrend Object are to be run.

Interfaces

The ProTrend Object provides both a graphical user interface and a programmatic automation interface. The ProTrend Object user interface displays the trend information to the end user and responds to various mouse actions while the programmatic interface allows the ProTrend Object to be controlled from the ProTrend Object's container.

Scrolling

The ProTrend Object allows the user to scroll through a longer time span than is visible at any given time. Scrolling is accomplished by moving the cursor to the left or right while holding down the Right Mouse button.



ATTENTION

When scrolling or zooming positions the visible portion of the trend so that it does not include the current time, the display remains static while current updates values accumulate in a buffer.

If a backward-scrolled trend is left on display for a long time (hours), when it is returned to current time – by forward scrolling or changing the EndTime – the trend may become non-responsive for a period of time while buffered data is trimmed and rearranged.

Zooming

The ProTrend user can zoom in on any portion of the trend by positioning the cursor to the top-left corner of the desired area then hold down the Left Mouse button while dragging the cursor to the desired lower-right corner.

A zoomed trend is returned to full scale by holding down the Left Mouse button while dragging the cursor on the diagonal going right-to-left or bottom-to-top.



ATTENTION

Zooming is only possible in Shared and Selected axis modes.

If the axis mode of a zoomed trend is changed, it is returned to its original scale.

When the trend is zoomed out to its original scale, the visible end time is probably older than the current time. Therefore, it may be necessary to scroll forward or reset the EndTime to view dynamically updated current values.

Supported Data Sources

The ProTrend Object's programmatic interface and the graphical user interface support the concurrent trending of data from the following data server types:

- TPN
 - TPN Server
 - Private TPN Server
- PHD Proprietary Interface
 - Historical
 - Real-Time

The Data Sources are configured using the GUSPro Configuration Tool.

The Default Data Source for History data is the PHD Server named in the GPRO_PHD_HOST environment variable.

The Default Data Source for Real-time updates is determined by the GPRO_TREND_RTSOURCE environment variable. If its value is "PHD" then the trend uses the same PHD Server as the History default. Otherwise, it is treated as the (TPS alias) name of a TPN Server. If it is NULL, the local private TPN Server is used.



ATTENTION

In order for the Trend data to update smoothly between History and Real-time sources, it is necessary that the PHD collector node and the TPN clocks be synchronized. If the PHD clock time is more recent than the real-time source, then there is a dead period following the fetch of any history data before any real-time data starts to appear. If the PHD clock time is older than the real-time source, there is a visible gap in the displayed trend data.

1. Introduction

1.2. ProTrend Object

Data Aggregates

Supported Data Aggregate Types

Data aggregates are values that are calculated for a set of historical data over a specified time interval. The ProTrend Object supports the following aggregate types:

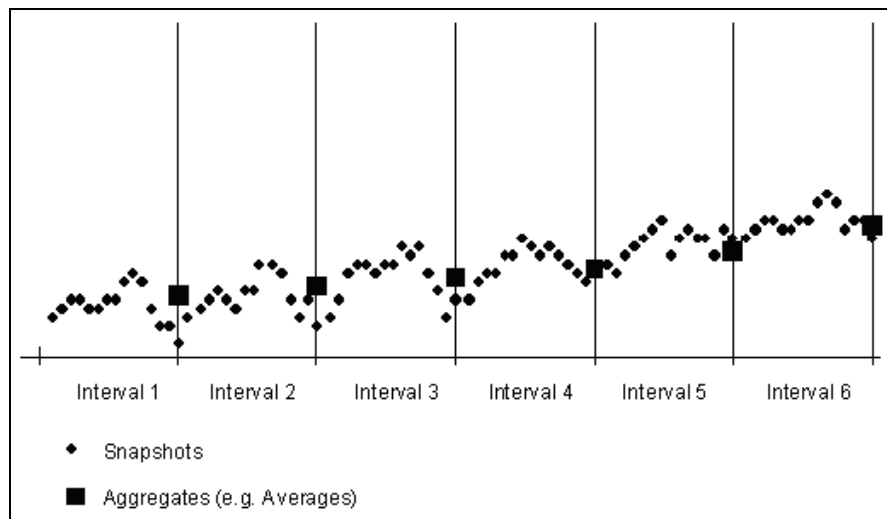
Aggregate Type	Description
Average	The numerical average of the samples in the interval.
Minimum	The lowest value among the samples in the interval.
Maximum	The highest value among the samples in the interval.
Start	The first sample in the interval.
End	The last sample in the interval.
Std.Deviation	The standard deviation for the values within the interval.
Range	The difference between the minimum and maximum values within the interval.
Delta	The difference between the first and the last values in the interval.

In addition to the data aggregates, raw and snapshot data can also be displayed. Raw data is retrieved directly from the data source without interpolation. Snapshot data is also retrieved directly from the data source but the values may be interpolated.

Aggregate Calculation

Figure 1 shows the relationship between aggregates and data samples. An aggregate value (such as the average) is calculated for the samples within the interval. For each interval, there is one aggregate value.

Figure 1 Snapshots and Aggregates



In most cases, the aggregates are supplied from a PHD data source. In this case, the aggregates are calculated by the Uniformance database system and accessed by the PHD OLE Automation Server.



ATTENTION

The TPN server does not support the full set of aggregate types. Therefore changing the data source from a PHD server to a TPN server may result in runtime errors.

Confidence Values

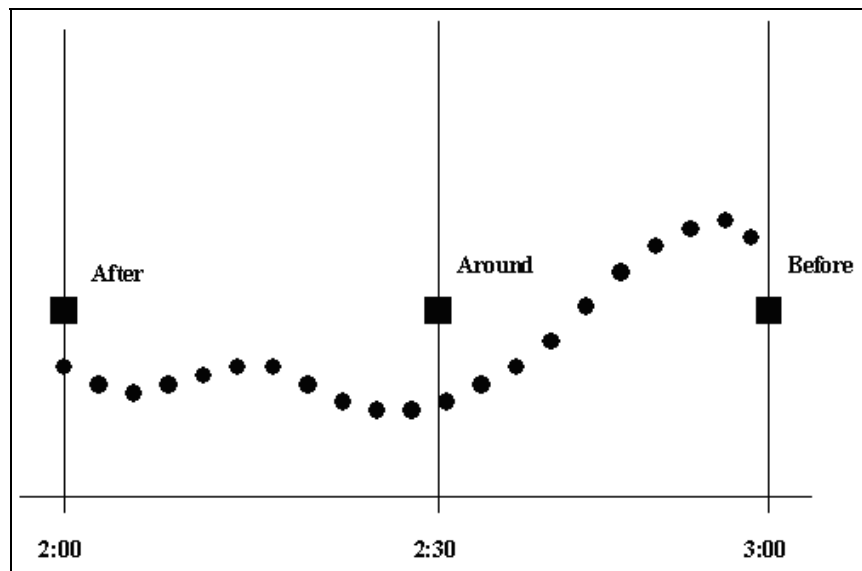
Aggregates have confidence level values, which indicate the percentage of good sample values from which the aggregate was calculated. For example, within an hour interval, 60 one-minute samples are collected. Of these 60 samples, 54 are good values and 6 are bad values. The confidence level of the hourly average of this interval is 90%.

Aggregate Offset

The *aggregate offset* determines the relationship between the time of the aggregate and the sample interval, as shown in Figure 2. In this example, the interval is one hour and the aggregate type is 'average'. The three options for aggregate offset are shown as:

Aggregate Offset Type	Description
After	The interval follows the time of the aggregate.
Before	The interval precedes the time of the aggregate.
Around	The time of the aggregate is in the middle of the interval.

Figure 2 Aggregate Offset



Aggregate and Sample Intervals

The *aggregate interval* is the total period over which the trend requests data from PHD. The *sample interval* defines how the aggregate interval is divided. By default, the aggregate interval is the same as the update rate.

1.3 ProTrend Tool

Functions and Features

The ProTrend Tool encapsulates the ProTrend Object in a standalone application. It packages the trend in a standard way that can be used as is. It allows the operator to dynamically build trends for monitoring and analysis, and to save them for future use on a console-wide basis.

Two versions of the ProTrend Tool are provided for different invocation scenarios.

A single instance of the standard ProTrend Tool may run on each node. This tool is integrated with ProMenus. If a Trend Tag Button on a ProMenu is selected, the trace for that tag is displayed on the standard ProTrend Tool.

The ProTrend Adhoc tool (gptt_adhoc.exe), allows the operator to run multiple copies of the ProTrend tool for tags that are added through its user interface. The functionality of the Adhoc tool is identical to that of the standard tool, except there is no interaction between ProMenus and the Adhoc tool.

The Adhoc tool can be invoked with the following optional command line parameters,

`gptt_adhoc.exe /H /L /P /F foldername /T trendname` where;

- `/F folder` -- sets the initial folder selection to the named string
- `/T trendname` - sets the initial trend / tagset selection to the named string.



TIP

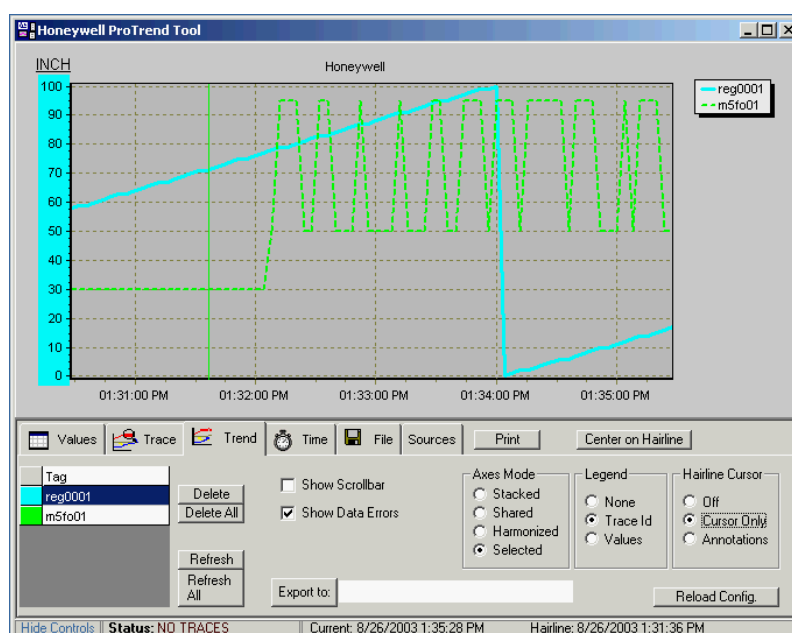
If both `/F` and `/T` are specified, then the specified file is loaded and started.

- `/H` -- causes the application to open with the Controls Hidden (so the Trend plot takes up the maximum space).
- `/L` -- causes the Controls to be Locked so that the operator cannot change the trend configuration. When the Tool is Locked, the operator is limited to viewing pre-defined trends.
- `/P` -- causes the Tool to be drawn without a title or axes, so the Plot Area gets the maximum share of the space.

1. Introduction

1.3. ProTrend Tool

Figure 3 ProTrend Tool



Data Displayed

The ProTrend Tool shows the following information:

- Trend trace for each trended tag
- Tag name and description for each tag
- Value axis scale for each tag with intermediate demarcations (y-axis)
- Time axis scale with intermediate demarcations (x-axis)
- Color key for each tag (if legend enabled)
- Hairline cursor read-out for each tag (if enabled)

Functions Available

The operator can perform the following functions from within the ProTrend Tool:

- Change time span
- Scroll back and forward in time
- Add and delete trend traces
- Select the trend display format (shared, stacked, harmonized and selected)
- Change the y-axis range of a selected trace
- Graphically zoom the trends with Shared or Selected axes
- Change the color for each trend trace
- Save a set of trends with a user-defined name on a console basis in user-defined folders
- Restore, rename and delete saved trend sets
- Enable and disable hairline cursor read-out
- Change the data aggregate specification, including aggregate type. (See Data Aggregates section for a description of the aggregate types.)
- Change the update rate
- Change the history and update tag names used
- Print a snapshot of the displayed trend

1. Introduction

1.3. ProTrend Tool

- Extract the data from the current trend to a comma-delimited text file (suitable for import into spreadsheets).

Configurable Defaults

The following list describes some of the defaults that can be set up (See ProTrend Tool Configuration Utility section for a complete list):

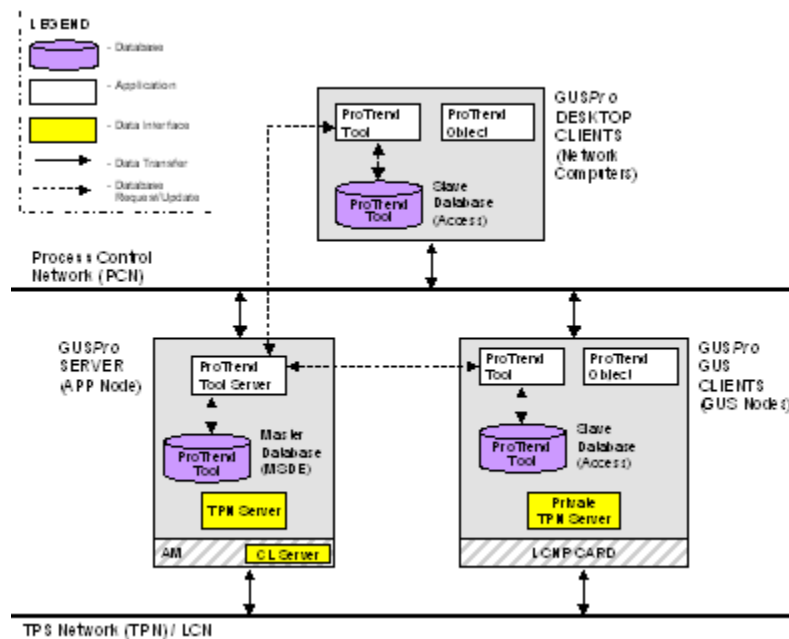
- Colors for the trend traces and for various trend display elements (for example, trend display background, legend, hairline cursor)
- PHD Server (for history and real time data)
- TPN Server (for real time data)
- Update/real time data source
- Update rates
- Time spans
- Tag naming conventions for both history and update data
- Aggregate types which may be selected for use from within ProTrend Tool at run time
- Minimum data confidence level
- Operator access privileges from within ProTrend Tool to various aggregate settings (aggregate interval, sample intervals, aggregate offset and update rate)

A programmatic interface is provided to allow the ProTrend Tool to be invoked for given saved trend set or for a given tag from a GUS display script.

ProTrend Tool Architecture

Figure 4 shows the relationship between the ProTrend Tool database and the ProTrend Tool application software. The ProTrend Tool Server software and the associated database are only required if the ProTrend Tool is going to be used. They are not needed on those systems that only use the ProTrend Object or the ProTrend Templates.

Figure 4 ProTrend Tool Architecture



On each computer where the ProTrend Tool is run, a local backup database (in Microsoft Access format) is also set up. This database is synchronized with the ProTrend Tool Server database every 20 minutes. If the connection between the client computer where the ProTrend Tool is being run and the ProTrend Tool Server is lost, the ProTrend continues to run but uses the trend settings in the local database instead. When the network connection is reinstated, the user can manually reconnect to the ProTrend Tool Server database (by clicking the **Connect** button in the File tab) or wait for the ProTrend Tool to reconnect automatically. When this happens the display settings stored in the two databases are synchronized automatically.

1.4 ProTrend Templates

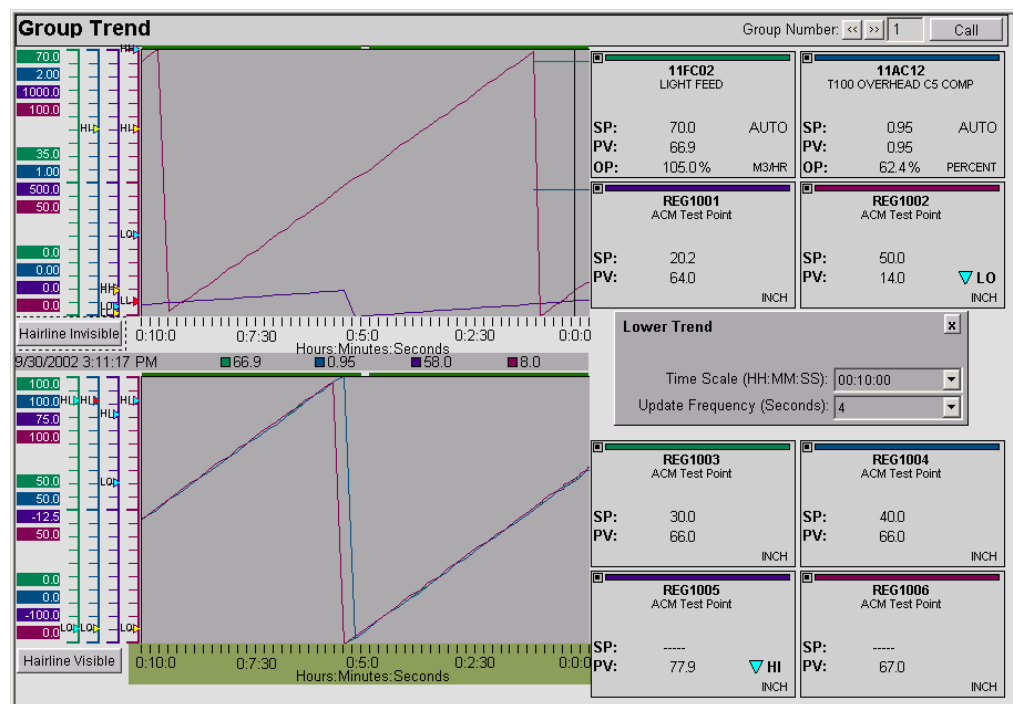
A set of GUS displays are supplied. These displays can be run as provided and they serve as templates from which build new trend displays.

Group Trend

The Group Trend is a display that allows the operator to monitor and display trends for configured groups as shown in Figure 5. The points are arranged in two trend graphs with four trend traces in each graph. See Group Trend section for a detailed description of the Group Trend template. NOTE: The Group Trend uses harmonized axes, so it does not support zooming.

The points trended correspond to the selected LCN group number and can be changed at runtime by changing the group number.

Figure 5 Group Trend Display





TIP

Place a copy of Trend_Group.pct into a directory in the GUS displays search path. With GUS340 or later, this enables a direct link from the GUS GroupDisplay to the Group Trend.

Overview Trend Templates

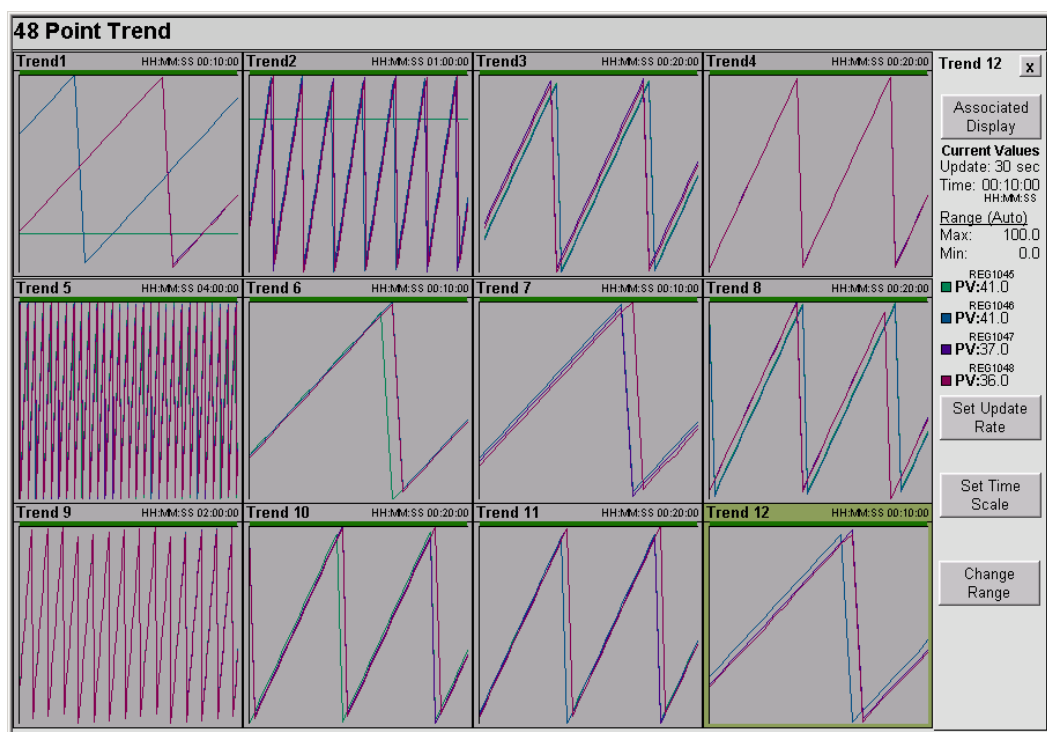
The Overview Trend templates allow the operator to monitor up to 48 points in a single display, as shown in Figure 6. The points may be arranged in 12 trend graphs with four trend traces in each graph or in 8 trend graphs with up to 6 traces in each graph. A user-defined description is shown for each trend graph. Each trend graph shows the time span configured for it. (See Overview Trend Template section for a detailed description of the Overview Trend templates.)

The Overview Trend templates may be copied, bound to specific tags, and used for custom trend displays. Binding to specific tags is accomplished by specifying the parameters of embedded displays.

1. Introduction

1.4. ProTrend Templates

Figure 6 Overview Trend Display



2. ProTrend Tool

2.1 Initial ProTrend Tool Configuration

Prior to running the ProTrend Tool or the ProTrend Tool Configuration utility, the following ProTrend Tool configuration steps must be completed:

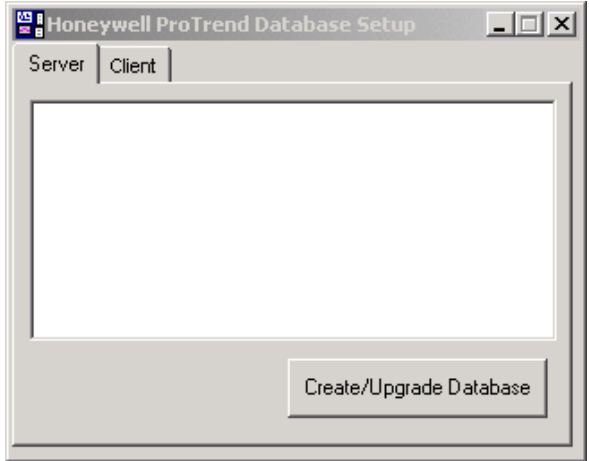
1. On each computer where the ProTrend Tool is going to run, configure the connection to the ProTrend Tool database (See Configure the Connection to the ProTrend Tool Database section).
2. Set up the ProTrend Tool database on the ProTrend Tool Server.

Set Up the ProTrend Tool Database

Step	Action
1	On the computer where the ProTrend Tool Server software was installed, launch the ProTrend Database Setup application from the Windows Start menu, by selecting Programs > Honeywell OPS > GUSPro > ProTrend DB Config. For <i>GUSPro</i> system installations, the ProTrend Tool Server software is generally installed on the <i>GUSPro</i> Server.

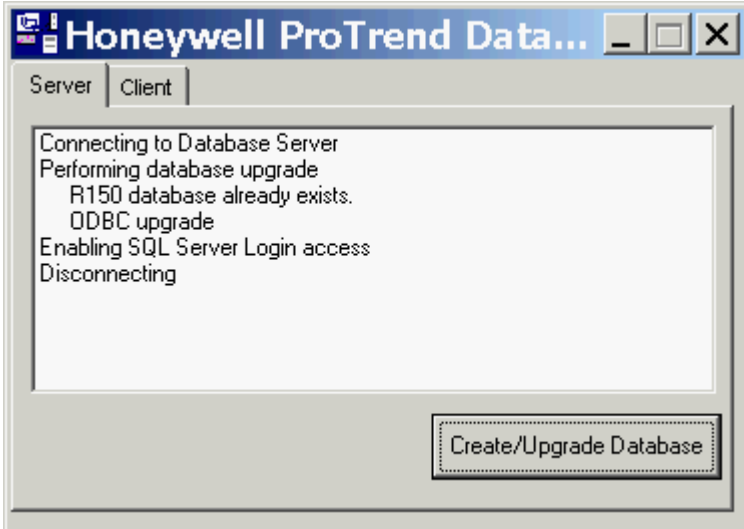
2. ProTrend Tool

2.1. Initial ProTrend Tool Configuration

Step	Action
2	In the Server tab, click Create/Upgrade Database to set up the ProTrend Tool database tables and to populate them with default values.  NOTE: If the ProTrend Database Setup application is run on a computer that is not intended to act as the ProTrend Tool Server, there is no need to run the Create/Upgrade Database function.
3	In order to create or upgrade the ProTrend Tool database, you may be required to provide a valid SQL Server user name and password for the MSDE or SQL Server database. (This would only be needed if the default SQL Server administrator, "sa", user name or password has been changed.) If the SQL Server Administrator Login dialog box appears, enter the current SQL Server user name and password, and click OK. 

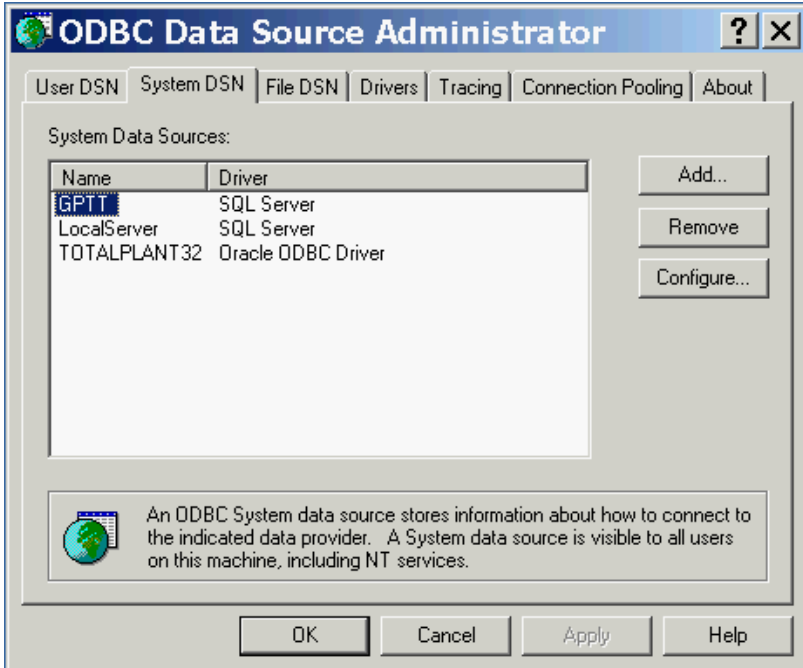
2. ProTrend Tool

2.1. Initial ProTrend Tool Configuration

Step	Action
4	The results of the database create or upgrade operation is displayed in the window. 
5	If the current computer is also going to be used to run the ProTrend Tool, be sure to create a PHD Host environment variable. See Configure the ProTrend Tool General Settings section for more information.

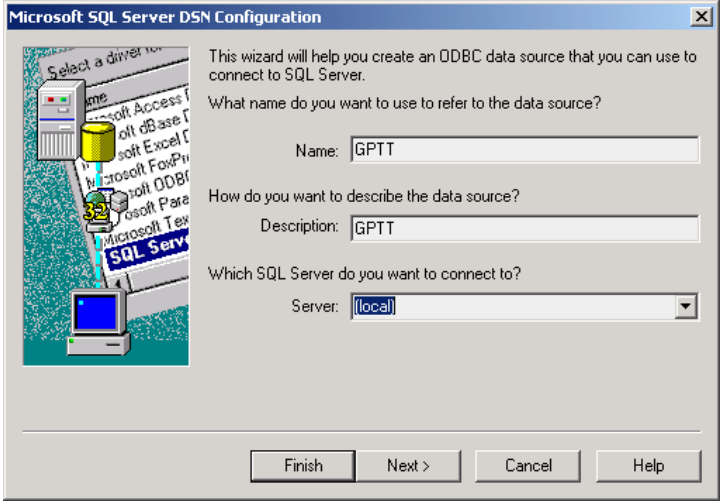
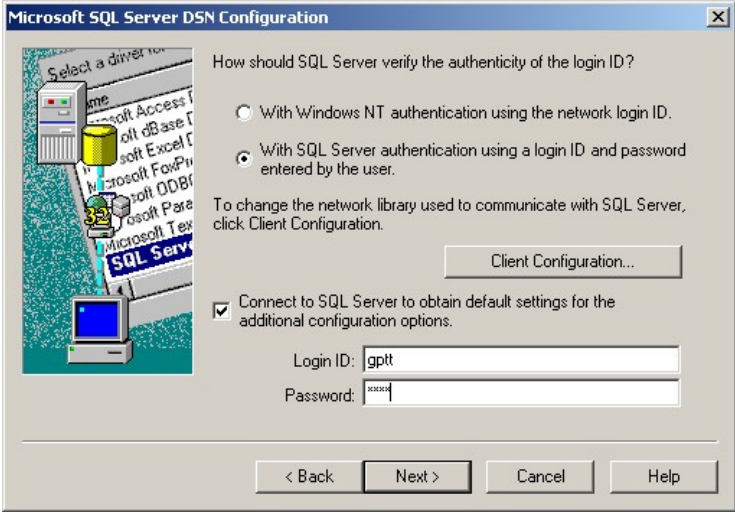
2. ProTrend Tool

2.1. Initial ProTrend Tool Configuration

Step	Action
6	Choose Settings > Control Panel > Administrative Tools from the Windows Start menu and then double click Data Sources (ODBC). Select the System DSN tab and then select GPTT in the System Data Sources field. Click Configure to continue. NOTE: The GPTT System Data Source is created when the Create/Upgrade Database button was clicked in Step 4. 

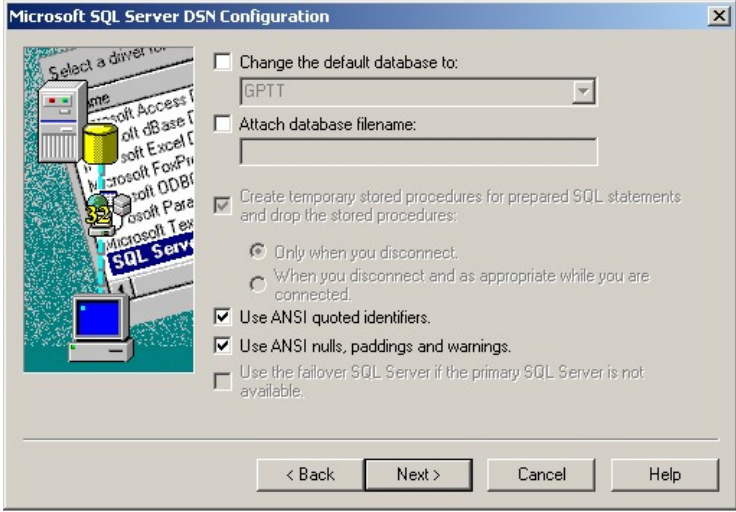
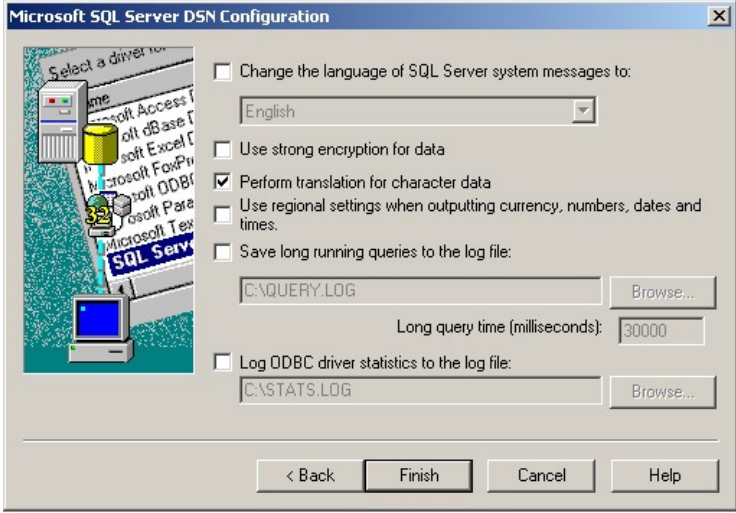
2. ProTrend Tool

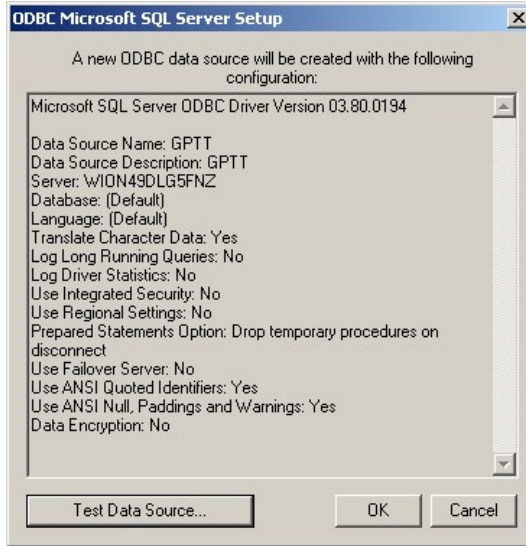
2.1. Initial ProTrend Tool Configuration

Step	Action
7	Click Next to continue. 
8	Select the With SQL Server authentication for login ID verification and check the Connect to SQL Server box. Enter "gptt" in the Login ID and Password fields and then click Next. 

2. ProTrend Tool

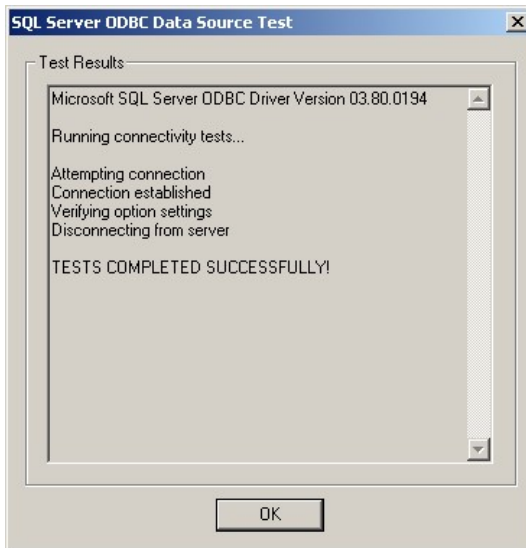
2.1. Initial ProTrend Tool Configuration

Step	Action
9	Make sure the Use ANSI quoted identifiers and Use ANSI nulls boxes are checked and then click Next to continue. Continue to click Next on each of the remaining screens to accept the default configuration. 
10	Click Finish when the final screen in the sequence appears. 

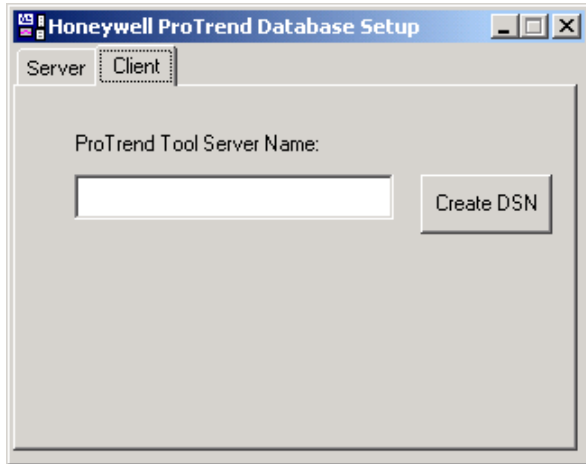
Step	Action
11	When the SQL Server Setup confirmation window appears, click Test Data Source to verify your setup. 

2. ProTrend Tool

2.1. Initial ProTrend Tool Configuration

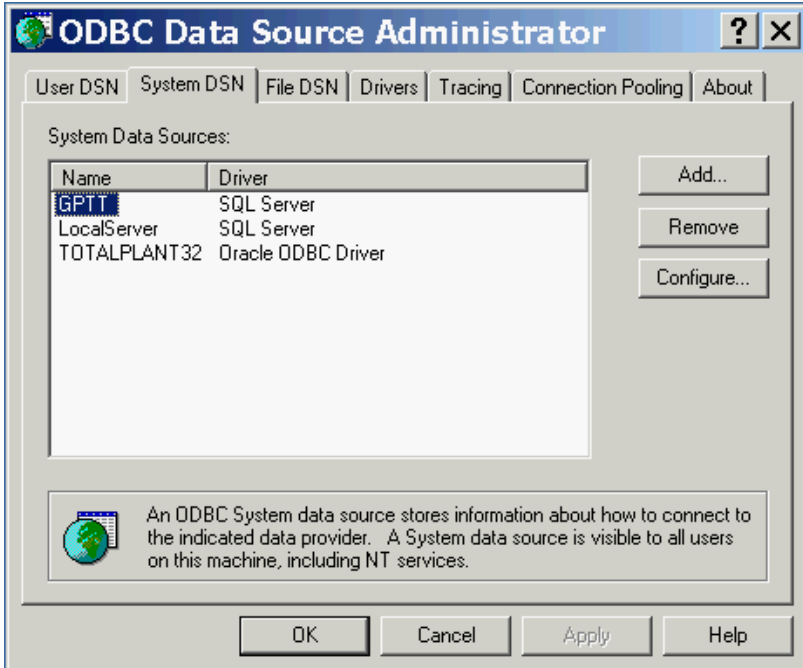
Step	Action
12	Click OK when the test completes successfully, or review and correct your configuration if there are problems. 

Configure the Connection to the ProTrend Tool Database

Step	Action
1	On each computer where the ProTrend Tool is going to be used, launch the ProTrend Database Setup application from the Windows Start menu, by selecting Programs > Honeywell OPS > GUSPro > ProTrend DB Config .
2	In the Client tab, enter the machine name for the computer where the ProTrend Tool database is installed (that is, on the ProTrend Tool Server node) in the ProTrend Tool Server Name text box. Next, click Create DSN. This creates the Data Source Name connection from the local Client node to the database on the Server. 

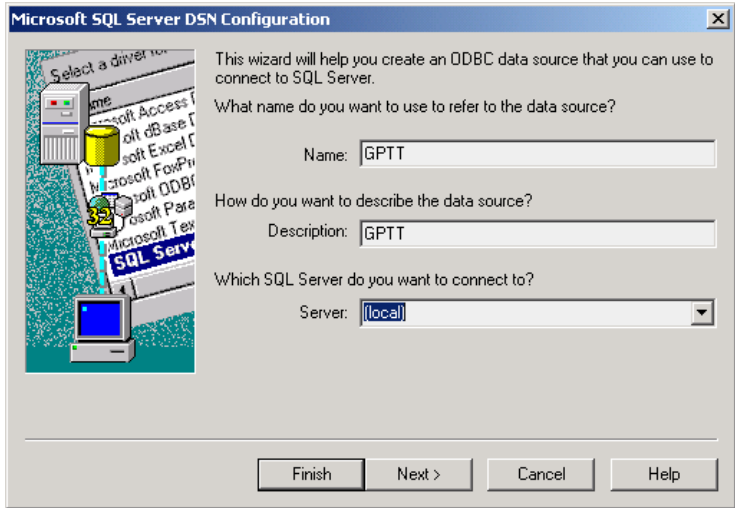
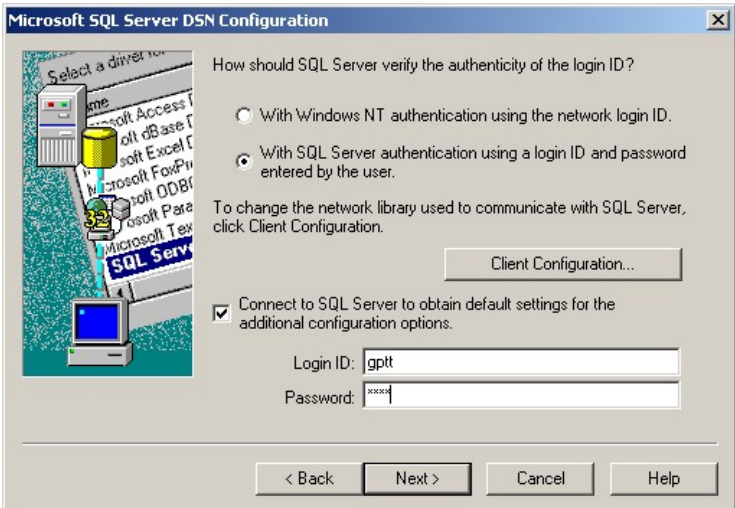
2. ProTrend Tool

2.1. Initial ProTrend Tool Configuration

Step	Action
3	Choose Settings > Control Panel > Administrative Tools from the Windows Start menu and then double click Data Sources (ODBC). Select the System DSN tab and then select GPTT in the System Data Sources field. Click Configure to continue. NOTE: The GPTT System Data Source is created when the Create DSN button was clicked in the preceding step. 

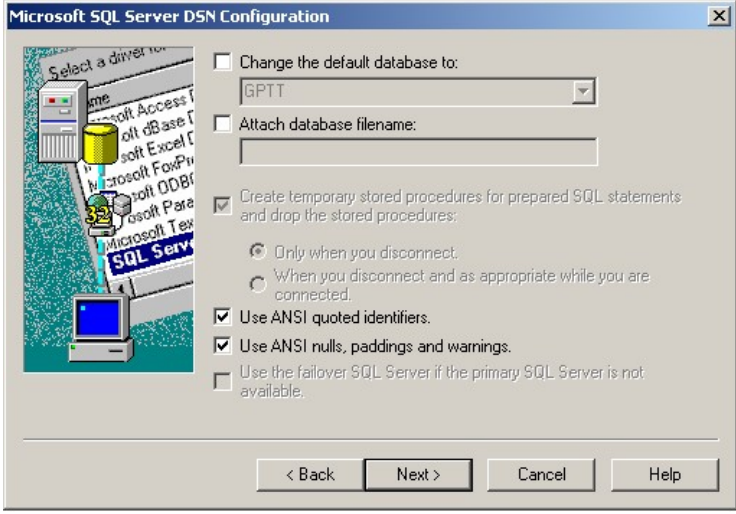
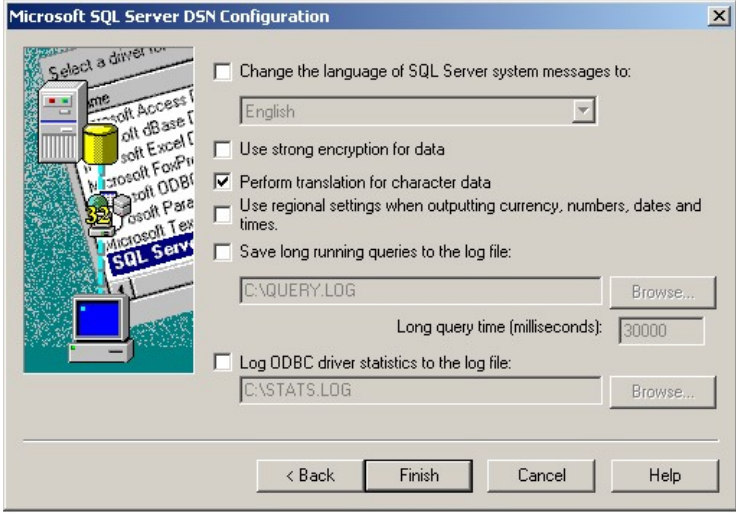
2. ProTrend Tool

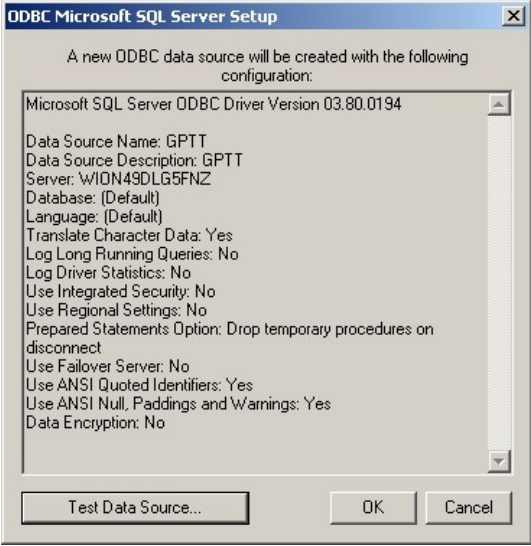
2.1. Initial ProTrend Tool Configuration

Step	Action
4	Enter the name of the node where ProServer is installed "(local)" if this node is the ProServer. Click Next to continue. 
5	Select the With SQL Server authentication for login ID verification and check the Connect to SQL Server box. Enter "gptt" in the Login ID and Password fields and then click Next. 

2. ProTrend Tool

2.1. Initial ProTrend Tool Configuration

Step	Action
6	Make sure the Use ANSI quoted identifiers and Use ANSI nulls boxes are checked and then click Next to continue. Continue to click Next on each of the remaining screens to accept the default configuration. 
7	Click Finish when the final screen in the sequence appears. 

Step	Action
8	When the SQL Server Setup confirmation window appears, click Test Data Source to verify your setup. 
9	Click OK when the test completes successfully, or review and correct your configuration if there are problems. 

2. ProTrend Tool

2.1. Initial ProTrend Tool Configuration

Step	Action
10	Reboot the node.
11	Select Program>Honeywell OPS>GUSPRO>ProTrend DB Configure
12	Select Create/Upgrade Database

2.2 ProTrend Tool Configuration Utility

Purpose

The ProTrend Tool Configuration utility (Figure 7) is used to configure the following ProTrend Tool settings:

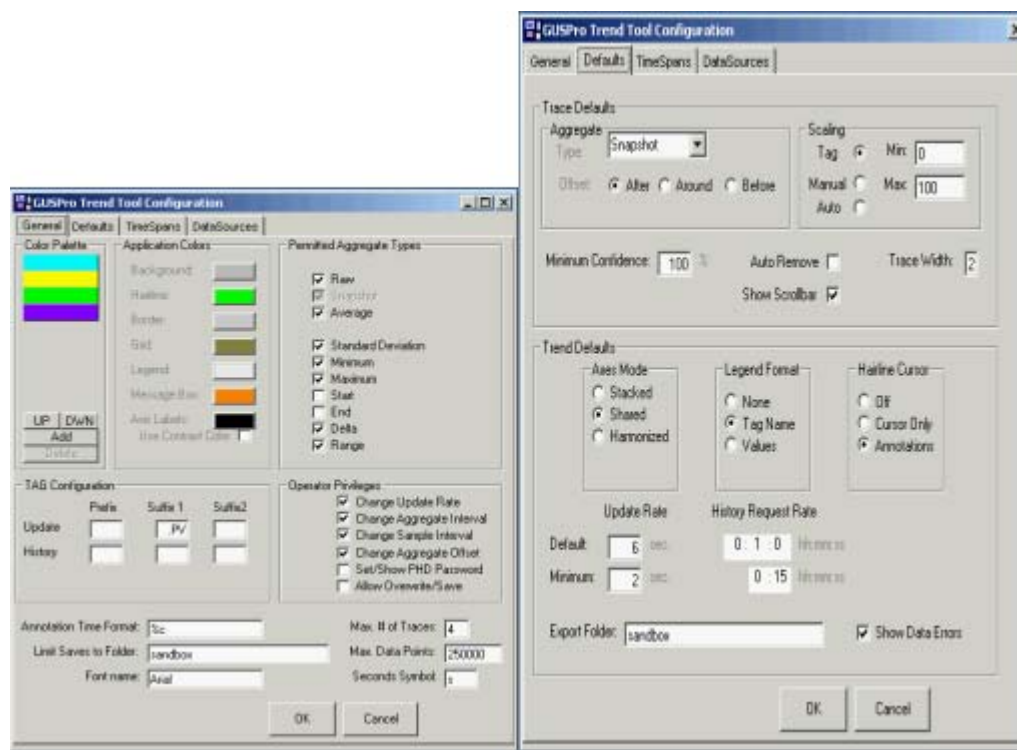
- Trace color palette
- Trend application colors
- Permitted aggregate types
- Real time and historical data tag configuration settings
- Operator privileges (allow online changes to update rate, aggregate interval, sample interval, aggregate offset or overwrite setting)
- Default historical data source (PHD Server)
- Default real time data source (TPN Server)
- Maximum number of traces per trend
- Maximum number of data points
- Default aggregate type and offset
- Default scaling
- Minimum confidence level and automatic removal
- Default update rate and source
- Default axes mode
- Default hairline cursor setting
- Default data error, overwrite and legend settings
- Time span and Aggregate interval options

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

The settings are stored in a console-wide ProTrend Tool database. The same ProTrend Tool display settings are used for every trend in the console.

Figure 7 ProTrend Tool Configuration Utility



2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

The image displays two side-by-side screenshots of the 'GUSPro Trend Tool Configuration' utility. The left window is on the 'TimeSpans' tab, showing settings for 'Visible Time Span' and 'Aggregate Samples'. The right window is on the 'DataSources' tab, showing settings for 'Update Source' and 'History Source'.

Visible Time Span (Left Window):

Title in Correlation	Time Span for Entry	Default
2 min	0 2	☐
5 min	0 5	☐
10 min	0 10	☐
20 min	0 20	☐
30 min	0 30	☐
1 hour	1 0	☒
4 hours	4 0	☐
8 hours	8 0	☐
1 day	1 0 0	☐
1 week	7 0 0	☐
	0 0	☐

Aggregate Samples (Left Window):

Title in Correlation	Time Span for Entry	Default
5 sec	0 0 5	☐
15 sec	0 0 15	☐
30 sec	0 0 30	☐
1 min	0 1 0	☒
6 min	0 6 0	☐
10 min	0 10 0	☐
20 min	0 20 0	☐
30 min	0 30 0	☐
1 hour	1 0 0	☐
4 hour	4 0 0	☐
8 hour	8 0 0	☐
1 day	24 0 0	☐

Update Source (Right Window):

☐ Local ☒ TPN Server ☐ PHD

TPN Alias:

Host (node):

ServerId:

History Source (Right Window):

PHD Host:

PHD Username:

PHD Password:

Access

To access the ProTrend Tool Configuration utility, be sure that you are logged on using an account with local administrator privileges. Launch the GUSPro Configuration utility from the Windows **Start** menu by selecting **Programs > Honeywell OPS > GUSPro > GUSPro Configuration**. In the GUSPro Configuration utility, click the **ProTrend Configuration** button.



ATTENTION

The ProTrend Tool Configuration utility must be run at least once prior to using the ProTrend Tool. Running this utility (and clicking OK or Apply) sets up the default ProTrend Tool database settings needed to run the ProTrend Tool.



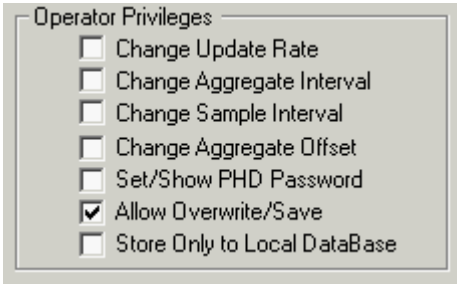
TIP

For consistency, all of the ProTrend Tool configuration tasks must be carried out on one of the GUSPro Client computers. By using the same computer for all of the configuration tasks, duplication of effort and configuration errors can be avoided.

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

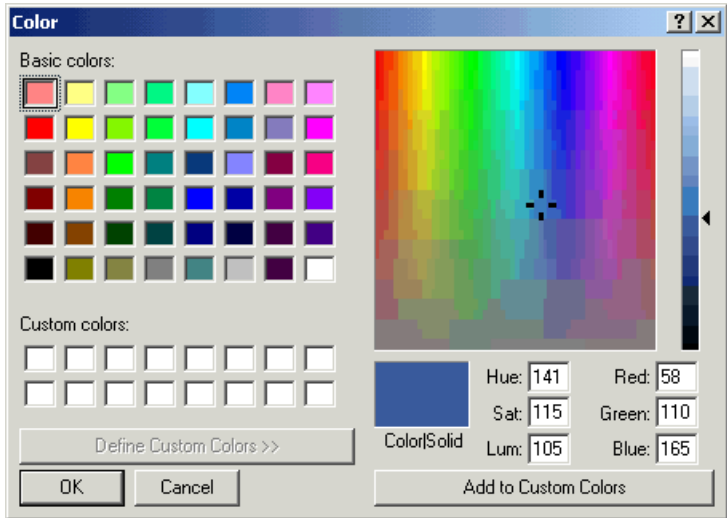
Configure the ProTrend Tool General Settings

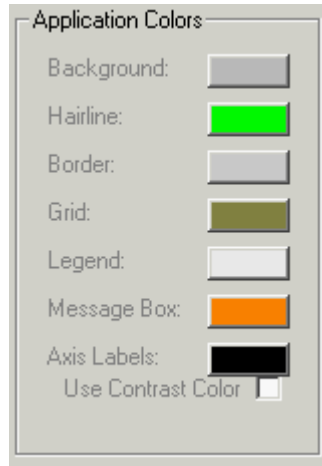
Step	Action
1	In the Operator Privileges region, select the ProTrend Tool adjustments that are permitted on-line.  When one of these options is deselected, it is not possible to change the associated settings from within the ProTrend Tool. NOTE: When these settings are turned off, no one, not even engineers or administrators, can change the associated settings in the ProTrend Tool. If operator access to any of these privileges is to be denied, be sure to set up and save the required trends first. Only after the trends have been created, would these settings be turned off. Set/Show PHD Password: When checked, the user has the option of specifying a user name and password when changing to a different PHD server. This option is relevant only for connections to PHD150. If all the PHD servers are R200 or later, this option must be unchecked. Allow Overwrite/Save: If NOT checked the user is not able to change or delete any stored Trendsets. Even if the option is checked, the common trendsets can be protected from changes/deletions by Limiting the saves to a specific folder. Store Only to Local DataBase: If this option is checked, the ProTrend Tool does not send updates to the remote ProTrend Server. Any files saved by the operator appears on the local node only.

Step	Action
2	In the Color Palette region, identify the standard set of colors for the traces that appear in each ProTrend Tool.  The colored cells represent the colors for each trace, with the first trace displayed in a trend using the top color, the second trace using the next color and so on for each new trace added to a trend. If no colors are defined here, the ProTrend Tool automatically picks colors for each new trace at runtime.
3	To change the trace color order, select a color cell which needs to be changed and click the UP or DWN buttons. The color cell moves one position up or down each time an UP or DWN button is clicked.

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

Step	Action
4	To change an existing trace color, double-click the trace color cell to open the Color dialog box.  You may select one of the basic colors or create a custom color using the color matrix. Once you have chosen the desired color, click OK to apply the new color to the selected trace or click Cancel to cancel the color change operation. NOTE: Duplicate color selections are not accepted.
5	To add a new trace color, click Add. A new color cell is added to the Color Palette and the Color dialog box is launched. Use the Color dialog box to set the new trace color. If the color selection operation is cancelled, the new color cell is deleted from the Color Palette. NOTE: Duplicate color selections are not accepted. The maximum number of traces does <i>not</i> limit the number of trace colors.
6	To delete a predefined trace color, select the color cell and click Delete.

Step	Action
7	In the Color Defaults region, identify the standard set of colors that are used for the trend background, hairline cursor, border, grid, legend and message box. The message box appears between the trend pane and the trend control area, whenever the ProTrend Tool generates an error or user input is required. See Trend Control Area section for details.  To change one of the default colors, click the associated color cell to launch the Color dialog box. Use the dialog box to select a new color. NOTE: If the “Use Contrast Color” box is checked, then the color of the Axis labels are determined at runtime to be either black or white (whichever provides the greatest contrast to the background/trace color).

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

Step	Action
8	In the Permitted Aggregate Types region, select the ProTrend Tool data aggregate types that are permitted for use. Permitted Aggregate Types ☒ Raw ☒ Snapshot ☒ Average ☒ Standard Deviation ☒ Minimum ☒ Maximum ☒ Start ☒ End ☒ Delta ☒ Range Data aggregates are values that are calculated for a set of historical data values over a specified time interval. See Data Aggregates section for a description of the available aggregate types. Snapshot is the default aggregate type and its use is always permitted. As a result, the associated checkbox cannot be deselected.

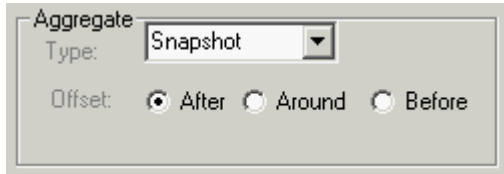
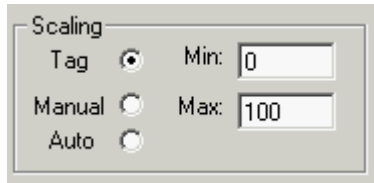
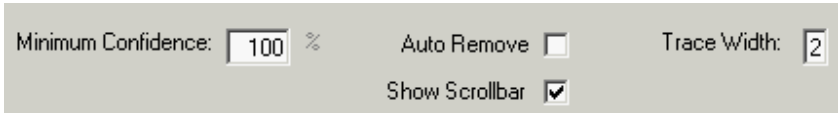
Step	Action												
9	The Tag Configuration region settings may be used to define default tag name prefixes and suffixes needed for the real-time (Update) or historical (History) data sources. For the ProTrend Tool, the real-time data source can be either PHD or a TPN data source like the LCN. The historical data source is always PHD. TAG Configuration <table><thead><tr><th></th><th>Prefix</th><th>Suffix 1</th><th>Suffix 2</th></tr></thead><tbody><tr><td>Update</td><td> </td><td> . </td><td> PV </td></tr><tr><td>History</td><td> </td><td> </td><td> </td></tr></tbody></table> When a new tag is added to a ProTrend Tool Trend display (in the Trend tab), the tag name is automatically altered to suit the tag naming conventions used by the update and history data sources. For example, if the update data source is an LCN and the tag IDs follow a standard point.parameter naming convention, then the Tag Configuration settings could be specified as shown in the figure above. If a new tag called ANLZ01 is added to the trend, the associated Update Tag is automatically identified as ANLZ01.PV and the History Tag ID is identified as ANLZ01. With this configuration, the new tags would always show the PV value for the selected tag. NOTE: Users can always manually change the update and history tag IDs after they are entered with the configured settings. (This is done in the ProTrend Tool - Trace tab.)		Prefix	Suffix 1	Suffix 2	Update		.	PV	History			
	Prefix	Suffix 1	Suffix 2										
Update		.	PV										
History													

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

Step	Action
10	In the lower portion of the General Tab, specify the following miscellaneous settings: Annotation Time Format: %c Max. # of Traces: 4 Limit Saves to Folder: sandbox Max. Data Points: 250000 Font name: Arial Seconds Symbol: \$ Annotation Time Format: Set the format string that determines how the timestamp appears on Hairline cursor annotations. “%c” uses the current system format. See 5.2.15 for more details. Limit Saves to Folder: This must be set to blank until the site’s common trendsets have been defined. When a folder name is set, users of the TrendTool are able to save and delete trends only within this folder. Trendsets in other folders are effectively Read-Only. Font name: Set the font to be used in the Trend Tool. This must be a TrueType font such as Arial. Maximum # of Traces: Set the maximum number of traces that can be shown on a Trend display. Maximum Data Points: Set the maximum number of data points the ProTrend Tool can collect. This is the total number of samples or snapshots in all traces. This feature is provided to limit the resources the ProTrend Tool can use so that, for example, if left running unattended on a GUS for several days it would not prevent other applications from working properly. When this value is reached, the ProTrend Tool automatically lowers the refresh rate by 10% to reduce the data point total. Seconds Symbol: Identify the symbol used to denote seconds in the time axis of the Trend display. The seconds symbol is limited to three characters.
11	Once the ProTrend Tool configuration settings have been defined, click Ok to write all changes to the Trend database and close the ProTrend Tool Configuration utility. Click Cancel to close the Configuration utility without applying any changes since the last apply.

Configure the ProTrend Tool Default Settings

Step	Action
1	In the Aggregate section of the Trace Defaults region, select a default aggregate type and offset.  See Data Aggregates section, for information about aggregates.
2	In the Scaling section of the Trace Defaults region, select the default trend scaling method.  See Set the Trace Details section, for information about trend scaling.
3	In the Trace Defaults region, enter a minimum confidence level, enable or disable the automatic removal and show scrollbar features, and set the width (in pixels) of trace lines.  See Set the Trace Details section, for information about the minimum confidence level and automatic trace removal.

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

Step	Action
4	In the Axes Mode section of the Trend Defaults region, select the default trend axes. Axes Mode ☐ Stacked ☒ Shared ☐ Harmonized See Trend Pane section, for more information about axes mode.
5	In the Legend Format section, select the contents of the Legend Legend Format ☐ None ☒ Tag Name ☐ Values
6	In the Hairline Cursor section of the Trend Defaults region, select the default hairline cursor setting. Hairline Cursor ☒ Off ☐ Cursor Only ☐ Annotations See Trend Pane section, for more information about the hairline cursor.

Step	Action															
7	In the Trend Defaults region, enter the default and minimum update rates. <table><thead><tr><th></th><th>Update Rate</th><th></th><th>History Request Rate</th><th></th></tr></thead><tbody><tr><td>Default:</td><td> 4 </td><td>sec.</td><td> 0 : 0 : 30 </td><td>hh:mm:ss</td></tr><tr><td>Minimum:</td><td> 2 </td><td>sec.</td><td> 0 : 15 </td><td>hh:mm:ss</td></tr></tbody></table> The Update Rates determine how frequently data is requested to update the traces on the trend. The History Request Rate is the frequency of data points requested from the History source (in hours:minutes:seconds). The user of the TrendTool may be able to change the update rate (based on the Operator Privileges), but those changes are limited to not allow a data collection rate more frequent than the configured Minimum values. See Set the Time Settings section, for more information about update rates.		Update Rate		History Request Rate		Default:	4	sec.	0 : 0 : 30	hh:mm:ss	Minimum:	2	sec.	0 : 15	hh:mm:ss
	Update Rate		History Request Rate													
Default:	4	sec.	0 : 0 : 30	hh:mm:ss												
Minimum:	2	sec.	0 : 15	hh:mm:ss												
8	In the Trend Defaults region, select the default Export Folder and data errors options. <table><tr><td>Export Folder:</td><td> c:\users </td><td>☒ Show Data Errors</td></tr></table> When the TrendTool user specifies a file name for an Export, that filename string are appended to the path configured as the Export Folder in the Windows file system.	Export Folder:	c:\users	☒ Show Data Errors												
Export Folder:	c:\users	☒ Show Data Errors														

Configure the ProTrend Tool Time Spans

Step	Action																																																								
1	In the Visible Time Span region of the TimeSpans tab, set the options that are available for specifying the trends time span. Visible Time Span <table><thead><tr><th>Title in ComboBox</th><th colspan="2">Time Span for Entry</th><th>Default</th></tr><tr><th></th><th>days</th><th>+ hh : mm</th><th></th></tr></thead><tbody><tr><td>2 min.</td><td></td><td>0 : 2</td><td>☐</td></tr><tr><td>5 min.</td><td></td><td>0 : 5</td><td>☐</td></tr><tr><td>10 min.</td><td></td><td>0 : 10</td><td>☐</td></tr><tr><td>20 min.</td><td></td><td>0 : 20</td><td>☐</td></tr><tr><td>30 min.</td><td></td><td>0 : 30</td><td>☐</td></tr><tr><td>1 hour</td><td></td><td>1 : 0</td><td>☒</td></tr><tr><td>4 hours</td><td></td><td>4 : 0</td><td>☐</td></tr><tr><td>8 hours</td><td></td><td>8 : 0</td><td>☐</td></tr><tr><td>1 day</td><td>1</td><td>0 : 0</td><td>☐</td></tr><tr><td>1 week</td><td>7</td><td>0 : 0</td><td>☐</td></tr><tr><td></td><td></td><td>0 : 0</td><td>☐</td></tr><tr><td></td><td></td><td>0 : 0</td><td>☐</td></tr></tbody></table> Defaults Clear	Title in ComboBox	Time Span for Entry		Default		days	+ hh : mm		2 min.		0 : 2	☐	5 min.		0 : 5	☐	10 min.		0 : 10	☐	20 min.		0 : 20	☐	30 min.		0 : 30	☐	1 hour		1 : 0	☒	4 hours		4 : 0	☐	8 hours		8 : 0	☐	1 day	1	0 : 0	☐	1 week	7	0 : 0	☐			0 : 0	☐			0 : 0	☐
Title in ComboBox	Time Span for Entry		Default																																																						
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8 hours		8 : 0	☐																																																						
1 day	1	0 : 0	☐																																																						
1 week	7	0 : 0	☐																																																						
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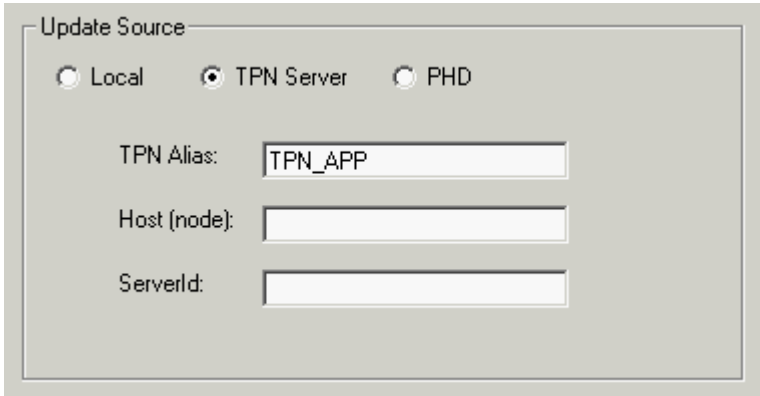
Step	Action																																							
	Up to 12 entries can be specified. Title in ComboBox is the string that the user see in the combo box to select the time span. Blank entries are ignored (not included in the combo box). For each non-blank Title, set the corresponding time period (in days, hours and minutes). Select one of the entries to be the default. See Set the Time Settings section, for more information about the time span.																																							
2	In the Aggregate Samples section, set the options that be available Aggregate Samples <table><thead><tr><th>Title in ComboBox</th><th>Time Span for Entry hh : mm : ss</th><th>Default</th></tr></thead><tbody><tr><td>5 sec.</td><td>0 : 0 : 5</td><td>☐</td></tr><tr><td>15 sec.</td><td>0 : 0 : 15</td><td>☐</td></tr><tr><td>30 sec.</td><td>0 : 0 : 30</td><td>☐</td></tr><tr><td>1 min.</td><td>0 : 1 : 0</td><td>☒</td></tr><tr><td>6 min.</td><td>0 : 6 : 0</td><td>☐</td></tr><tr><td>10 min.</td><td>0 : 10 : 0</td><td>☐</td></tr><tr><td>20 min.</td><td>0 : 20 : 0</td><td>☐</td></tr><tr><td>30 min.</td><td>0 : 30 : 0</td><td>☐</td></tr><tr><td>1 hour</td><td>1 : 0 : 0</td><td>☐</td></tr><tr><td>4 hour</td><td>4 : 0 : 0</td><td>☐</td></tr><tr><td>8 hour</td><td>8 : 0 : 0</td><td>☐</td></tr><tr><td>1 day</td><td>24 : 0 : 0</td><td>☐</td></tr></tbody></table> Defaults Clear	Title in ComboBox	Time Span for Entry hh : mm : ss	Default	5 sec.	0 : 0 : 5	☐	15 sec.	0 : 0 : 15	☐	30 sec.	0 : 0 : 30	☐	1 min.	0 : 1 : 0	☒	6 min.	0 : 6 : 0	☐	10 min.	0 : 10 : 0	☐	20 min.	0 : 20 : 0	☐	30 min.	0 : 30 : 0	☐	1 hour	1 : 0 : 0	☐	4 hour	4 : 0 : 0	☐	8 hour	8 : 0 : 0	☐	1 day	24 : 0 : 0	☐
Title in ComboBox	Time Span for Entry hh : mm : ss	Default																																						
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30 min.	0 : 30 : 0	☐																																						
1 hour	1 : 0 : 0	☐																																						
4 hour	4 : 0 : 0	☐																																						
8 hour	8 : 0 : 0	☐																																						
1 day	24 : 0 : 0	☐																																						

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

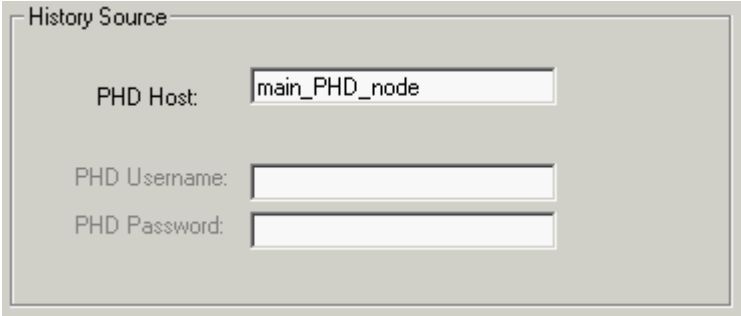
Step	Action
	Up to 12 entries can be specified. Title in ComboBox is the string that the user sees in the combo box to select the Aggregate size. Blank entries are ignored (not included in the combo box). For each non-blank Title, set the corresponding time period (in hours, minutes and seconds_. Select one of the entries to be the default. See Data Aggregates section, for information about aggregates.

Configure the ProTrend Tool Data Sources

Step	Action
1	In the Update Source section of the DataSources tab, select the default Update data source.  Local selects the private TPN server. TPN Server selects a public TPN/HCI server. PHD selects the History Source to also be used as the Update Source TPN Alias specifies a public TPN server. Host and ServerId can be used to specify a TPN Server. The Host can be either the computer name or its IP address. For a TPN server, if the ServerId is left blank, the Trend Tool defaults to "Hci.TPNServer". See Set the Trend Details section, for more about the default data source. NOTE: If the TPN Alias is specified, it can be used to resolve the Host node at runtime. The Host and ServerID are only used if the TPN Alias field is left blank.

2. ProTrend Tool

2.2. ProTrend Tool Configuration Utility

Step	Action
2	In the History Source section of the DataSources tab, select the default History data source.  If the PHD Host text box is left blank, the PHD Server identified in the local PHD_HOST environment variable are used instead. PHD_HOST is already defined on each ProTrend Tool client computer if PHD has been installed on that machine; otherwise, create it manually, if it is to be used. For PHD150, the PHD Username and PHD Password can be specified.
	ATTENTION: If there is more than one PHD Server in the system, the PHD_HOST environment variable can be used to identify which PHD Server is to be used by the local version of the ProTrend Tool. In this case, leave the PHD Host setting blank or else the PHD_HOST environment variable is ignored. The PHD Host setting does not affect the PHD_HOST environment variable.

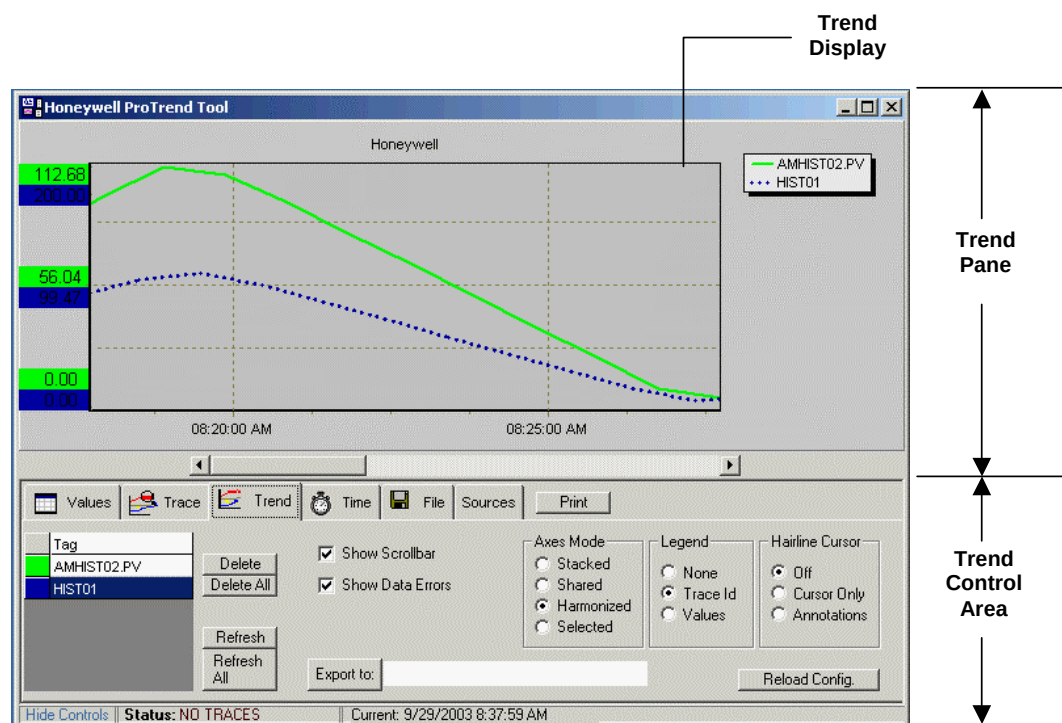
2.3 ProTrend Tool User Interface

Purpose

The ProTrend Tool, shown in Figure 8, provides the following functions and features:

- Acts as a container for the ProTrend Object, which displays the trend process data.
- Provides the operator interface for adding and deleting traces, and for updating their properties.
- Provides an interface that can be used by other software components to add traces. This interface is documented in scripting examples provided in Section ProTrend Tool Interface Examples.

Figure 8 ProTrend Tool



2. ProTrend Tool

2.3. ProTrend Tool User Interface

Access

The ProTrend Tool can be accessed in any of the following ways:

- From the ProMenu for a selected tag, the operator can trend that tag in the ProTrend Tool, by selecting the Trend Tag Link button (if configured).
- From the Windows **Start** menu select **Programs > Honeywell OPS > GUSPro > ProTrend Tool**.

Only one instance of the ProTrend Tool can run on a given node at a time. If there is a request to trend a tag while the ProTrend Tool is running, the tag is added to the current Trend display. If a new ProTrend Tool file is requested, the current trend is replaced by the requested trend.

Display Components

The ProTrend Tool display is broken down into two sections: Trend Pane and Trend Control Area, which are described in the following sections.

Trend Pane Functions and Features

The Trend pane displays a single instance of the ProTrend Object, including the trend traces, axes, and legend. The ProTrend Object provides the following functions and features:

Zoom

The Zoom feature may be used to get a closer look at portions of the traces in the trend with Shared or Selected axes. This is done using the left mouse button to drag a rectangle, down and to the right, over the area you wish to zoom in on. When the left mouse button is released, the trend is redrawn to match the selected area. The time axis and y-axis scales are updated to show the new time and value ranges.

While the Trend display is zoomed, a ZoomOut button is visible in the top-right of Trend Control Area. Tag values continue to be collected, but the Trend display does not scroll.

To Zoom out, click on the “Zoom Out” button or use the left mouse button to drag a rectangle up and to the left. This action causes the Trend display to return to 100%. The y-axis scales return to the settings that existed before the Trend display was zoomed in. In addition, the “Zoom Out” button resets the right side of the trend to the current time.



TIP

Zooming out by dragging with the left mouse button returns the times what they were when the zoom was done, this is no longer the current time. It is necessary to scroll forward to see the dynamic trace updates again.

Time Axis Scrolling

To view historical trend data beyond that which is currently visible in the Trend display, the entire Trend display can be scrolled backward and forward in time. To do this, right-click inside the Trend display and then, while holding the right mouse button down, drag the Trend display in the desired direction. When the right mouse button is released, the Trend display remains in its current position.

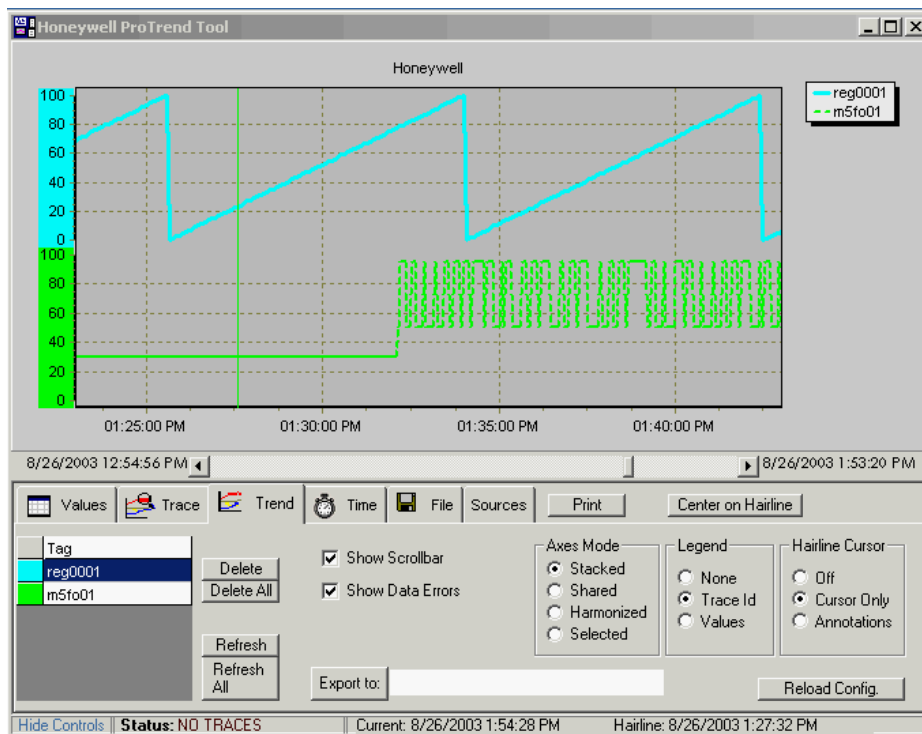
When the trend has been scrolled so that the current time is not display, a “Now” button is visible. Click on that button to jump the display to the current time.

The optional time scroll bar, shown below, can also be used to scroll the Trend display and to load additional history data. See Set the Time Settings section for details.

2. ProTrend Tool

2.3. ProTrend Tool User Interface

When a time increment in the time axis crosses a date boundary, the date is displayed.

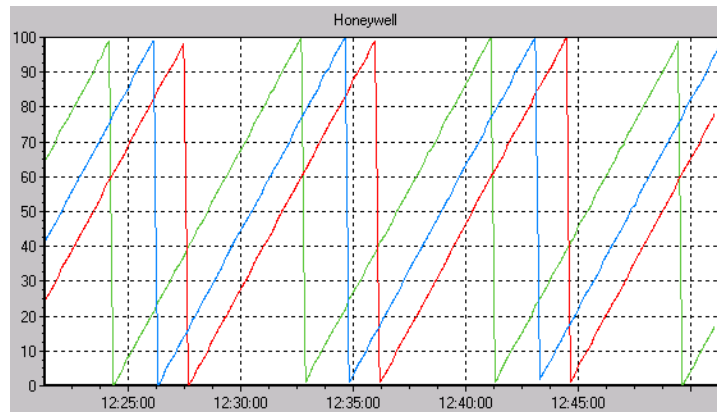


Axes Modes

Users may change the way that multiple trend traces are displayed by changing the axes mode for the Trend display. (Axes mode selection is provided in the Trend tab in the Trend Control Area of the ProTrend Tool.)

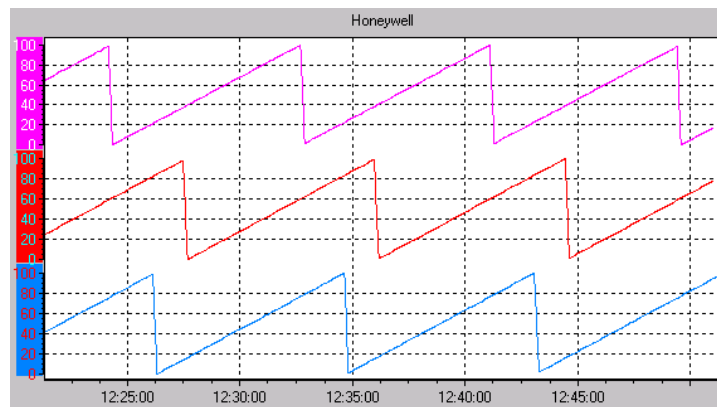
Axes Mode - Shared

By default, the axes mode for a new trend is **Shared**. In this mode, all of the traces share a common y-axis, which is automatically scaled to suit the currently displayed values.



Axes Mode - Stacked

When the **Stacked** axes mode is selected, each trace occupies its own horizontal zone, with its equal share of the available Trend display space. A separate, color-coded y-axis is displayed for each trace. Stacked mode must not be used when there are more than ten traces on the trend.



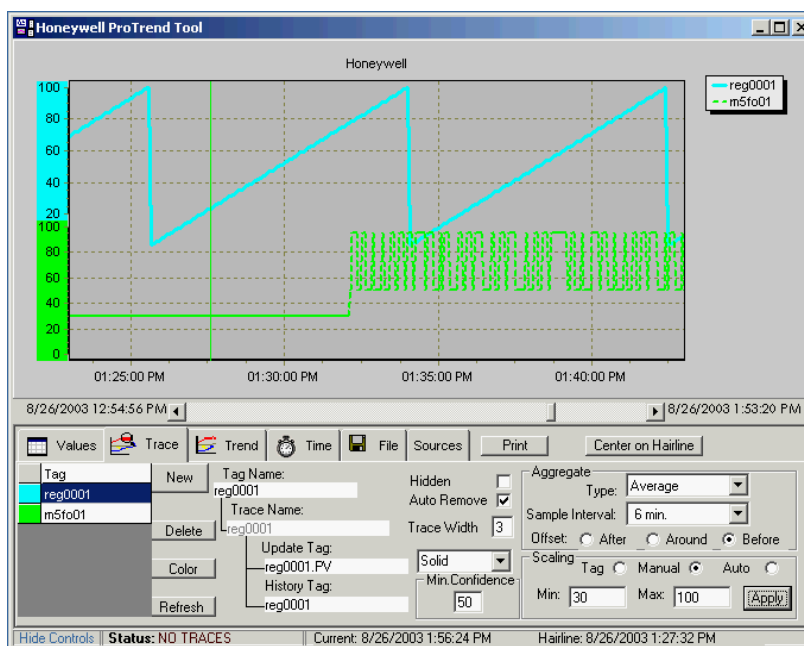
TIP

When AxesMode is set to Stacked, and a Trace is set to Manual scaling with a range that is smaller than the plotted values, the plot for that trace is extended beyond its stacked area, potentially overlapping with another trace (see example below).

If this effect is not wanted, set the scaling mode to automatic or set the manual range to values that are larger than the range of points being plotted.

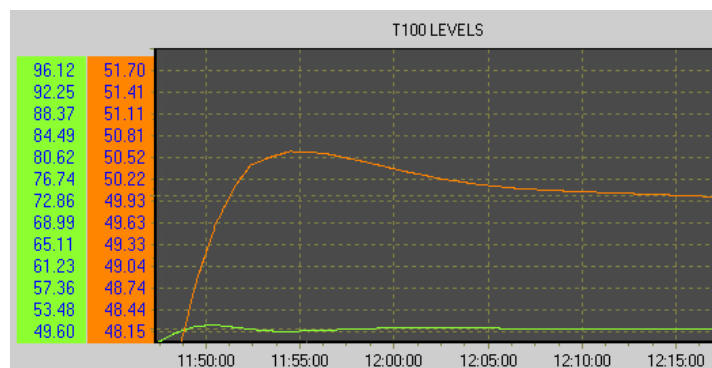
2. ProTrend Tool

2.3. ProTrend Tool User Interface

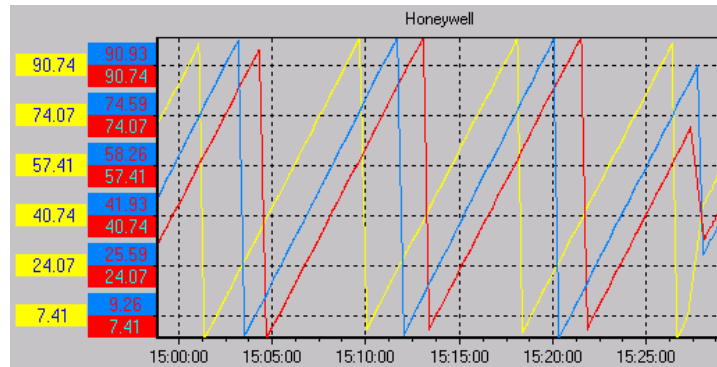


Axes Mode - Harmonized

When the **Harmonized** axes mode option is selected, each displayed trace is scaled so that it can share the same grid lines with the other visible traces in the Trend display. A separate, color-coded, y-axis is displayed for each trace, with scale values appearing to the left of each major tick mark.

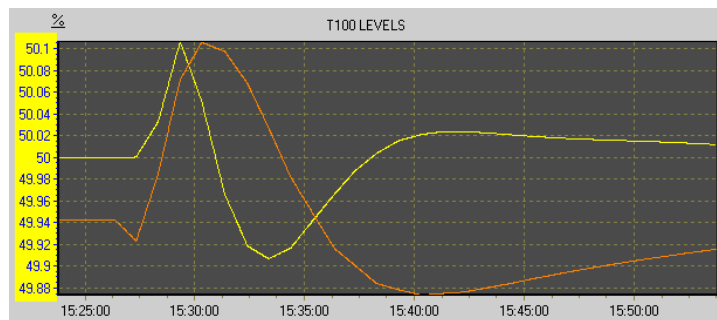


When there are more than two traces, the y-axes are replaced by color-coded value boxes corresponding to each y-axis grid line, as shown in the following figure.



Axes Mode - Selected

If the **Selected** axes mode option is selected, only the y-axis for the currently selected tag is displayed. The y-axis background color is changed to match the color of the selected trace. In addition, the engineering units for the point are displayed above the y-axis scale. The other traces in the display are redrawn to suit the new scale.



The tags may be selected in either the Trace Details or the Trend tabs.

Hairline Cursor

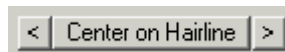
As an aid for determining exact point values within the Trend display, a hairline cursor feature is provided. When the hairline cursor is turned on, a vertical cursor appears in the Trend display and the values at the current hairline position are displayed in the Values tab.

2. ProTrend Tool

2.3. ProTrend Tool User Interface

The hairline cursor is positioned on the trend by a single left mouse click. To move of the hairline simply move the cursor over the trend area. The hairline follows the cursor while it is within the Trend display.

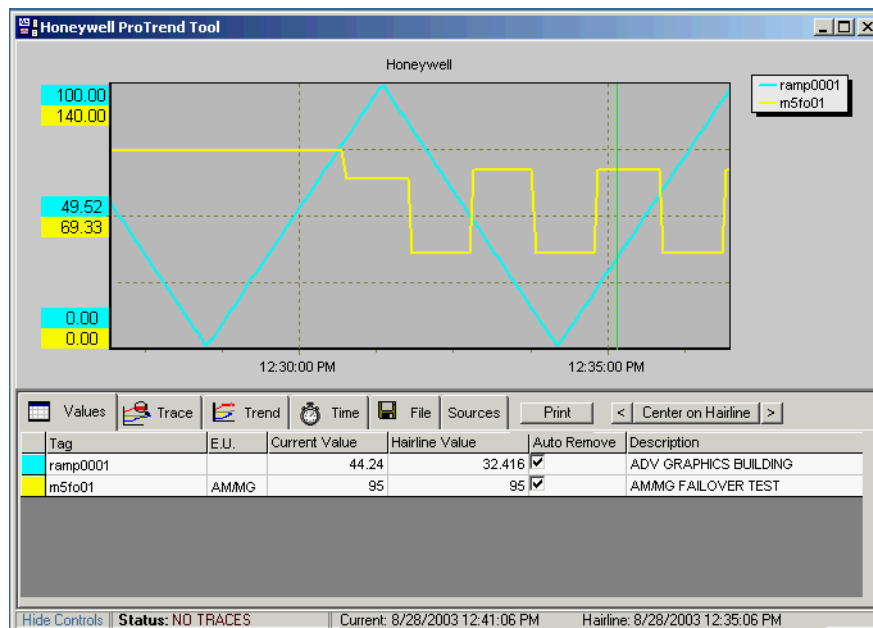
While the hairline cursor is active, a set of “Hairline” buttons is visible in the top-right portion of the Trend Control Area.



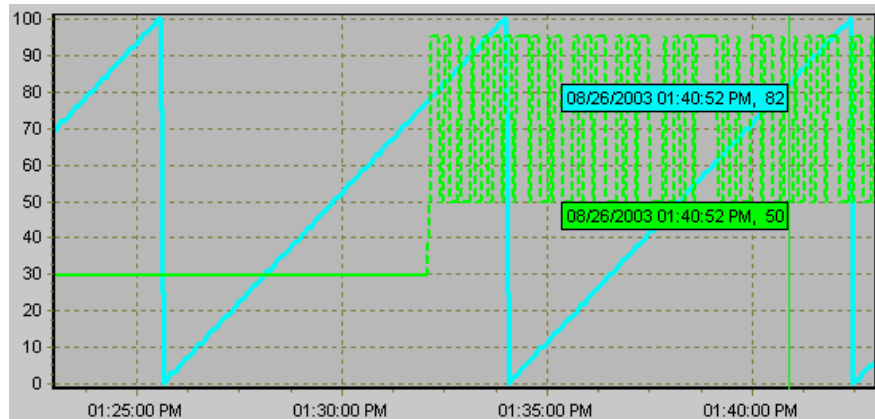
Clicking the “<” or “>” button moves the cursor to the next data point on the left of right respectively. Alternately **PageUp** moves the cursor left and **PageDown** moves it right.

Clicking the “Center on Hairline” button causes the Trend display to be redrawn with the Hairline cursor time for the selected trace centered in the visible time span.

NOTE: Hairline cursor selection is provided in the Trend tab, in the Trend Control Area of the ProTrend Tool.



If the hairline cursor annotations option is selected, the date and time at the hairline is displayed next to the trace, as well as the values for each point that existed at the hairline time.



If the point values for two or more tags are very close together at the hairline, the annotations overlap.

To move of the hairline simply move the cursor over the trend area. The hairline follows the cursor while it is within the Trend display.

While the hairline cursor is active, a “Center on Hairline” button is visible in the top-right portion of the Trend Control Area. Clicking this button causes the Trend display to be redrawn with the time at the Hairline cursor centered in the visible time span.

Trend Control Area

The Trend Control Area provides the tools used to control the contents of the Trend pane, including adding or removing trend traces, controlling the trend appearance, changing the aggregate type and associated settings. The tabs in the Trend Control Area are described in detail in the following subsections.

Print Button

The Print button sends a bitmap image of the entire TrendTool window to the default printer.

Common Tab Components

The Trend Control Area includes the status bar shown below.

Figure 9 Trend Status Bar

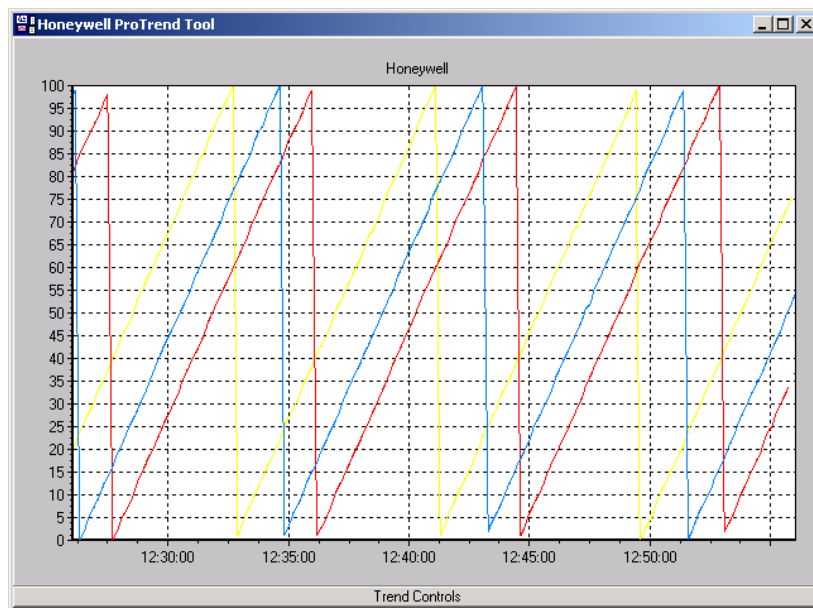


2. ProTrend Tool

2.3. ProTrend Tool User Interface

The status bar shows the overall trend status, the current trend time and in some cases the hairline value for a selected trace. The **Hide Controls** link removes the Trend Controls, allowing the Trend Pane to expand to fill the entire window. When this occurs, a **Trend Controls** button appears just above the status bar at the bottom of the display. Clicking this button restores the Trend Control Area.

Figure 10 Hidden Trend Controls

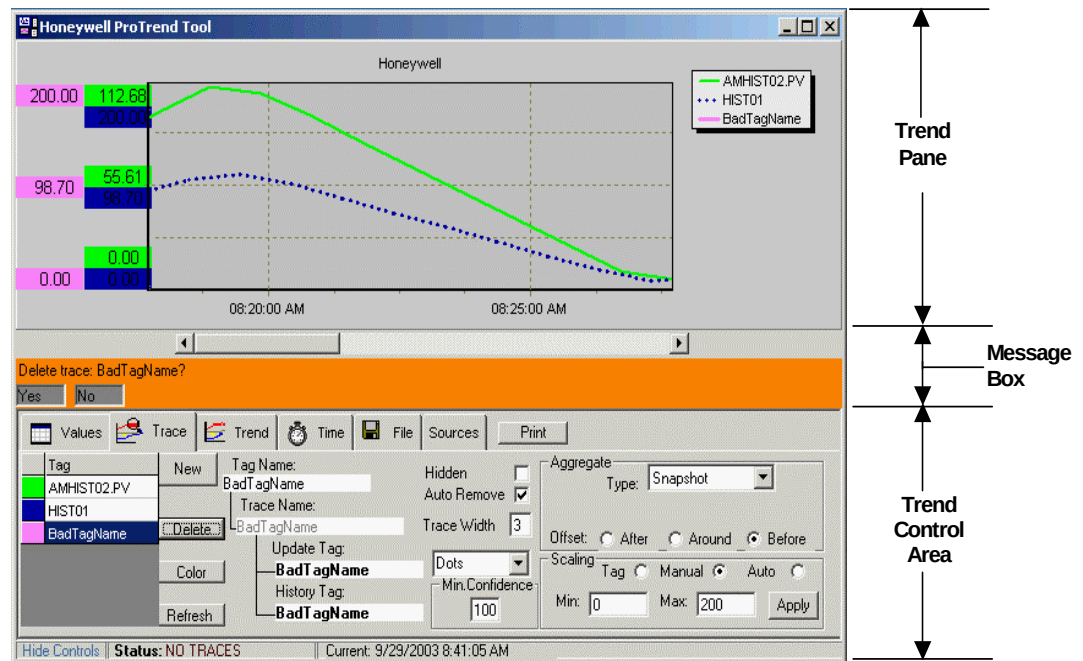


TIP

When the trend controls are hidden, the status bar still appears at the bottom of the trend tool. There is no button appearing with the words "Trend Controls". Instead, the user can click on the status bar on the text "Show Controls" to view the trend controls again.

Whenever the ProTrend Tool raises an error or the user needs to provide input, the error or message text appears between the Trend pane and the Trend Control Area as shown below. Once the message has been acknowledged, the message area disappears.

Figure 11 Trend Message Area



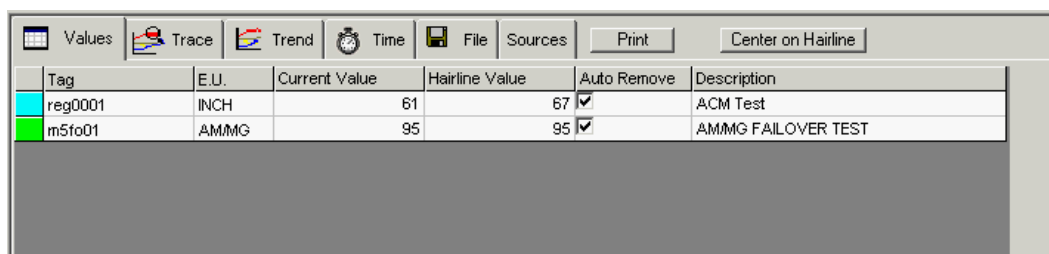
2. ProTrend Tool

2.3. ProTrend Tool User Interface

Review the Current Trend Trace Status

The Values tab shows detailed information for each trace currently displayed in the Trend pane.

Figure 12 Value Tab



Tag	E.U.	Current Value	Hairline Value	Auto Remove	Description
reg0001	INCH	61	67	☒	ACM Test
m5fo01	AMMG	95	95	☒	AMMG FAILOVER TEST

The following information is displayed for each trace:

Display Item	Description
Tag	The tag ID for the trace, as entered on the Trend tab. NOTE: The cell to the left of the tag name shows the trace color that corresponds to that tag in the Trend pane.
E.U.	The Engineering Units for the Tag, if available.
Current Value	The latest value for the tag. If the Trend display has been zoomed in or scrolled to show historical data, this value may not be visible in the Trend display.
Hairline Value	Current tag value at the hairline, shown only if the Hairline Cursor option is selected in the Trend tab.

Display Item	Description
Auto Remove	Determines whether or not the trace can be deleted when a new trace is requested in the Trend display. Normally when a new trace is requested it is simply added. However, if the maximum number of traces for the current Trend display has been reached, one of the traces must be deleted before the new trace can be added. The Auto Remove setting for each trace in the Trend display is used to determine which trace is deleted. Those traces that have Auto Remove selected (that is, the associated check box is checked) are deleted first, beginning at the top of the list of tags. Those traces that do not have Auto Remove selected (that is, the associated check box is not checked) are not deleted automatically. If Auto Remove is turned off for all of the traces in the display, an error is generated when an existing trace needs to be removed.
Description	Tag description, if available.

In addition to the individual trace information, the following information for the entire trend is also displayed:

Rightmost Time	Shows the date and time for the newest trend value displayed. For live trends, the Rightmost Time is the current date and time. For trends that display historical data, the Rightmost Time is the date and time that corresponds to the last value in the Trend (that is, the value at the furthest right)
Hairline Time	Shows the date and time at the hairline cursor's current position in the Trend display. The Hairline Time is only displayed if the Hairline Cursor option is selected in the Trend tab.

2. ProTrend Tool

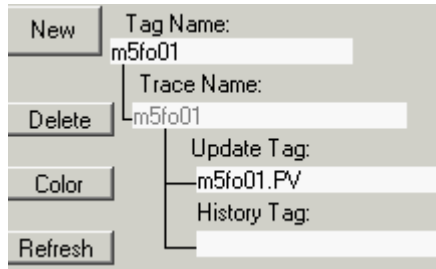
2.3. ProTrend Tool User Interface

Set the Trace Details

The Trace tab may be used to alter the settings for each trace currently displayed in the Trend pane.

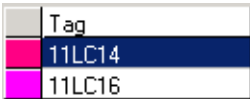
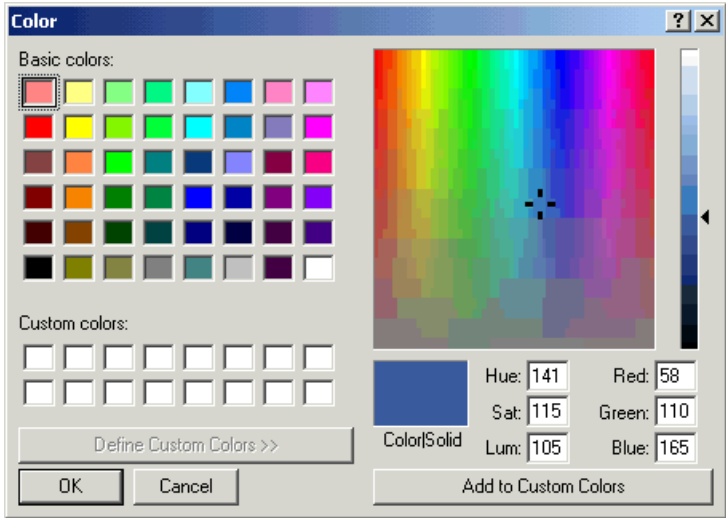
Figure 13 Trace Tab

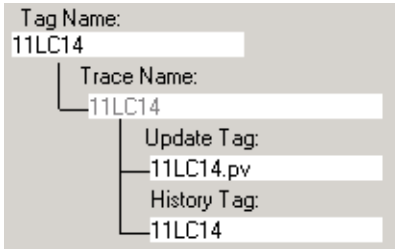
The screenshot shows the 'Trace' tab in the ProTrend Tool. The interface includes a menu bar with 'Values', 'Trace', 'Trend', 'Time', 'File', and 'Sources'. Below the menu bar, there are buttons for 'Print' and 'Center on Hairline'. The main area is divided into several sections: a list of tags on the left, a 'New' button, a 'Delete' button, a 'Color' button, and a 'Refresh' button. The 'Tag' list contains 'reg0001' and 'm5fo01'. The 'New' button is labeled 'Tag Name: m5fo01'. The 'Delete' button is labeled 'Trace Name: m5fo01'. The 'Color' button is labeled 'Update Tag: m5fo01.PV'. The 'Refresh' button is labeled 'History Tag:'. The right side of the interface contains settings for 'Aggregate Type' (set to 'Average'), 'Sample Interval' (set to '6 min.'), 'Offset' (radio buttons for 'After', 'Around', and 'Before'), 'Dashes' (a dropdown menu), 'Min. Confidence' (set to '50'), and 'Scaling' (radio buttons for 'Tag', 'Manual', and 'Auto').

Step	Action
1	To add a new trace tag, proceed as follows:  - If the top left button is New, click the New button. This clears the tag name area and change the button options to Add and Cancel. - Enter the tag ID in the Tag text box. When you enter a name in the Tag text box, the same text automatically appears in the Trace Name text box and the Update Tag and History Tag entries are automatically generated based on the Tag Configuration settings configured in the ProTrend Tool Configuration utility. See ProTrend Tool Configuration Utility section for details - You can change the Trace Name entry if you want the legend entry for this tag to be different from the tag ID. - Confirm that the Update Tag, History Tag and all of the other options are set as desired. It is best to make any changes before Adding the trace. - Click Add. The identified tag is added to the Tag list. A corresponding trace is added to the Trend display, using the next available trace color (as configured in the ProTrend Tool Configuration utility). The Trace Name text appears is added to the legend, if the legend is visible. The cancel button cancels the pending addition without adding a trace. NOTE: Once you click the Add button, the only way to change the Trace Name that appears in the legend is to delete the trace and then add it again with the new name.

2. ProTrend Tool

2.3. ProTrend Tool User Interface

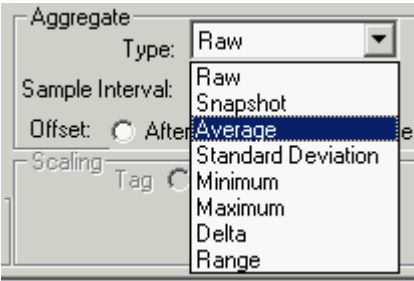
Step	Action
2	To change the trace settings for a particular tag, click on the tag name to select it.  When a tag trace is selected, the rest of the settings in the Trace tab apply to that tag.
3	To delete the currently selected tag, click the Delete button.
4	To redraw the currently selected tag with a fresh read of history data, click the Refresh button.
5	To change the trace color, click the Color button to open the Color dialog box.  You may select one of the basic colors or create a custom color using the color matrix. Once you have chosen the desired color, click OK to apply the new color to the selected trace or click Cancel to cancel the color change operation. NOTE: Duplicate trace color selections are not accepted.

Step	Action
6	To change the tag IDs for the selected tag, change the displayed Update Tag or History Tag entries as required.  The Update Tag is the tag ID used to access real-time data from the Update Source (either PHD or LCN) identified in the Trend tab. The History Tag is the tag ID used to access historical data from the historical data source. For the ProTrend Tool, the historical data source is always PHD. The Update Tag and History Tag entries are automatically generated from the Tag identified in the Trend tab and any Tag Configuration settings configured in the ProTrend Tool Configuration utility. (See ProTrend Tool Configuration Utility section for details.)
7	You may hide a trace by checking the Hidden checkbox. A hidden trace is not shown on the Values tab nor drawn on the trend area, but the data continues to be collected.
8	Use the Auto Remove check box to control whether or not the trace can be deleted when a new trace is requested in the Trend display. (This check box is identical to the Auto Remove check box in the Values tab.)
9	You may change the appearance of the trace line by entering a new Trace Width and/or selecting a different Line Style from the combo box.

2. ProTrend Tool

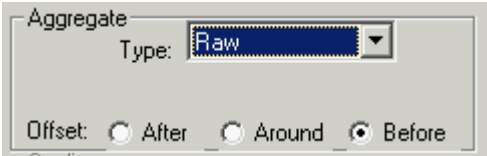
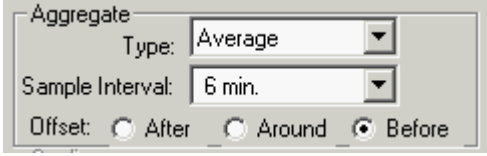
2.3. ProTrend Tool User Interface

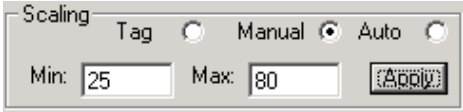
Step	Action
10	Each time a value is collected for a tag from its assigned data source, the data source provides an indication of the quality of that value as a percent. When a value is deemed to be correct, it is assigned the highest possible quality value of 100% confidence. For example, this would be the case for a real-time data value collected directly from a measuring device in the field. If an estimated value is supplied by the data source, the percent confidence for that value is reduced. This would be the case for historical data sources that may supply an interpolated value due to missing or uncertain values. In order to ensure that the values displayed in a given trace are of reasonable quality, the Minimum Confidence text box shows the lowest allowable confidence value (that is, the confidence threshold) for the supplied tag data. If a supplied data value has a percent confidence that is less than the minimum confidence value, its value is not displayed in the Trend display. The valid portion continues to be displayed and the last valid value is displayed on the Values tab. Once the valid data values are generated, the trace is restarted, leaving a gap in the trace. To change the minimum confidence level for the tag, type a number in the Minimum Confidence text box. If you enter a number with decimals, it is automatically rounded to the nearest integer value. A default minimum confidence level for all traces is defined in the ProTrend Tool Configuration utility. (See the ProTrend Tool Configuration Utility section for details.)

Step	Action
11	If access is permitted (as specified with the ProTrend Tool Configuration utility), configure the aggregate settings for the trace.  Use the Type list box to select the desired aggregate type for the trace. Data aggregates are values calculated for a set of historical data values over a specified period. See Data Aggregates section, for information. NOTE: The available aggregate types are defined in the ProTrend Tool Configuration utility. Whenever the aggregate type is changed, the change goes into effect immediately and the trace is redrawn. Aggregate type changes only apply to the selected tag.
	TIP When plotting a trend of static PHD data with PHD as the update source, you may get a PHD error message, if the aggregate type is set to Average or Raw. The cause of the error may be that PHD is not collecting the data unless it changes. If you ask the trend to collect the data for the current time, no data exists on PHD. Use a snapshot aggregate type or set the PHD system to refresh the data at an appropriate period. An appropriate setting to avoid the error message would be for the PHD parameter MAX_STOREDELTA to be less than or equal to the Update Period of the trend. Otherwise, ignore the error message.

2. ProTrend Tool

2.3. ProTrend Tool User Interface

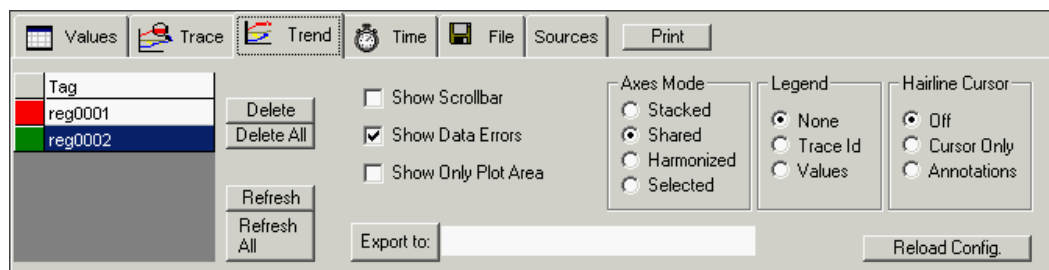
Step	Action
12	If access is permitted, Depending on the aggregate type selected, configure the aggregate interval settings as follows: For Raw and Snapshot data collection, the data supplied is not averaged in any way. The current value at data collection time is supplied as is.  For Average data aggregates, specify a Sample Interval. The sample interval defines how the aggregate interval is divided.  For all of the other data aggregate types, specify both a Sample Interval and an aggregate interval (Interval). Depending on how the ProTrend Tool has been configured, the other possible data aggregate types are: Minimum, Maximum, Start, End, Delta, Standard Deviation and Range.
13	If access is permitted, select one of the following aggregate offset options: After The interval follows the time of the aggregate. Around The time of the aggregate is in the middle of the interval. Before The interval precedes the time of the aggregate.

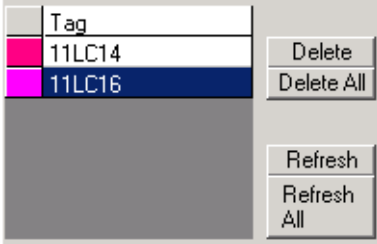
Step	Action
14	To change the trace scaling settings for the selected tag, select one of the scaling options.  Tag The scale is set to the trace's RetrievedRangeLow and RetrievedRangeHigh property values. (See Assigning RetrievedRangeHigh and RetrievedRangeLow Properties section for a detailed description of how the ProTrend Object determines these values based on the current data source type.) Manual The scale is set based on the values entered in the Min and Max text boxes. Auto The scale is set automatically based on the currently displayed values. NOTE: The Auto scaling mode is not appropriate for steady state. If the axes mode, in the Trend tab is set to Shared the ProTrend Object determines y-axis scale values are determined automatically. When this happens, the trace scaling settings cannot be changed, the Scaling label is dimmed and the scaling options cannot be changed.

Set the Trend Details

The Trend tab may be used to add traces to or remove traces from the Trend display. It may also be used to alter the display settings for the entire Trend display.

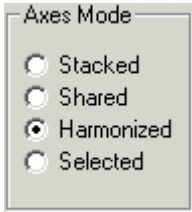
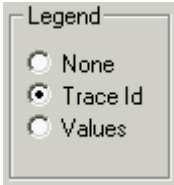
Figure 14 Trend Tab



Step	Action
1	To delete a single trace tag, select the tag and click Delete. To delete all of the tags currently assigned to the Trend display, click Delete All.  Similarly, to redraw a single trace, select the associated tag and click Refresh. To redraw all of the current traces, click Refresh All.
2	To turn the time scroll bar on or off, check or uncheck the Show Scrollbar box. When the time scroll bar is displayed, the dates and times corresponding to the beginning and end of the tag data currently stored in the buffer is displayed at each end of the scroll bar. The date and time to the right of the scroll bar is the date and time corresponding to the earliest data collected. The scroll bar may be used to scroll back and forth between the history data and current update data displayed in the trend.
3	To suppress any data errors generated by the embedded ProTrend Object, deselect the Show Data Errors check box. Selecting Ignore in the Message Box when such an error occurs, also deselects this check box.
4	To give the maximum space for the plotted traces, check Show Only Plot Area. This suppress the title, labels, and legend.

2. ProTrend Tool

2.3. ProTrend Tool User Interface

Step	Action						
5	To change how the trend traces are displayed, select one of the Axes Modes. (See Trend Pane section for a description of the axes modes.)  The default axes mode for all new trends is Shared.						
6	To change the Trend pane legend, select one of the Legend options.  <table><tr><td>None</td><td>No Legend is displayed.</td></tr><tr><td>Trace Id</td><td>Legend is displayed with the name of each trace.</td></tr><tr><td>Values</td><td>Legend is displayed with the latest value for each trace.</td></tr></table>	None	No Legend is displayed.	Trace Id	Legend is displayed with the name of each trace.	Values	Legend is displayed with the latest value for each trace.
None	No Legend is displayed.						
Trace Id	Legend is displayed with the name of each trace.						
Values	Legend is displayed with the latest value for each trace.						

Step	Action
7	To control the hairline cursor, select one of the Hairline Cursor options. (See Trend Pane section for a description of the hairline cursor modes.) Off Turns the hairline cursor off. Cursor Only Shows the cursor only, without any value annotations in the Trend pane. The hairline values appear in the Values tab. Annotations Shows both the cursor and the value annotations in the Trend pane. The hairline values appear in the Values tab. By default, the hairline cursor is turned off for all new trends.
8	To Export the data from the trend to a text file, enter the name of the output file in the text field to the right of the Export to button then click on the Export to button. NOTE: To use the Export to File function, the user must have permission to create and modify small work files in the "Users" directory on the system drive. While the export operation is in progress, the Export to button is disabled. Upon completion, the specified text file exists within the configured Export Folder. The Export file is a comma-delimited text file suitable for importing into spreadsheets. The first record contains the names of each of the traces. For each timestamp in the trend, there is an record containing the timestamp and the value for each trace at that time. If a particular trace does not have a valid numeric value matching a given timestamp, its entry in that record is null.

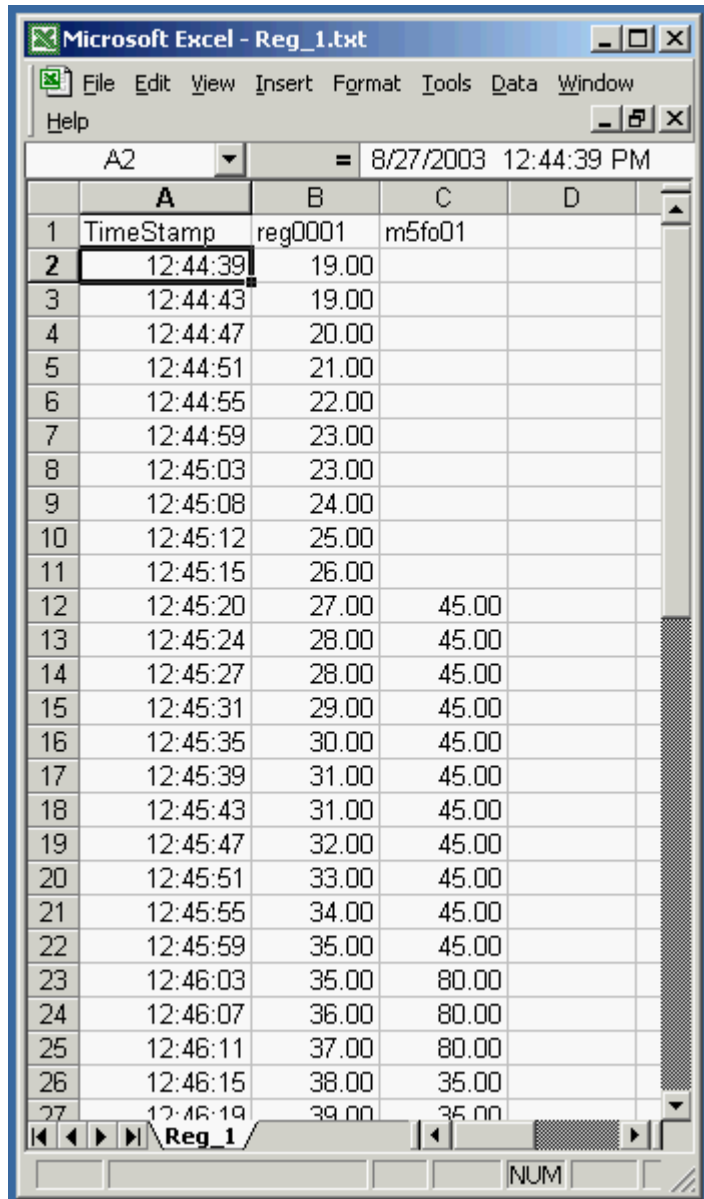
2. ProTrend Tool

2.3. ProTrend Tool User Interface

Step	Action
	NOTE: The timestamps are recorded in internal format. After importing into Excel, the Timestamp cells can be formatted into any desired representation.

2. ProTrend Tool

2.3. ProTrend Tool User Interface

Step	Action																																																																																																																																												
	 The screenshot shows a Microsoft Excel window titled "Microsoft Excel - Reg_1.txt". The menu bar includes File, Edit, View, Insert, Format, Tools, Data, Window, and Help. The formula bar shows "A2" and the value "8/27/2003 12:44:39 PM". The worksheet contains a table with the following data: <table><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>1</td><td>TimeStamp</td><td>reg0001</td><td>m5fo01</td><td></td></tr><tr><td>2</td><td>12:44:39</td><td>19.00</td><td></td><td></td></tr><tr><td>3</td><td>12:44:43</td><td>19.00</td><td></td><td></td></tr><tr><td>4</td><td>12:44:47</td><td>20.00</td><td></td><td></td></tr><tr><td>5</td><td>12:44:51</td><td>21.00</td><td></td><td></td></tr><tr><td>6</td><td>12:44:55</td><td>22.00</td><td></td><td></td></tr><tr><td>7</td><td>12:44:59</td><td>23.00</td><td></td><td></td></tr><tr><td>8</td><td>12:45:03</td><td>23.00</td><td></td><td></td></tr><tr><td>9</td><td>12:45:08</td><td>24.00</td><td></td><td></td></tr><tr><td>10</td><td>12:45:12</td><td>25.00</td><td></td><td></td></tr><tr><td>11</td><td>12:45:15</td><td>26.00</td><td></td><td></td></tr><tr><td>12</td><td>12:45:20</td><td>27.00</td><td>45.00</td><td></td></tr><tr><td>13</td><td>12:45:24</td><td>28.00</td><td>45.00</td><td></td></tr><tr><td>14</td><td>12:45:27</td><td>28.00</td><td>45.00</td><td></td></tr><tr><td>15</td><td>12:45:31</td><td>29.00</td><td>45.00</td><td></td></tr><tr><td>16</td><td>12:45:35</td><td>30.00</td><td>45.00</td><td></td></tr><tr><td>17</td><td>12:45:39</td><td>31.00</td><td>45.00</td><td></td></tr><tr><td>18</td><td>12:45:43</td><td>31.00</td><td>45.00</td><td></td></tr><tr><td>19</td><td>12:45:47</td><td>32.00</td><td>45.00</td><td></td></tr><tr><td>20</td><td>12:45:51</td><td>33.00</td><td>45.00</td><td></td></tr><tr><td>21</td><td>12:45:55</td><td>34.00</td><td>45.00</td><td></td></tr><tr><td>22</td><td>12:45:59</td><td>35.00</td><td>45.00</td><td></td></tr><tr><td>23</td><td>12:46:03</td><td>35.00</td><td>80.00</td><td></td></tr><tr><td>24</td><td>12:46:07</td><td>36.00</td><td>80.00</td><td></td></tr><tr><td>25</td><td>12:46:11</td><td>37.00</td><td>80.00</td><td></td></tr><tr><td>26</td><td>12:46:15</td><td>38.00</td><td>35.00</td><td></td></tr><tr><td>27</td><td>12:46:19</td><td>39.00</td><td>35.00</td><td></td></tr></table> The status bar at the bottom shows "Reg_1" and "NUM".		A	B	C	D	1	TimeStamp	reg0001	m5fo01		2	12:44:39	19.00			3	12:44:43	19.00			4	12:44:47	20.00			5	12:44:51	21.00			6	12:44:55	22.00			7	12:44:59	23.00			8	12:45:03	23.00			9	12:45:08	24.00			10	12:45:12	25.00			11	12:45:15	26.00			12	12:45:20	27.00	45.00		13	12:45:24	28.00	45.00		14	12:45:27	28.00	45.00		15	12:45:31	29.00	45.00		16	12:45:35	30.00	45.00		17	12:45:39	31.00	45.00		18	12:45:43	31.00	45.00		19	12:45:47	32.00	45.00		20	12:45:51	33.00	45.00		21	12:45:55	34.00	45.00		22	12:45:59	35.00	45.00		23	12:46:03	35.00	80.00		24	12:46:07	36.00	80.00		25	12:46:11	37.00	80.00		26	12:46:15	38.00	35.00		27	12:46:19	39.00	35.00	
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2. ProTrend Tool

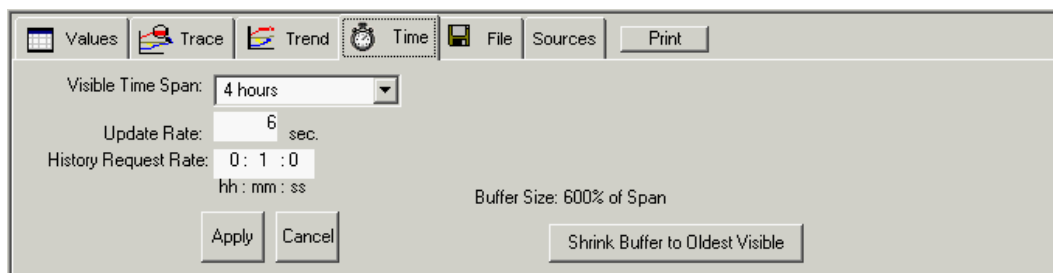
2.3. ProTrend Tool User Interface

Step	Action
9	To Reload the Configuration, click on the Reload Config. Button. This refreshes all of the information with the current default configuration.

Set the Time Settings

The Time tab may be used to alter the time related settings for the current Trend display.

Figure 15 Time Tab



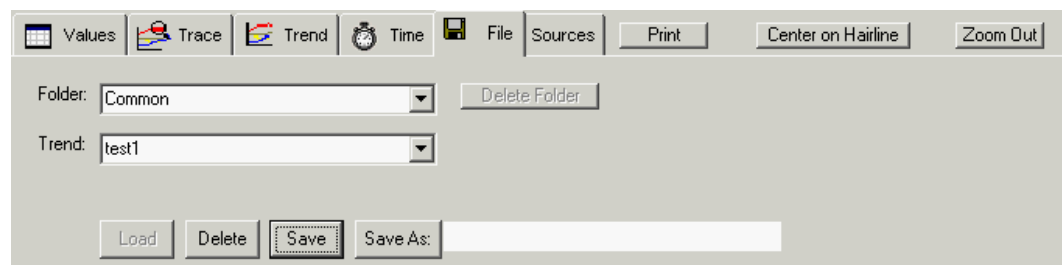
Step	Action
1	Set the Time Span displayed in the trend by selecting one of the values in the combo box. Then click Apply (which applies the new rates and time span).
2	If access is permitted, configure the trend Update Rate , in seconds. User access to the update rate settings is controlled in the ProTrend Tool Configuration utility. Then click Apply (which applies the new rates and time span).NOTE: The actual update rate is limited by the data source being used. Each trace queries its data source for up to 3 parameters.
	TIP If the ProTrend Tool reaches the configured maximum number of data points, it will automatically lower the update rate by 10% to reduce the data point total. The configured update rate remains the same, however placing the cursor above the configured value displays the actual, reduced value in a tooltip. Under normal circumstances no tooltip appears. See Configure the ProTrend Tool General Settings section, for more information about the maximum number of data points.

Step	Action
	ATTENTION Requesting data beyond that which PHD has collected results in false extrapolated data being returned from PHD. To avoid this when in snapshot mode, make sure the update rate specified for the trend is at least twice as much as the update period on PHD.
3	If access is permitted, configure the trend History Request Rate , in hours, minutes and seconds. User access to the history rate settings is controlled by the "Change Aggregate Interval" privilege in the ProTrend Tool Configuration utility. Then click Apply (which applies the new rates and time span).
4	If changes have been entered but not applied, a click on the Cancel button returns the displayed selections to their current values.
5	In order to provide for smooth scrolling in the time axis, tag data for each displayed trace is buffered. The amount buffered depends on the computer's available memory. The current buffer size is displayed as a percent of the visible time span. The Shrink Buffer to Oldest Visible button reduces buffer by discarding older data (making the corresponding memory available again, either for more buffering or to other applications).

Save Load or Delete the Trend Settings

The File tab may be used to save the current trend settings for future use, or to load, or delete, a previously defined trend. It may also be used to reconnect to the ProTrend Tool Server following a connection failure.

Figure 16 File Tab



2. ProTrend Tool

2.3. ProTrend Tool User Interface

Step	Action
1	To save the current trend settings for future use as a new trend, proceed as follows: a) In the Folder list box, enter or select the name of the target folder in the ProTrend Tool database where the trend file is to be saved. NOTE: If the user is limited to saving to a specific folder, the Save As button is disabled until that folder is selected b) Enter the name for the trend to be used as the Trend title in the field to the right of the Save As. c) Click Save As. NOTE: All of the settings from the Trend, Trace and Time tabs are saved to the file.
2	To load the trend settings for a previously defined trend into the ProTrend Tool, identify the trend and the database folder where the trend settings are stored, in the Trend and Folder list boxes, and click Load. When this happens, the ProTrendTool resets its current settings to match the saved file.
3	To save modifications to the previously loaded file, verify that the desired Folder and Trend names are selected, and click Save. NOTE: The Save button is not visible unless overwrite permission is configured.
4	To rename an existing database folder name, select the folder in the Folder list box and press F2. Once the folder name is in edit mode, modify the folder name to suit and press ENTER.
5	To delete the trend settings for a previously defined trend, identify the trend and the database folder where the trend settings are stored, in the Trend and Folder list boxes, and click the Delete button at the bottom of the tab. NOTE: Deleting a previously defined trend has no effect on the trend which is currently displayed. NOTE: The Delete button is not visible unless overwrite permission is configured.

Step	Action
6	To delete a previously defined database folder name, select the folder in the Folder list box and click the Delete Folder button to the right of the Folder list box. NOTE: This function can only be used if the selected database folder is empty (that is, there are no trends stored in it) and overwrite permissions are configured.
7	As described in the ProTrend Tool Architecture section, the ProTrend Tool trend display settings are stored in the central ProTrend Tool Server database. A local backup copy of this database is maintained so that the ProTrend Tool application can continue to run in the event of ProTrend Tool Server connection failure. When a failure occurs the text “(Local Database)” appears in the ProTrend Tool title bar and a Connect button appears in the File tab. When the network connection is reinstated, the user can reconnect to the ProTrend Tool Server database by any of the following methods: • Click Connect in the File tab. • Restart the ProTrend Tool. • Wait 20 minutes for the automatic backup cycle to occur.



TIP

If the Axes Mode was set to **Selected** when the trend settings were saved, the y-axis for a different tag trace being displayed when the trend is reloaded. On load, the first tag in the tag list is selected by default.

Set the Data Source Settings

The Sources tab may be used to alter the data sources for the current Trend display.

Figure 17 Sources Tab

Step	Action
1	Select the data source for the real-time data for the current trend, by selecting the appropriate Update Source option (either Local, TPN, or PHD). The Local update source uses the private TPN server installed on the local machine. The TPN update source uses the TPN Server specified either by its TPS Alias name or by the entered Host (node name or IP Address) and ServerId. If the ServerId is blank, it defaults to "Hci.TPNServer". The PHD update source uses the PHD Server specified by the entered Host (node name). Depending on the configuration, a UserName and Password may also be entered for connecting to the PHD Server.
	ATTENTION The TPN servers do not support the full set of aggregate types, so changing the data source from a PHD server to LCN or TPN Server may result in runtime errors. In addition, if PHD is configured to use converted engineering units it does not make sense to mix PHD history data with real-time data from a different update source.

Step	Action
2	Select the data source for the history data on the current trend, by specifying a PHD server. the PHD Server is specified by the entered Host (node name). Depending on the configuration, a UserName and Password may also be entered for connecting to the PHD Server.
3	Changes to the Data Sources become effective when you click on the Apply button. Click on the Cancel button to replace any pending changes with the actual current configuration. Click on the Defaults button to set the source information to the configured defaults. (This action does NOT apply the defaults.) NOTE: Changing the History Source does not update the existing data. In order to replace the previously collected history data with values from the new History source, click on the "Refresh All" button on the Trend tab.

2. ProTrend Tool

2.3. ProTrend Tool User Interface

3. ProTrend Templates Reference

3.1 General Trend Template Information

The trend templates supplied with ProTrend are designed to provide the user with examples of how ProTrend Object components can be used with standard GUS tools. Two, load-and-go GUS displays have been built with the ProTrend Object:

- Group Trend is a GUS display with two trends, which displays the PVs for up to 8 points. The points are found based on the groups defined in the LCN. Access to the Group Trend is limited to one instance on a given client station at a time
- Overview Trend is a GUS display with twelve trends, which displays the PVs for 48 points.

A set of displays and subpictures are provided as examples of GUS scripting for the ProTrend Object in

\Program Files\Honeywell\GUSPro\RAC\ProTrend folder.

These are reusable components that can be inserted as is into custom displays. See the accompanying .txt files for specific scripting guidance.

For all subpictures, the font and character set are inherited from the containing display.

Color settings are stored in the registry. (See Color Registry Settings for list of the color registry settings supplied.)

Data Access

For all trends, real-time updates are collected from the TPN and history data is collected from PHD. The trend control itself (contained in the trend subpictures) gets TPN data from HCI and from the TPN Server, if installed on a GUS node. All real-time values shown in the trends are collected from the Private TPN Server. For the Group Trend, all other values shown on the display (with the exception of hairline values) come directly from the LCN through HOPC.

3. ProTrend Templates Reference

3.2. Group Trend

Error Handling

All errors that occur in the template subpicture scripts are displayed in the error section of the title bar. Clicking on the error section of the title bar clears the current error. Only the most recent error is displayed. The exception to this error handling method occurs when an error is generated while attempting to display LCN parameter values in a text box. These errors are shown as characters in the text box in the following manner:

Error Number	Error Description	Value Displayed in Text Box	Reason for Error
1051	HOPC_Communication_Error	?????	Data owner shut down.
1052	HOPC_Configuration_Error	@@@@@	Bad entity ID.
1053	HOPC_Value_Error	-----	Bad value or not a number.
1054	HOPC_Store_Error	!!!!	Node initializing or indirection to entity that does not have the matching parameter.
1056	HOPC_Dynamic_Index_Error	^^^^	Index not resolved yet.

3.2 Group Trend

Purpose

The Group Trend (Trend_Group.pct) may be used as a Group Trend display with two trends and up to four points in each.



ATTENTION

Only one copy of the Group Trend, or other display built using its subpictures must, be run at a time..

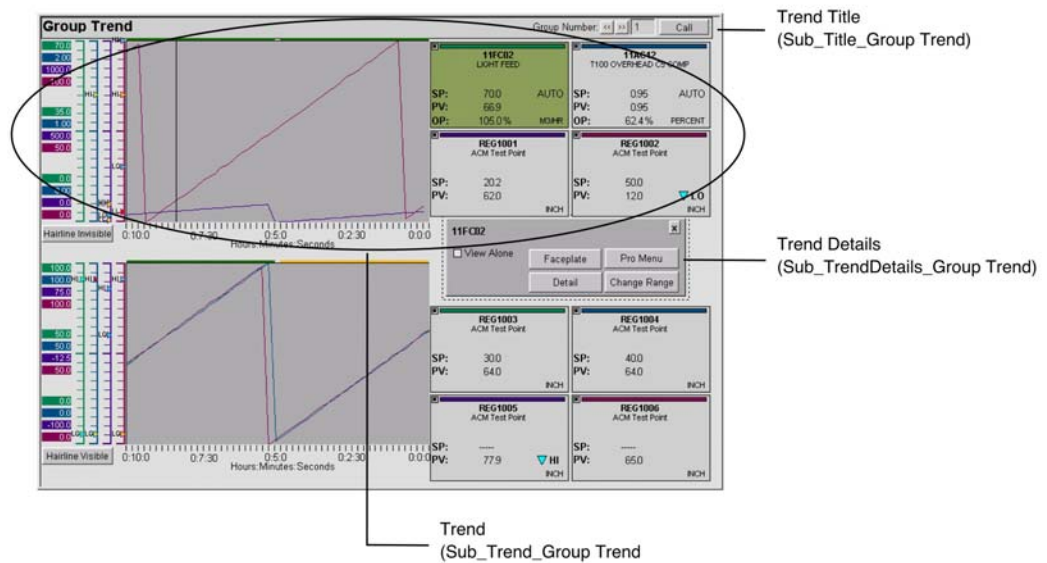
By default, the Trend_Group.pct starts with the last previously displayed group. Any Group can be called up as the initial display from GUS scripts by using the Disp_parameter argument on the InvokeDisplay statement. For example:

```
InvokeDisplay "C:\Program  
Files\Honeywell\GUSPro\RAC\ProTrend\Trend_group", "", 25
```

Loads the group number 25 during display startup.

The points trended correspond to the selected LCN group number. Tags are assigned to display database entity variables. The display database entities used are dispdb.ent245 through dispdb.ent252.

Figure 17 Group Trend Display



Subpictures

The template is comprised of the following subpictures:

- Sub_Title_Group Trend (Trend Title)
- Sub_Trend_Group Trend (Trend)
- Sub_TrendDetails_Group Trend (Trend Details)

The individual subpictures are described in detail in the following subsections.

3. ProTrend Templates Reference

3.2. Group Trend

Supported Point Types

Although this trend is similar to the Native Window group trend, and can be invoked for a group, not all point types are currently supported. The following point types are supported in the Group Trend:

- | | | | |
|------------|------------|------------|------------|
| • ANINNIM | • COUNTAM | • NUMERCAM | • REGPVNIM |
| • ANLCMPHG | • COUNTHG | • NUMERHG | • TIMERAM |
| • ANLINHG | • CUSTOMAM | • REGAM | • TIMERHG |
| • ANLOUTHG | • FLAGHG | • REGCLNIM | • TIMERNIM |
| • ANOUTNIM | • NUMERAM | • REGHG | |

Alarm Behavior

In the Group Trend's Point Detail subpictures, alarms are indicated with text for the specific alarm that has been triggered, as well as with a shape to indicate the priority of the alarm. This is demonstrated in the following examples:



LL Indicates that an Emergency priority, Low-Low alarm was triggered. A red box is always used to indicate Emergency priority alarms.



HI Indicates that a High priority, High alarm was triggered. A yellow triangle pointing up is always used to indicate High priority alarms.



HH Indicates that a Low priority, High-High alarm was triggered. A cyan triangle pointing down is a Low priority alarm.

Template Settings

Some settings for the group trend are set in the registry. See the Appendix, 13.2, for details.



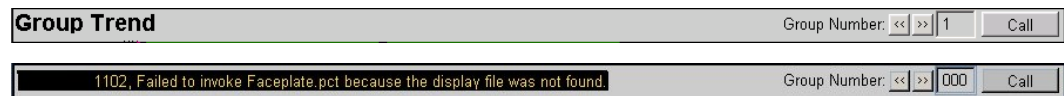
TIP

PHD tags are constructed using the TagPrefix, the tagname and the TagSuffix. For example, a PV value with a TagPrefixPV of "", a tagname of FIC100 and a TagSuffixPV of ".PV" the PHD tag accessed is FIC100.PV.

Sub_Title_Group Trend (Trend Title)

The Trend Title subpicture, Sub_Title_Group Trend, displays the title of the graphic, as well as any error messages. It is also used to select the LCN group to be trended in the display.

The first figure below shows the Trend Title with the LCN group call feature. The second figure shows how the Trend Title appears when an error is generated in the Group Trend display.



Functions and Features

Feature	Description
Group Number Select	The title includes a Group Number text box and Call button that may be used to select the group number for the set of tags to be trended.
Error Message Display	If an error is generated by any of the trend subpicture scripts, it is displayed on the left side of the title bar. Clicking on the displayed error message clears the error.

Display Parameters

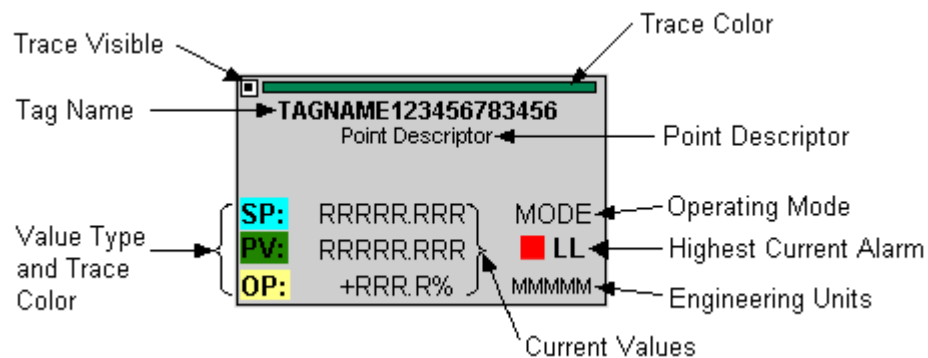
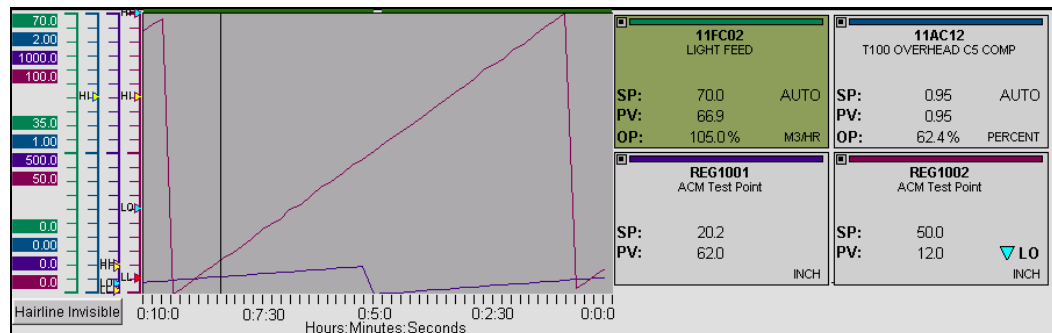
Parameter	Description
Label	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the title that appears at the top of the Group Trend display.

3. ProTrend Templates Reference

3.2. Group Trend

Sub_Trend_Group Trend (Trend)

The Trend subpicture, Sub_Trend_Group Trend, is used to display the trend.



Functions and Features

Feature	Description
Trend Area	Displays the color-coded, trend traces for up to four points. By default, each trace shows the PV values for a single point. The two horizontal bars above the trace area show the connection status for the historical (left) and real-time (right) data sources, as follows: Black Not finished database connection yet Red Error in database; no data is collected Orange Database overrun; can not keep up to requests Green No problems with the database To display the trends for a single point only, select the associated Point Detail and then, in the Trend Details subpicture Point view, select the View Alone option. In this mode, you may chose to display the point's SP, PV and/or OP values in the trend area.
Time Axis	A single time axis is displayed, which is common to all of the traces in the trend. To change the time scale, right-click the time scale to invoke the Trend Details subpicture Time view. The Time view controls may be used to adjust the time scale. When this happens the time scale is highlighted to indicate which trend is affected by the time scale change.

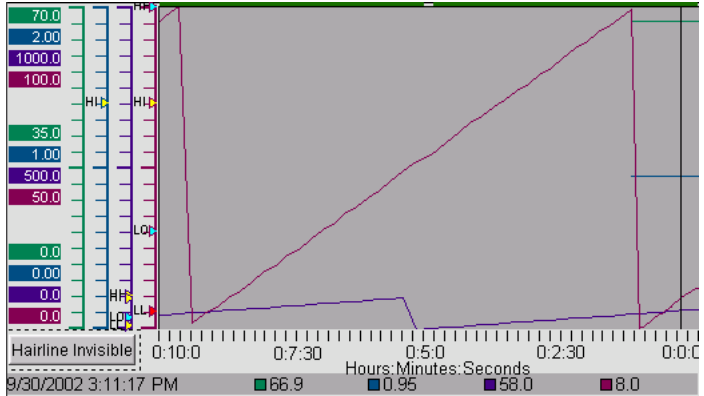
3. ProTrend Templates Reference

3.2. Group Trend

Feature	Description
Y-Axes (Sub_Trend _Group Trend_Axis Subpicture)	Up to four, color-coded, y-axis scales are displayed, one per trace. In addition, the upper, middle and bottom values for each scale are displayed in color-coded boxes next to the y-axes. The scales on the axes are harmonized as a percent of range. (Therefore, zooming is not supported.) The initial range for each trace is the point's PVEUHI and PVEULO values. To change the range, select the associated Point Detail, and use the Change Range feature in the Trend Details subpicture. When a selected point is in View Alone mode, the PV value y-axes are replaced by color-coded, y-axes scales for each type of value requested (that is, SP, PV and/or OP). Axes are always visible (even if a trace is made invisible) unless the associated point is one that cannot be trended.
PV Alarm Trip Points 	Within the PV value, y-axis scale for each trace, the configured PV alarm trip points are shown. The color of the triangle corresponds to the configured alarm priority: red for Emergency priority, yellow for High priority and cyan for Low priority alarms. Only those PV alarm trip points (HH, HI, LO or LL) that are configured for each point are displayed.

3. ProTrend Templates Reference

3.2. Group Trend

Feature	Description
Hairline Cursor	The Hairline Visible button is used to turn the hairline cursor on and off. When the hairline cursor is turned on, a vertical cursor appears in the trend area and the values at the current hairline position are displayed below the time axis.  The hairline values include the date and time at the hairline, as well as the values for each point that existed at the hairline time. Each point value is displayed with a color-coded box that identifies the corresponding trace. To change the position of the hairline click the mouse on the section of the trend where you want the hairline to appear.
Trace Visible	Indicates whether the point's trace is currently visible in the Trend subpicture. It is also used to hide or view the point's trace. To hide a trace, click the trace visible indicator in the subpicture. When a point's trace is hidden, the trace visible box is empty, and the associated trace is removed from the trend – the axis for the trace remains visible. To view a hidden trace, click the empty trace visible box. When the trace is visible, the trace visible box is filled in.
Trace Color	Displays the color used for the point's trend trace for PV and in the associated y-axis scale.

3. ProTrend Templates Reference

3.2. Group Trend

Feature	Description
Tag Name	Displays the tag name for the point. Up to 19 characters can be displayed.
Point Descriptor	Displays a description of the point, if there is one available. Up to three lines of description, text can be displayed.
Value Type and Trace Color	In the default mode, the SP, PV and OP text simply act as labels for the current values displayed in the Point Detail subpicture. In view alone mode, the SP, PV and OP labels are displayed in color-coded boxes that correspond to the trace color and y-axis scale color in the trend.
Current Values	Displays the current SP, PV and OP values for the point, if available.
Operating Mode	Displays the current operating mode (MODE) for the point. The text is backlit if the value is off normal.
Highest Current Alarm	Displays the current, highest priority, tripped alarm for the point, if there is an outstanding alarm.
Engineering Units	Displays the engineering units (EUDESC) for the point.
Faceplate Display Request	Left clicking on the Point Detail invokes a faceplate display, called "faceplate.pct", for the point. A registry entry is used to pass the tag name to the faceplate display. The Faceplate.pct must include OnDisplayStartup script that uses one of the following two commands: - FP.tagname = GetDisplatParameter or - FP.tagname = GetSetting ("GUSPro","GroupTrend","SelectedTag","") Although "faceplate.pct" is the default name of the faceplate display, its actual name is configurable as a parameter on the subpicture.
Trend Details Request	Right clicking on the Point Detail invokes the Trend Details subpicture Point view. This subpicture may be used to adjust the trend settings for that point and to request other displays associated with the point.

Feature	Description
Point Detail Highlight	When the Point Detail is selected (with either a right or left-click), the Point Detail's background color is changed. This helps to identify which point is altered by any trend display changes or which point the new faceplate display that has been launched.

Display Parameters

Parameter	Description
TrendPosition	A string parameter that is assigned upon insertion of the trend into the trend display. It must be set to either "UPPER" or "LOWER".
Faceplate	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It must contain the name of the faceplate display.

Sub_TrendDetails_Group Trend (Trend Details)

The Trend Details subpicture, Sub_TrendDetails_Group Trend, is used to adjust the trend settings or to view detailed information for a selected point.



Functions and Features

The Trend Details subpicture consists of the Point view and the Time view.

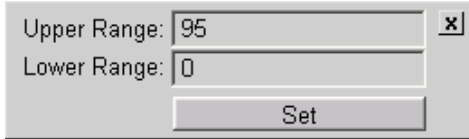
Point View



3. ProTrend Templates Reference

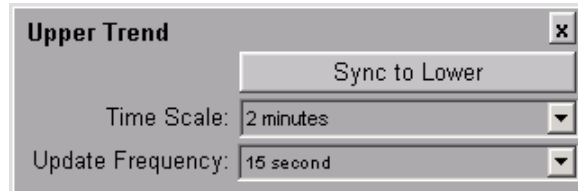
3.2. Group Trend

The Point view is accessed by right-clicking a Point Detail subpicture.

Feature	Description
Selected Point Name	Displays the tag name for the currently selected point. Any changes or selections made in the Point view apply to that point.
View Alone View SP View PV View OP Options	Selecting the View Alone option causes the associated Trend subpicture to show the trends for the selected point only. In this mode, you may chose to display the point's SP, PV and/or OP values, by selecting the desired View options. To clear the View Alone mode, either deselect the View Alone option by clicking it or select a different trace by clicking on the alternate trace.
Faceplate Button	Invokes a faceplate display, called "faceplate.pct", for the selected point. A registry entry is used to pass the tag name to the faceplate display.
ProMenu Button	Launches the ProMenu graphic display for the selected point.
Detail Button	Invokes the Native Window Point Detail display for the selected point. It is assumed that SafeView is configured to handle this.
Change Range Button	Opens a window directly above the Trend Details subpicture, that may be used to enter a new high and low y-axis scale range for the selected point.  Simply enter the desired Upper and Lower Range values and click Set . Click the X button to cancel the change range request operation.
Close (X) Button	Hides the Trend Details subpicture.

Time View

The Time view allows the user to adjust the time scale and update frequency for one of the trends.



The Time view is accessed by right-clicking the time scale in one of the Trends.

Feature	Description
Trend Indicator	Identifies the trend (Upper Trend or Lower Trend) that is affected by any time setting changes.
Time Scale List Box	Used to adjust the time scale in the selected trend.
Update Frequency List Box	Used to adjust the update frequency for the selected trend.
Sync to Upper Sync to Lower Button	If the time scales for the two trends are not synchronized, the Sync to Upper or Sync to Lower button may be used to make the time scales the same for both trends. Selecting this button causes the time scale of the selected trend to be set the same as the non-selected trend. If timescales and update rates are already in sync, this button does not appear.
Close (X) Button	Hides the Trend Details subpicture.

Display Parameters

Parameter	Description
FaceplateDisplay	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It must contain the name of a GUS display that is in a folder within each node's Search Path.

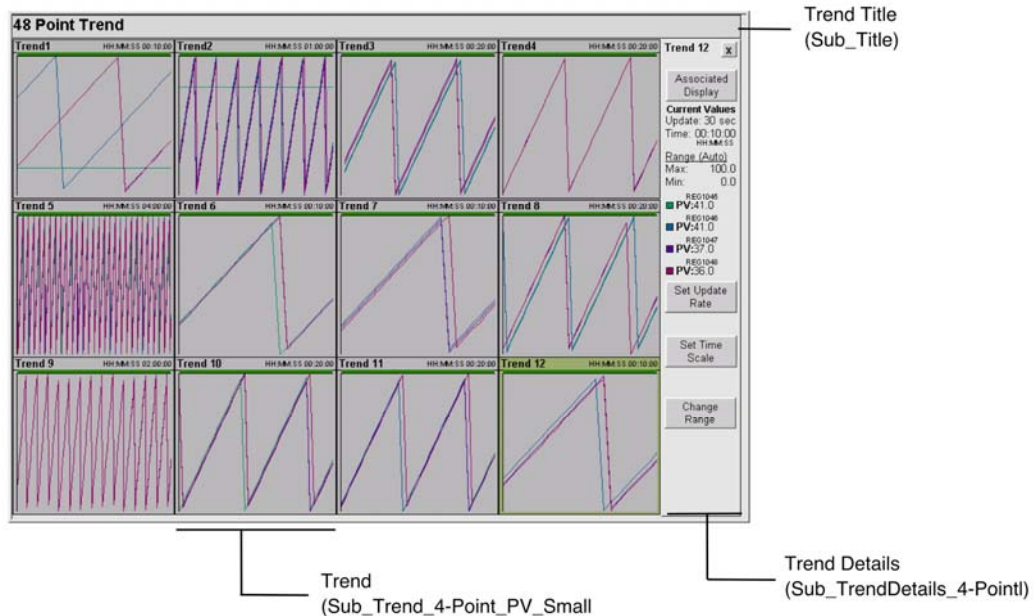
3.3 Overview Trend Template

Purpose

The Overview Trend templates (Trend_48_Point.pct) may be used to create an Overview Trend displays. Trend_48_Point.pct has twelve trends and four points in each. Mtrend_48Point.pct has eight trends with up to 6 points in each trace. Additional Trend displays are documented within the RAC folder.

Each Overview Trend display must have the names of the trend tags configured before validation. The tag names reside in inline parameters on each trend subpicture.

Figure 18 Overview Trend



Subpictures

The template is comprised of the following subpictures:

- Trend Title: Sub_Title.pct or subMT_Title.pct
- Trend Graphs: Sub_Trend_4-Point_PV_Small and variants.
- Trend Details: Sub_TrendDetails_4-Point and variants.

The individual subpictures are described in detail in the following subsections.

Template Settings

Some settings for the overview trend are set in the registry. See the Appendix, 13.3, for details.

Sub_Title / subMT_Title(Trend Title)

The Trend Title subpicture, Sub_Title, displays the title of the graphic, as well as any error messages.

48 Point Trend

Functions and Features

Feature	Description
Error Message Display	If an error is generated by any of the trend subpicture scripts, it is displayed on the right side of the title bar. Clicking on the displayed error message clears the error.

Display Parameters

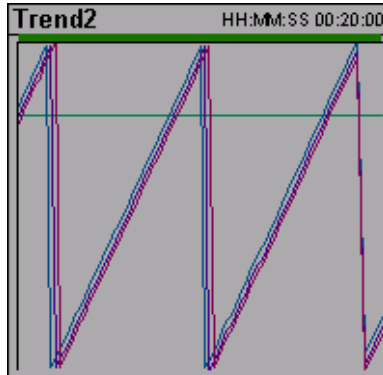
Parameter	Description
Label	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the title that appears at the top of the Overview Trend display.

3. ProTrend Templates Reference

3.3. Overview Trend Template

Trend Graphs: Sub_Trend_4-Point_PV_Small

The Trend subpicture, Sub_Trend_4-Point_PV_Small, trends the PV values for four points. Variants are provided in RAC that allow graphing 1 to 8 traces.



Functions and Features

Feature	Description
Trend Title	A name is given to the set of tags that are trended together in the trend subpicture. The horizontal bar above the trace area shows the connection status for the data sources, as follows: • Black: Connection in progress • Green: Connection OK • Red: Connection problem Left clicking in the trend title area invokes the Trend Details subpicture for the selected trend.
Time Indication	Since there is no time axis displayed in the trend area, the time span covered by the trend is displayed next to the trend title. The default time span is 2 hours, but this value may be reconfigured in the registry. The trace values displayed extend over the specified time span, with the most recent real-time ("update") value displayed the right side of the trend. To change the time span for the trend, left-click in the trend title area to open the associated Trend Details subpicture and use the Set Time Scale controls to set the desired time span.

3. ProTrend Templates Reference

3.3. Overview Trend Template

Feature	Description
Trend Area	Each trace in the trend area shows the PV values for a single point. The y-axis is auto-scaled. Since there is no y-axis displayed in the trend area and the traces are not identified, the user must refer to the associated Trend Details subpicture for this information. (In the Trend Details, the displayed Max. and Min. values correspond to the current upper and lower limits for the trend. The Trend Details subpicture can be used to change the y-axis scale for the trend values. Use the Change Range controls to set the value range.
Selected Trend	When the Trend is selected (with a left-click in the trend title area), the Trend's background color is changed. This helps to identify which trend is altered by any trend display changes.

Display Parameters

Parameter	Description
TrendLabel	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the name of the trend that appears at the top of the trend area.
Associated Display	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It must contain the name of the display that is to be invoked for this trend in the trend details subpicture.
Tagx	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the name of the tag to be trended. Each tag is identified separately in the parameters Tag1, Tag2, Tag3 and Tag4.
TagxPHDprefix	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the text string corresponding to the prefix used to identify tags in PHD, if required. A separate prefix string must be specified for each tag, such as Tag1PHDprefix and Tag2PHDprefix. This parameter can be left blank.

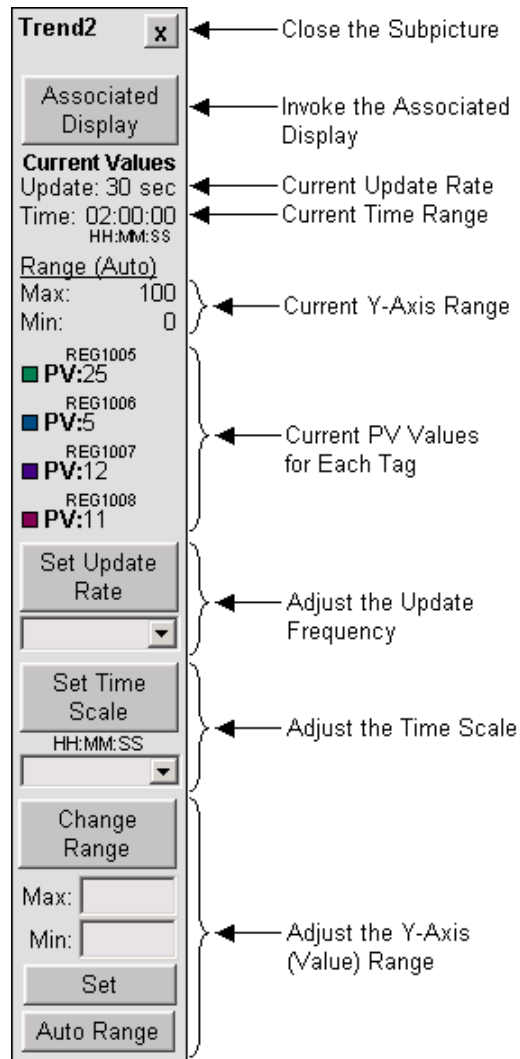
3. ProTrend Templates Reference

3.3. Overview Trend Template

Parameter	Description
TagxPHDSuffix	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the text string corresponding to the suffix used to identify tags in PHD, if required. A separate suffix string must be specified for each tag, such as Tag1PHDSuffix and Tag2PHDSuffix. This parameter can be left blank.
TagxPHDParameter	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the text string corresponding to the PHD parameter for the tag. A separate PHD parameter must be specified for each tag, such as Tag1PHDParameter and Tag2PHDParameter. This parameter can be left blank.
TagxLCNParameter	An inline parameter that is assigned upon insertion of the subpicture into the trend display. It contains the text string corresponding to the LCN parameter to be trended for the tag. A separate LCN parameter must be specified for each tag, such as Tag1LCNParameter and Tag2LCNParameter. The default is .PV. This parameter must not be blank.

Trend Details: Sub_TrendDetails_4-Point

The Trend Details subpicture, Sub_TrendDetails_4-Point, shows the current values for the points in the selected trend and may be used to adjust the trend settings. Other Variants in the RAC folder include data for up to 8 traces and options reporting on values at a Hairline cursor or displaying a Faceplate for a selected trace.



3. ProTrend Templates Reference

3.3. Overview Trend Template

Functions and Features

Feature	Description
Trend Title	Displays the name of the selected trend.
Close (X) Button	Hides the Trend Details subpicture.
Associated Display Button	Invokes the associated display for the selected trend.
Current Update Rate	The Update value corresponds to the current update frequency for the displayed values. NOTE: The actual update rate is limited by the data source being used. Each trace queries its data source for up to 13 parameters.
Current Time Range	The Time value corresponds to the time span covered by the selected trend. The default time span is 2 hours, but this value may be reconfigured in the registry.
Current Y-Axis Range	The Max and Min values correspond to the current upper and lower limits for the selected trend's y-axis. By default, the y-axis ranges are initially set to the highest PVEUHI and lowest PVEULO values found for the set of points displayed in the trend. If none of the points in the trend has PVEUHI or PVEULO values, then the y-axis is auto-scaled. When the default y-axis scale is in effect, the Range label shows Range (Auto) .
Point Identification and Current PV Values	For each point in the selected trend, the color used for the point's trend trace and the point's tag name are displayed, along with the current PV value. This value corresponds to the rightmost value visible in the trend for each point.
Set Update Rate Controls	To change the update frequency to be used in the selected trend, click the Set Update Rate button and select the desired update frequency from the list box displayed. The set of update frequencies that appear in the list box are configured in the registry. (See Group Trend Registry Settings section for the associated registry settings.)

3. ProTrend Templates Reference

3.3. Overview Trend Template

Feature	Description
Set Time Scale Controls	To change the time scale to be used in the selected trend, click the Set Time Scale button and select the desired time span from the list box displayed. The set of time spans that appear in the list box are configured in the registry. (See Group Trend Registry Settings section for the associated registry settings.)
Change Range Controls	To change the y-axis scale to be used in the selected trend, click the Change Range button, enter the desired Max and Min limit values in the text boxes displayed and click Set. Alternately, the Auto Range button, which is displayed when the Change Range button is clicked, may be used to reset the default y-axis range.

3. ProTrend Templates Reference

3.3. Overview Trend Template

4. ProTrend Object Scripting

4.1 Programmatic Interface

The purpose of this section, and those that follow it, is to describe how GUS displays may be scripted to present ProTrends. See `Trend_2tags_sub.pct` (and its associated `.txt` file) in the RAC directory for an example of scripting the ProTrend in GUS.

Due to the large number of objects supplied with the ProTrend Object, the detailed descriptions of the various ProTrend object properties, methods and events are broken down into the following sections:

- GProTrend Object (Section 5)
- GProDataSources Object (Section 6)
- GProDataSource Object (Section 7)
- GProTag Object (Section 8)
- GProTagPair Object (Section 9)
- GProTraces Object (Section 10)
- GProTrace Object (Section 11)

These sections describe the programmatic interface of the ProTrend Object.

The following general information applies in all cases:

1. Whenever strings are supplied for searching OLE collections, such searching is case-insensitive.
2. Collections are iterated in the order of their items' numerical indexes. As items are removed, any remaining items that follow in the index order are moved so that the nearest one occupies the vacated index. Items are added to the end of the collection and occupy previously unoccupied indexes.
3. For GUS Picture Builder, every enumeration is implemented as a long integer.

4. ProTrend Object Scripting

4.1. Programmatic Interface

The ProTrend Object Model

Object Relationships

The various ProTrend OCX objects are related to each other as follows:

1. Each GProTrend has a GProDataSources object.
2. In turn, every GProDataSources object references the GProTrend object that has it.
3. Each GProTrend also references two GProDataSource objects, which serves as the default for GProTag objects when the user does not explicitly assign their GProDataSource references.
4. Each GProDataSources object has GProDataSource objects, each of which has a reference back to the GProTrend object that owns it (albeit, indirectly through the GProDataSources object).
5. Each GProTrend has a GProTraces object.
6. In turn, every GProTraces object references the GProTrend object that has it.
7. Each GProTraces has GProTrace objects, each of which has a reference back to the GProTrend object that owns it (albeit, indirectly through the GProTraces object).
8. Each GProTrace has two GProTagPair objects. One GProTagPair provides history data while the other provides real-time update data. Each GProTagPair has a reference back to the GProTrace object that owns it.
9. Each GProTagPair object contains a single GProTag object, which can provide data to the GProTrace that owns the GProTagPair. The GProTag object has a reference back to the GProTagPair that owns it.
10. Each GProTag object references two GProDataSource objects. One provides the data for the high range and low range, while the other provides all other data, including the value to be trended.

OLE Collections

This object model exposes multiplicity as collections to GPB scripts.

- The GProDataSources object is an OLE collection that contains GProDataSource objects.
- The GProTraces object is an OLE collection that contains GProTrace objects.

4.2 Data Source Types

There are different kinds of data sources, and not every ProTrend Object property and method applies to every data source type, or applies to every data source type in the same way. Throughout the sections that follow, the properties and methods are described as they apply to an abstract, generalized data source type. Any data source specific details or restrictions are included where required.

TPN Data Source

The TPN data source represents a specialization of the generic data source in which specific code, that is used to access data, and the methods and properties, are specific for accessing a TPN through a TPN Server or a Private TPN Server.

If the full TPN Server is not available, then the Private TPN Server is automatically used in its place, imposing its limitations, such as maximum group size, relative to the full TPN Server. Multiple data source objects, configured to the same TPN Server, may be employed to circumvent maximum group size.

PHD Data Source

A PHD data source is used to access data from a PHD Server. This data source uses PHD's own non-COM interface. The PHD interface handle appears concealed within a C-language variable of module scope and, therefore, it is possible to access only a single PHD Server for an entire trend display.

4. ProTrend Object Scripting

4.2. Data Source Types

5. GProTrend Object

5.1 Introduction

Purpose

The GProTrend object is the ProTrend Object itself. This section describes the GProTrend properties and methods, as well as the events fired by the GProTrend object.

Example ProTrend OCX Name

For the sake of clarity, assume that the name of the GProTrend OCX is “ProTrend”.

This name is used in the GProTrend property and method syntax, and in the examples throughout this section.

5.2 GProTrend Properties

Name

Syntax

ProTrend.Name

Description

The Name property contains a string with the name of the trend. The Name property maps into the container’s extended Name property. This is the name as it is known to the container.

Comments

- The Name property does not take any parameters.
- It is persistable.
- It has a default of “OleObjectN”.
- It is read-only.

5. GProTrend Object

5.2. GProTrend Properties

Example

No example needed as this property is not used in scripting.

Description

Syntax

ProTrend.Description

Description

The Description property returns a string with a description of the trend object for documentation purposes.

Comments

- The Description property does not take any parameters.
- It is persistable.
- It has a default of “”.

Example

No example needed as this property is not used in scripting.

Common Errors

Error Number	Error Description	Possible Causes
-2147024882	Out of Memory	

Title

Syntax

ProTrend.Title

Description

The Title property returns a string with the trend title.

The Title string appears in the top middle portion of the trend display. Changing this property results in the immediate update of the display. When the display width is narrower than the title, the title is truncated, until the display is resized to be wide enough to accommodate the width of the title text.

Comments

- The Title property does not take any parameters.
- It is persistable.
- It has a default of "Honeywell".

Example

```
ProTrend.Title = "Flow Controller Trend"
```

Common Errors

Error Number	Error Description	Possible Causes
-2147024882	Out of Memory	

Font

Syntax

ProTrend.Font

Description

The Font property returns a standard Font object. This is the default font for the text objects within the ProTrend. It may be used to alter the standard font properties including .Name and .Size.

5. GProTrend Object

5.2. GProTrend Properties

NOTE: Some of the internal text objects overrides the Font.Size and Font.Color properties based on their own context.

Comments

- The Font property does not take any parameters.
- It is persistable.
- It has a default determined by system settings.

Example

```
ProTrend.Font.Name = "Arial"
```

UpdatePeriod

Syntax

ProTrend.UpdatePeriod

Description

The UpdatePeriod property returns a double value that is the period (in seconds) between data fetches.

At each data fetch, the GProTrend object requests that each data source object get data for each tag that references it as its data source. When each data source has assembled all of the tag data, it sends the data to the tags, which send the data to the traces for display in the trend display area.

If the system clock is advanced, the timer assumes that the clock was behind, and requests the data sources to return data for the times that were apparently missed. This is relevant for historical data sources. For real-time sources, this causes a harmless increase in the data request and data plot density.

If the system clock is set backwards, the timer assumes that the clock was too far in advance and that data may have been requested before it was recorded by historical or real-time data source, and that it must set its next-trigger-time to be the current system time plus the time interval. It then requests data for the system time at which it triggered, and continues triggering at the configured update interval, using successive timestamps, consistent with the changed system time, for its successive requests.



ATTENTION

Requesting data beyond that which PHD has collected results in false extrapolated data being returned from PHD. To avoid this when in snapshot mode, make sure the update rate specified for the trend is at least twice as much as the update period on PHD.

Comments

- The UpdatePeriod property does not take any parameters.
- It is persistable.
- It has a minimum value of 1 (second).
- It has a default value of 5.
- The actual update rate is limited by the data source being used. Each trace queries its data source for up to 13 parameters.

Example

'Set the update period to 5 seconds

UpdatePeriod = 5

Common Errors

Error Number	Error Description	Possible Causes
-2147024809	Invalid Argument	Attempt to set the UpdatePeriod to a value less than 1 second.

VisibleTimeSpan

Syntax

ProTrend.VisibleTimeSpan

Description

The VisibleTimeSpan property returns a date for the length of the time scale that is to be visible on the trend display. The scaling on the trend time axes is adjusted accordingly.

5. GProTrend Object

5.2. GProTrend Properties

When this property is set, the ProTrend Object queries its historical data sources, as appropriate, in order to provide data for the interval, ending at the date specified by the EndTime property, for the interval indicated by VisibleTimeSpan, after which, the ProTrend Object continues to update the trend display according to the UpdatePeriod property.

For times less than 24 hours, GPB scripts can automatically convert a string, of the form “hh:mm:ss”, and assign it to the VisibleTimeSpan property. In other circumstances, it is necessary to use the TimeSpanToDate method of the trend object.

Comments

- The VisibleTimeSpan property does not take any parameters.
- It is persistable.
- It has a default of 1 hour.

Example 1

```
'Getting the VisibleTimeSpan
'Need to get the current time span and display it

ProTrend.DateToTimeSpan _
    ProTrend.VisibleTimeSpan,years,months, _
    days,hours,minutes,seconds

'Because seconds are returned with accuracy up to
'1 millisecond - need to ignore all but the second
'value - so convert it to an integer

intSeconds = cint(seconds)

'Occasionally the years are included in the months

While Months > 12
    Months = Months - 12
    years = years + 1
Wend

'Now display the visible time span.

txtVisibleTimeSpan.text = "Visible Time Span: " & years _
```

```
& ":" & months & ":" & days & ":" & hours & ":" & _  
minutes & ":" & intSeconds
```

Example 2

```
'Setting the Visible Time Span  
'If just setting hours, minutes and seconds:  
  
ProTrend.VisibleTimeSpan = "02:00:00" '2 hours
```

Example 3

```
'If setting years, months and days as well  
  
ProTrend.VisibleTimeSpan = _  
    ProTrend.TimeSpanToDate(0,0,1,8,0,0)  
  
'This sets it to one and a half days
```

BufferTimeSpan

Syntax

```
ProTrend.BufferTimeSpan
```

Description

The BufferTimeSpan property returns a date for the maximum time interval for which retrieved values are retained in each trace.

Whenever retrieved values are older than indicated by this interval they are discarded. If the trend is scrolled to a time interval for which these retrieved values would be discarded, the ProTrend Object attempts to retrieve them for display using the history tag pair. Scrolling to a time interval for which values have not been discarded, allows the values to be displayed without having to request the data from a history data source again, thereby providing faster and smoother scrolling.

If the VisibleTimeSpan property is set larger than the BufferTimeSpan property, then the BufferTimeSpan property automatically increases to be the same as the VisibleTimeSpan property.

For times less than 24 hours, GPB scripts can automatically convert a string, of the form “hh:mm:ss”, and assign it to the BufferTimeSpan property. In other circumstances, it is necessary to use the TimeSpanToDate method of the trend object.

Comments

- The BufferTimeSpan property does not take any parameters.
- It is persistable.
- It has a default of 1 hour.

Example

```
Sub OnEnter()  
  
    Dim sBufferTimeSpan As String  
    sBufferTimeSpan = me.text  
    ProTrend.BufferTimeSpan = sBufferTimeSpan  
  
End Sub
```

EndTime

Syntax

ProTrend.EndTime

Description

The EndTime property returns a date for the latest moment of time to appear on the trend display. For example, if the EndTime is 27-Jul-02 23:59:59, then the right most visible portion of any trace represents the values at times before or equal to 27-Jul-02 23:59:59.

When this property is set, the ProTrend Object queries its historical data sources, as appropriate, to provide data for the interval, ending at the date specified by the EndTime property, for the interval indicated by the VisibleTimeSpan property. After this, the ProTrend Object updates the trend display according to the UpdatePeriod property.

Comments

- The EndTime property does not take any parameters.
- It is persistable.
- It has a default of 1 hour.

Example

'To set the EndTime back to now and bring the trend back
'to the latest data

```
ProTrend.BeforeNow = True  
ProTrend.EndTime = 0
```

'To set the EndTime to a specific time

```
ProTrend.EndTime = _  
    ProTrend.TimeSpanToDate(Years,Months,Days,Hours, _  
        Minutes,Seconds)
```

'The before now property will determine if the EndTime is
'relative or absolute. If BeforeNow is true the EndTime
'is relative. If BeforeNow is false the EndTime is
'an actual date

BeforeNow

Syntax

ProTrend.BeforeNow

Description

The BeforeNow property returns a Boolean which indicates whether the EndTime property functions as an absolute time (False), or whether it functions as a time relative to the system time of the ProTrend Object's host computer (True).

This property supports functionality available through PHD through the "Now – TimeString" idiom.

When this property is set, the ProTrend Object queries its historical data sources, as appropriate, in order to provide data for the interval, ending at the date specified by the EndTime property, for the interval indicated by the VisibleTimeSpan property, after which the ProTrend Object continues to update the trend display according to the UpdatePeriod property.



TIP

To have the trend set the end time exactly equal to the system time, set the BeforeNow property to True and set the EndTime property to 0. This is equivalent to the PHD "Now - 0" idiom.

Comments

- The BeforeNow property does not take any parameters.
- It is persistable.
- It has a default of True.

Example

`ProTrend.BeforeNow = True`

VisibleEarliestTime

Syntax

ProTrend.VisibleEarliestTime

Description

The VisibleEarliestTime property returns a date that contains the earliest date and time that is visible on the trend. This value changes when the trend is side-scrolled.

Comments

- The VisibleEarliestTime property does not take any parameters.
- It is not persistable.
- It has a default of Epoch.
- It is read-only.



TIP

Epoch is the first (smallest) date and time that can be expressed by the date/time system.

Example

```
Sub OnUpdateCollected()  
  
    txtVisibleEarliestTime.text = "Earliest Time: " _  
        & ProTrend.VisibleEarliestTime  
  
End Sub
```

VisibleLatestTime

Syntax

ProTrend.VisibleLatestTime

Description

The VisibleLatestTime property returns a date that contains the latest date and time that is visible on the trend. This value changes when the trend is side-scrolled.

Comments

- The VisibleLatestTime property does not take any parameters.
- It is not persistable.
- It has a default of Epoch.
- It is read-only.

Example

```
Sub OnUpdateCollected()  
  
    txtVisibleLatestTime.text = "Latest Time: " _  
        & ProTrend.VisibleLatestTime  
  
End Sub
```

SelectedTrace

Syntax

ProTrend.SelectedTrace

Description

The SelectedTrace property returns a GProTrace object, which corresponds to the trace that was selected for the kSelected axis mode (specified in the AxesMode property, described in the Axes Modes section). This property is set to NULL, if the trace is deleted.

Comments

- The SelectedTrace property does not take any parameters.
- It is persistable.
- It has a default of NULL.
- Although the default of this property is NULL it can only be set to a trace object. It cannot be set to NULL or nothing.

Example

'In the following example oTrace1 is a Trace object that
'was created when the trace was added to the trend

```
Set ProTrend.SelectedTrace = oTrace1
```

'All of the properties of the selected trace can be
'accessed by referencing the ProTrend.SelectedTrace
'property

```
txtSelectedTraceName.text = ProTrend.SelectedTrace.Name
```

AxesMode

Syntax

ProTrend.AxesMode

Description

The AxesMode property returns the GProAxesMode that defines the format for the trend display. The GProAxesMode can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kShared	0	All of the traces share a common y-axis, which is automatically scaled to suit the currently displayed values.

5. GProTrend Object

5.2. GProTrend Properties

Constant	Numerical Value	Description
kStacked	1	The traces are displayed, stacked one atop the other, in the trend display.
kHarmonized	2	The traces are displayed on the same axes, but value-axis ticks are labeled with a separate calibration for each trace. Each trace has a separate y-axis scale.
kSelected	3	Only the selected trace is shown on the trend display. The value-axis ticks are labeled only for the displayed trace. The trace is selected by placing its interface in the SelectedTrace property.

Comments

- The AxesMode property does not take any parameters.
- It is persistable.
- It has a default of 0 (kShared).
- When AxesMode is set to Stacked, and a Trace is set to Manual scaling with a range that is smaller than the plotted values, the plot for that trace extends beyond its stacked area, potentially overlapping with another trace. For more information, refer to Axes Mode - Stacked section.

Example

```
'This script exists on a button that has 4 interlocking
'radio buttons

Sub OnLButtonUp()

    If Button_AxesMode.ButtonState(0) = true Then
        'Harmonized
        ProTrend.AxesMode = 2
    ElseIf Button_AxesMode.ButtonState(1) = true Then
        'Selected
        ProTrend.AxesMode = 3
    ElseIf Button_AxesMode.ButtonState(2) = true Then
        'Shared
        ProTrend.AxesMode = 0
    End If
End Sub
```

```

ElseIf Button_AxesMode.ButtonState(3) = true Then
    'Stacked
    ProTrend.AxesMode = 1
End If

End Sub

```

HairlineCursor

Syntax

ProTrend.HairlineCursor

Description

The HairlineCursor property returns the GProHairlineCursorMode that sets the hairline cursor mode. The GProHairlineCursorMode can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kOff	0	When the hairline cursor is off, the cursor is not visible and no hairline cursor events are fired.
kCursorOnly	1	In this mode, the hairline cursor is visible, but no annotations are displayed. Whenever a hairline value changes, the OnHairlineCursor event is fired from the associated trace object, as well as from the trend object.
kWithAnnotations	2	In this mode, the hairline cursor is visible, and events are fired, just as with the kCursorOnly mode. In addition, an annotation is displayed for each trace. An annotation is an ordered pair consisting of the date and time at the current hairline cursor position, followed by the value at the current hairline cursor position. The ordered pair is displayed with a background that is the same color as the trace. The annotation follows both the trace and the cursor as the cursor is moved right or left. The text color is determined by the AxesColor property.

5. GProTrend Object

5.2. GProTrend Properties

Constant	Numerical Value	Description
		The date and time are displayed in the format specified by the HairlineTimeFormat property. The value either appears in e-format or not, based on whichever format is more compact for the given value and precision. The e-format is used only when the exponent of the value is less than -4, or is greater than or equal to its precision. Trailing zeros do not appear, and the decimal point appears only if one or more digits follow it.

Comments

- The HairlineCursor property does not take any parameters.
- It is persistable.
- It has a default of 0 (kOff).

Example

```
'This script exists on a button that has 3 interlocking
'radio buttons

Sub OnLButtonUp()

    If Button_HairlineCursor.ButtonState(0) = true Then
        ProTrend.HairlineCursor = 0 'kOff
    ElseIf Button_HairlineCursor.ButtonState(1) = true Then
        ProTrend.HairlineCursor = 1
    Else
        ProTrend.HairlineCursor = 2
    End If

End Sub
```

HairlineTimeFormat

Syntax

ProTrend.HairlineTimeFormat

Description

The YScaleVisible property returns a string, which determines the format of the timestamp displayed on the annotations for the Hairline cursor. This format string generally includes the delimiter characters (/ - :) and one or more of the following substitution symbols:

- o %c – date and time representation for the current locale
- o %X – time representation for current locale
- o %y – 2 digit year
- o %Y – 4 digit year
- o %b – Abbreviated month name
- o %m – 2 digit Month as decimal number
- o %#m – Month without any leading zero
- o %d – 2 digit Day of the month
- o %#d – Day of the month without any leading zero
- o %#j – Julian day of the year (without any leading zero)
- o %H – Hour 00-23
- o %#H – Hour without any leading zero
- o %I – Hour 1-12 (without any leading zero)
- o %p –AM/PM indicator for current locale
- o %M – Minute
- o %S -- Second

Comments

- The HairlineTimeFormat property does not take any parameters.
- It is persistable.
- It has a default of “%c” Locale-based format.

Example

ProTrend.HairlineTimeFormat = "" 'do not include any timestamp on the hairline annotations.

YScaleVisible

Syntax

ProTrend.YScaleVisible

Description

The YScaleVisible property returns a Boolean which determines whether the y-axis scale values are visible (True) or not (False).

Comments

- The YScaleVisible property does not take any parameters.
- It is persistable.
- It has a default of True.

Example

```
Global lCurAxesMode As Long

'This script exists on a checkbox button

Sub OnLButtonUp()

    If Button_YScaleVisible.ButtonState(0) = true Then
        ProTrend.AxesMode = lCurAxesMode
        ProTrend.YScaleVisible = true
    Else
        'Need to set the axis mode to shared - this is
        'a work-around because the turning off of the
```

```
        'YScale does not work for harmonized and  
        'stacked axis modes  
  
        lCurAxesMode = ProTrend.AxesMode  
        ProTrend.AxesMode = 0  
        ProTrend.YScaleVisible = false  
    End If  
  
End Sub
```

TimeScaleVisible

Syntax

ProTrend.TimeScaleVisible

Description

The TimeScaleVisible property returns a Boolean which determines whether the time-axis scale values are visible (True) or not (False).

Comments

- The TimeScaleVisible property does not take any parameters.
- It is persistable.
- It has a default of True.

Example

```
Sub OnLButtonUp()  
  
    If Button_TimeScaleVisible.ButtonState(0) = true Then  
        ProTrend.TimeScaleVisible = true  
    Else  
        ProTrend.TimeScaleVisible = false  
    End If  
  
End Sub
```

5. GProTrend Object

5.2. GProTrend Properties

LegendPosition

Syntax

ProTrend.LegendPosition

Description

The LegendPosition property returns the GProLegendPosition that sets the position of the legend on the trend display. The legend is a display region that displays the color of each trace beside the trace's update tag pair's active tag's tag ID.

The GProLegendPosition can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kOff	0	The legend is not displayed.
kLeft	1	The legend is displayed to the left of the trend display.
kRight	2	The legend is displayed to the right of the trend display.
kTop	3	The legend is displayed above the trend display.
kBottom	4	The legend is displayed below the trend display.

Comments

- The LegendPosition property does not take any parameters.
- It is persistable.
- It has a default of 2 (kRight).



ATTENTION

Setting Legend position to 1 (left side) results in having the legend obscured by the labels for the Y-axis.

Do not set LegendPosition = 1 (left) unless YScaleVisible = False.

Example

```
'This script exists on a button that five interlocking  
'radio buttons  
  
Sub OnLButtonUp()  
  
    If Button_LegendPosition.ButtonState(0) = true Then  
        ProTrend.LegendPosition = 0 'kOff  
    ElseIf Button_LegendPosition.ButtonState(1) = true Then  
        ProTrend.LegendPosition = 1 'kLeft  
    ElseIf Button_LegendPosition.ButtonState(2) = true Then  
        ProTrend.LegendPosition = 2 'kRight  
    ElseIf Button_LegendPosition.ButtonState(3) = true Then  
        ProTrend.LegendPosition = 3 'kTop  
    Else  
        ProTrend.LegendPosition = 4 'kBottom  
    End If  
  
End Sub
```

LegendStyle

Syntax

ProTrend.LegendStyle

Description

The LegendStyle property returns a long value for the legend area content.

The valid values are:

0 = Trace Name

1 = Latest Value

NOTE: The trend object responds to some other values for LegendStyle, but they are not meaningful for line graphs.

5. GProTrend Object

5.2. GProTrend Properties

Comments

- The LegendStyle property does not take any parameters.
- It is persistable.
- It has a default of 0 (Trace Name).

Example

```
ProTrend.LegendColor = 50331647 'White
```

PlotAreaOnly

Syntax

ProTrend.PlotAreaOnly

Description

The PlotAreaOnly property returns a Boolean which determines whether the trend control border is visible (False) or not (True).

The border is the area where the trend axes, legend and title are displayed. (This is the area surrounding the plot area.) Hiding the border area (by setting this property to True) is appropriate when only the trace display area must be visible. This can be used by containers that provide their own calibrations and other decorations.

Comments

- The PlotAreaOnly property does not take any parameters.
- It is persistable.
- It has a default of False.

Example

```
'This script exists on a latching button

Sub OnLButtonUp()

    If Button_Decorations.ButtonState(0) = true Then
        ProTrend.PlotAreaOnly = false
    Else
        ProTrend.PlotAreaOnly = true
    End If

End Sub
```

MinimumDataChangePeriod

Syntax

ProTrend.MinimumDataChangePeriod

Description

The MinimumDataChangePeriod property returns a double value that sets the minimum time interval (in seconds) in which the OnUpdateCollected and OnHistoryCollected events can fire.

Comments

- The MinimumDataChangePeriod property does not take any parameters.
- It is persistable.
- It has a minimum value of 1 second.
- It has a default value of 1 (second).

5. GProTrend Object

5.2. GProTrend Properties

Example

```
Sub OnEnter()  
  
    'Although the trend will still update according to  
    'the UpdatePeriod set the OnUpdateCollected event  
    'will only fire every four seconds now  
  
    ProTrend.MinimumDataChangePeriod = 4  
  
End Sub
```

Common Errors

Error Number	Error Description	Possible Causes
-2147024809	Invalid Argument	Attempt to assign a value less than 1 second.

LastUpdateTime

Syntax

ProTrend.LastUpdateTime

Description

The LastUpdateTime property returns the date and time corresponding to the latest firing of the OnUpdateCollected event.

Comments

- The LastUpdateTime property does not take any parameters.
- It is not persistable.
- It has a default of Epoch.
- It is read-only.

Example

```
'OnUpdateCollected is a logical place to put a call to  
'this property  
  
Sub OnUpdateCollected()  
  
    txtLastUpdateTime.text = ProTrend.LastUpdateTime  
  
End Sub
```

ColorPallet

Syntax

ProTrend.ColorPallet

Description

The ColorPallet property returns a safe array of long values for color.

Whenever the ProTrend Object needs to choose a default color for a new trace, the array is scanned, starting at the first element, until an unused color is found. This unused color is used. The array index is zero based. If the number of colors in the color pallet is zero, then the default color is selected automatically.

Comments

- The ColorPallet property does not take any parameters.
- It is persistable.
- It has default array value of red, green, blue, yellow, magenta, cyan, white, black, dark red, dark green, dark blue, light brown, dark magenta, dark cyan, gray and light gray.

Example

```
Global Index As Integer  
Global Count As Integer  
  
'This script will cycle through all the colors in the  
'ColorPallet
```

5. GProTrend Object

5.2. GProTrend Properties

```
Sub OnPeriodicUpdate()  
  
    'Only run this script every 2 seconds  
  
    If count = 4 Then  
        If Index > ProTrend.Colors - 1 Then  
            Index = 0  
        End If  
        ColorIndex.text = Index  
        Colors.fillcolor = ProTrend.ColorPallet(Index)  
        Index = Index + 1  
    End If  
  
    If Count < 4 Then  
        count = count + 1  
    Else  
        count = 1  
    End If  
  
End Sub
```

Colors

Syntax

ProTrend.Colors

Description

The Colors property returns a long value equal to the number of colors in the ColorPallet array.

Comments

- The Colors property does not take any parameters.
- It is persistable.
- It has default value of 16.
- It is read-only.

Example

See ColorPallet example in ColorPallet section.

GridColor

Syntax

ProTrend.GridColor

Description

The GridColor property returns a long value for the plot area's grid color.

Comments

- The GridColor property does not take any parameters.
- It is persistable.
- It has a default of dark gray.

Example

```
ProTrend.GridColor = 45857723 'Gray
```

BackgroundColor

Syntax

ProTrend.BackgroundColor

Description

The BackgroundColor property returns a long value for the plot area's background color.

Comments

- The BackgroundColor property does not take any parameters.
- It is persistable.
- It has a default of light gray.

Example

```
ProTrend.BackgroundColor = 33554432 'Black
```

HairlineColor

Syntax

ProTrend.HairlineColor

Description

The HairlineColor property returns a long value for the hairline cursor color.

Comments

- The HairlineColor property does not take any parameters.
- It is persistable.
- It has a default of black.

Example

```
ProTrend.HairlineColor = 50331647 'White
```

BorderColor

Syntax

ProTrend.BorderColor

Description

The BorderColor property returns a long value for the border area color. (This is the area surrounding the plot area.)

Comments

- The BorderColor property does not take any parameters.
- It is persistable.
- It has a default of light gray.

Example

```
ProTrend.BorderColor = 48094685 'Gray
```

LegendColor

Syntax

ProTrend.LegendColor

Description

The LegendColor property returns a long value for the legend area color.

Comments

- The LegendColor property does not take any parameters.
- It is persistable.
- It has a default of white.

Example

```
ProTrend.LegendColor = 50331647 'White
```

AxesColor

Syntax

ProTrend.AxesColor

Description

The LegendColor property returns a long value for the color of the text labels on the Axes and Hairline annotations. The special value of –1 causes the color to be set to either black or white at runtime, depending on which provides the greatest contrast with the Trace color.

Comments

- The AxesColor property does not take any parameters.
- It is persistable.
- It has a default of –1 (contrast).

Example

```
ProTrend.AxesColor = 0 'Black
```

TraceWidth

Syntax

ProTrend.TraceWidth

Description

The TraceWidth property returns a long value for the default width (in pixels) of traces added to the trend.

NOTE: changing this value does not affect any current traces, that is done by setting the TraceWidth property on the Trace object.

Comments

- The TraceWidth property does not take any parameters.
- It is persistable.
- It has a default of 2.

Example

```
ProTrend.TraceWidth = 1
```

ZoomEnabled

Syntax

ProTrend.ZoomEnabled

Description

The ZoomEnabled property returns a Boolean which determines whether the trend control allows Zooming (True) or not (False). This property has no effect while the AxesMode is Stacked or Harmonized.

NOTE: If the trend is Zoomed in at the time ZoomEnabled is set to False, it is “frozen” in its current Zoom position.

Comments

- The PlotAreaOnly property does not take any parameters.
- It is persistable.
- It has a default of True.

Example

```
'This script exists on a latching button

Sub OnLButtonUp()

    If Button_AllowZoom.ButtonState(0) = true Then
        ProTrend.ZoomEnabled = true
    Else
        ProTrend.ZoomOut
        ProTrend.ZoomEnabled = false
    End If

End Sub
```

5.3 GProTrend Methods

Traces

Syntax

ProTrend.Traces

Description

The Traces method returns the GProTraces object associated with the GProTrend object. The GProTraces object is an OLE collection and is described in the GProTraces Object section.

Comments

The Traces method does not take any parameters.

Example

```
'This script cycles through the Traces on the trend and if
'the name matches the name being searched for it indicates the
'status of the trace in a small circle

Dim Count As Integer
Sub OnPeriodicUpdate()

    Dim oDS As Object
    For Each oDS in ProTrend.Traces
        If oDS.Name = "Flow1" Then
            If oDS.hResult = 0 Then
                indTrace1Status.fillcolor = TDC_Green
            Else
                indTrace1Status.fillcolor = TDC_Red
            End If
        End If
    Next

End Sub
```

CollectHistory

Syntax

ProTrend.CollectHistory (StartTime, EndTime)

Description

Calling the CollectHistory method causes each trace to collect historical data from its active history data source. The starting time of the history collection is the earliest whole sample interval that includes the time of the StartTime parameter. This is done using each history tag pair's AggregateSampleInterval property and going back in time from the EndTime property. The ending time of the history collection is the latest whole sample interval that includes the time of the EndTime parameter, using each history tag pair's AggregateSampleInterval property and going back in time from the starting time.

If history is unavailable, no event is fired and no error is raised. The script checks for success by examining each trace's LatestTime and EarliestTime properties.

Comments

The CollectHistory method takes the following parameters:

Parameter	Description
StartTime	A date that specifies the start time for history data collection.
EndTime	A date that specifies the end time for history data collection.

Example

```
ProTrend.CollectHistory ProTrend.VisibleEarliestTime, _  
    ProTrend.VisibleLatestTime
```

InsertColor

Syntax

ProTrend.InsertColor (Index, NewColor)

Description

The InsertColor method inserts a new color into the ColorPallet array in front of the index specified. For example, if the index is zero, then the new color becomes that of element zero in the ColorPallet array. To insert after the last element in the array, specify an element that is one larger than the last element's index.

Comments

The InsertColor method takes the following parameters:

Parameter	Description
Index	A long value that specifies the position in the ColorPallet array where the color is to be added.
NewColor	A long value for the color to be added to the ColorPallet array.

Example

'Insert a color at the beginning of the color pallet

```
ProTrend.InsertColor 0, 50331647 'White
```

DeleteColor

Syntax

ProTrend.DeleteColor (Index)

Description

The DeleteColor method removes the color at the specified index in the ColorPallet array. Each color in the array, which follows the deleted color, is moved up one position, so that the newly vacant array element is filled in.

Comments

The DeleteColor method takes the following parameter:

Parameter	Description
Index	A long value that specifies the position in the ColorPallet array for the color, which is to be deleted.

Example

```
'Delete the first color in the color pallet
```

```
ProTrend.DeleteColor 0
```

DateToTimeSpan

Syntax

ProTrend.DateToTimeSpan (RelativeDate, Years, Months, Days, Hours, Minutes, Seconds)

Description

The DateToTimeSpan method is a convenience method that converts a relative DATE into a time span.

NOTE: The time span is relative to January 1, 1900 (which converts to 0, 0, 0, 0, 0, 0).

Comments

The DateToTimeSpan method takes the following parameters:

Parameter	In/Out	Description
RelativeDate	In	A date value that is the relative time span that is represented in a DATE variable.
Years	Out	A short value for the number of years in the time span.
Months	Out	A short value for the number of months in the time span.
Days	Out	A short value for the number of days in the time span.

5. GProTrend Object

5.3. GProTrend Methods

Parameter	In/Out	Description
Hours	Out	A short value for the number of hours in the time span, in 24-hour format.
Minutes	Out	A short value for the number of minutes in the time span.
Seconds	Out	A single value for the number of seconds (and fractions of a second) in the time span.

Example

See examples in VisibleTimeSpan property (See VisibleTimeSpan section).

TimeSpanToDate

Syntax

ProTrend.TimeSpanToDate (Years, Months, Days, Hours, Minutes, Seconds)

Description

The TimeSpanToDate method is a convenience method that returns a Date converted from a time span. This method is intended be used to translate **relative** times into a date to pass into properties that require a date.

Comments

The TimeSpanToDate method takes the following parameters:

Parameter	In/Out	Description
Years	In	A short value for the number of years in the time span. NOTE: Rounding errors may occur if this value is greater than 100.
Months	In	A short value for the number of months in the time span.
Days	In	A short value for the number of days in the time span.
Hours	In	A short value for the number of hours in the time span, in 24-hour format.

Parameter	In/Out	Description
Minutes	In	A short value for the number of minutes in the time span.
Seconds	In	A single value for the number of seconds and fractions thereof in the time span.

Example

See examples in VisibleTimeSpan property (See VisibleTimeSpan section).

ZoomOut

Syntax

ProTrend.ZoomOut

Description

The ZoomOut method removes any zooming from the trend. It also returns the rightmost time to the current time.

Comments

The ZoomOut method does not take any parameters.

Example

```
Sub OnLButtonClick()  
    ProTrend.ZoomOut  
End Sub
```

ShiftHairlineLeft

Syntax

ProTrend.ShiftHairlineLeft

Description

The ShiftHairlineLeft method shifts the hairline left by one data point.

Comments

The ShiftHairlineLeft method does not take any parameters.

Example

```
Sub OnLButtonClick()  
    If ProTrend.HairlineCursor <> 0 then  
        ProTrend.ShiftHairlineLeft  
    End if  
End Sub
```

ShiftHairlineRight

Syntax

ProTrend. ShiftHairlineRight

Description

The ShiftHairlineRight method shifts the hairline right by one data point.

Comments

The ShiftHairlineRight method does not take any parameters.

Example

```
Sub OnLButtonClick()  
    If ProTrend.HairlineCursor <> 0 then  
        ProTrend.ShiftHairlineRight  
    End if  
End Sub
```

CenterHairline

Syntax

```
bReturn = ProTrend.CenterHairline()
```

Description

The CenterHairline method moves the hairline cursor to the middle of the visible time span.

Comments

The returned Boolean value reports whether or not the operation was successful.

Example

```
Sub OnLButtonUp()  
    Dim dtemp As Date  
    Dim d2 As Date  
    dim bret as Boolean  
    'adjust the end time so that the HairlineDateTime will be  
    'in the center of the displayed trend area:  
        dtemp = Trend.SelectedTrace.HairlineDateTime  
        Trend.BeforeNow = FALSE  
        Trend.EndTime = dtemp + CDBL((Trend.VisibleLatestTime  
-  
Trend.VisibleEarliestTime)/2.0)  
        bret =Trend.CenterHairline()  
End Sub
```

ExportTrace

Syntax

```
bReturn = ProTrend.ExportTrace( TraceName, Filepath, outFormat)
```

Description

The ExportTrace method generates a file containing the timestamps and values for a specified trace.

TraceName is a string identifying a current trace.

Filepath is a string identifying the output file. If the file already exists, it is overwritten.

OutFormat is a Long value specifying the format of the output file. The choices are

- 0 : comma-delimited text file
- 1 : Excel file
- 2 : XML file

Comments

The returned Boolean value reports whether or not the operation was successful. The result is false if the TraceName does not match a current trace, if the Filepath cannot be written to, or if OutFormat is not one of the supported values.

Example

```
Sub OnLButtonUp()  
    Dim bret as boolean  
    Dim ExportFile as string  
    ExportFile = InputBox ("path for export file:", "Export  
Trace")  
    Bret = ProTrend.ExportTrace (ProTrend.SelectedTrace.Name,  
    —  
                                ExportFile, 1)  
  
End Sub
```

5.4 GProTrend Events

OnUpdateCollected

Syntax

OnUpdateCollected

Description

The OnUpdateCollected event is fired whenever any new update data arrives from a data source, with the constraint that this event is not fired any more often than once per the time period configured in the MinimumDataChangePeriod property, described in MinimumDataChangePeriod section. The successful retrieval means that the data source functioned adequately to collect the tags configured to it, but a tag's value could still be occluded by, for example, a failure of a field instrument. For this reason, event-handling code must check the tag's hResult property to ensure that the tag's value has sufficient quality.

Although this event is fired as a result of operations by the data collection thread, this event is fired from the main UI thread.

Comments

The OnUpdateCollected event does not return any parameters.

Example

Examples exist on several other properties and methods.

```
Sub OnUpdateCollected()  
  
    Dim years As Integer  
    Dim months As Integer  
    Dim days As Integer  
    Dim hours As Integer  
    Dim minutes As Integer  
    Dim seconds As Single  
    Dim intSeconds As Integer  
    Dim TimeScale As String  
  
    'Get current values back from the trend  
  
    ProTrend.DateToTimeSpan _  
        ProTrend.BufferTimeSpan, years, months, days, _  
        hours, minutes, seconds  
    intSeconds = cint(seconds)  
  
    'Sometimes 1 year will be returned as 12 months  
  
    While Months > 12  
        Months = Months - 12  
        years = years + 1  
    Wend  
  
    txtActualBufferTimeSpan.Text = "Buffer Time Span: " _  
        & years & ":" & months & ":" & days & " " & _  
        hours & ":" & minutes & ":" & intseconds  
  
    'Need to get the current time span and display it  
  
    ProTrend.DateToTimeSpan _  
        ProTrend.VisibleTimeSpan, years, months, days, _  
        hours, minutes, seconds  
    intSeconds = cint(seconds)  
    While Months > 12  
        Months = Months - 12
```

```
        years = years + 1
    Wend

    txtVisibleTimeSpan.text = "Visible Time Span: " & years _
        & ":" & months & ":" & days & ":" & hours & ":" _
        & minutes & ":" & intSeconds
    txtVisibleEarliestTime.text = "Earliest Time: " & _
        ProTrend.VisibleEarliestTime
    txtVisibleLatestTime.text = "Latest Time: " & _
        ProTrend.VisibleLatestTime
    txtEndTime.text = "End Time: " & ProTrend.EndTime
    txtLastUpdateTime.text = "Last Update Time: " & _
        ProTrend.LastUpdateTime

End Sub
```

OnHistoryCollected

Syntax

OnHistoryCollected

Description

The OnHistoryCollected event is fired whenever history data arrives from a data source, with the constraint that this event is not fired any more often than once per the time period configured in the MinimumDataChangePeriod property, described in MinimumDataChangePeriod section. The successful retrieval means that the data source functioned adequately to collect the tags configured to it, but an individual tag's value could still be occluded by, for example, a failure of a field instrument. For this reason, event-handling code must check the tag's hResult property to ensure that the tag's value has sufficient quality.

Although this event is fired as a result of operations by the data collection thread, this event is fired from the main UI thread.

Comments

The OnHistoryCollected event does not return any parameters.

5. GProTrend Object

5.4. GProTrend Events

Example

```
Sub OnHistoryCollected()  
    msgbox "History Collected"  
End Sub
```

OnDataCollectError

Syntax

OnDataCollectError (pIDataSource, hResult)

Description

The OnDataCollectError event is fired whenever a data source fails an attempt to retrieve history or update data. The OnDataCollectError event originates only from data sources since these are the only objects that operate on their own threads and asynchronously. All other operations are synchronous so errors can be caught with "On Error Goto..." statements.

Comments

The OnDataCollectError event returns the following parameters:

Parameter	Description
pIDataSource	A GProDataSource object for the data source.
hResult	A long value for the error number.

Example

```
'This subroutine will display the data error collected in a  
'text box
```

```
Sub OnDataCollectError(pIDataSource As Object, _  
    hResult As Long)  
    DataCollectError.text = "Data Error in " & _  
        pIDataSource.Name & ", error " & hResult  
    DataCollectError.visible = true  
End Sub
```

OnHairlineCursor

Syntax

OnHairlineCursor (pITrace, HairlineDateTime, HairlineValue, HairlineDataValid As Boolean)

Description

The OnHairlineCursor event fires whenever a trace's HairlineValue property changes. This event can occur as a result of any of the following:

- The user has moved the hairline cursor.
- The trace moves beneath the hairline cursor so that a different retrieved value becomes relevant.

Comments

The OnHairlineCursor event returns the following parameters:

Parameter	Description
pITrace	A GProTrace object, which contains the trace.
HairlineDateTime	A date for the position of the hairline cursor in the time axis.
HairlineValue	A double value for the trace value at the hairline cursor.
HairlineDataValid	A Boolean indicating if the value being returned is valid.

Example

```
'This subroutine writes the values of 4 traces into  
'text boxes. It also writes the date and time to another  
'text box.
```

```
Global sTrace1TagID As String  
Global sTrace2TagID As String  
Global sTrace3TagID As String  
Global sTrace4TagID As String
```

```
Sub OnHairlineCursor(Trace As Object, _  
    HairlineDateTime As Date, _  
    HairlineValue As Double, _
```

5. GProTrend Object

5.4. GProTrend Events

```
        HairlineDataValid As Boolean)
Dim sTagID As String
Dim oTag As Object

HairlineDateTimeValue.text = HairlineDateTime

Set oTag = Trace.UpdateTagPair.PrimaryTag

sTagID = oTag.TagID

Select Case sTagID
    Case sTrace1TagID
        If HairlineDataValid then
            Trace1HairlineValue.text = HairlineValue
        Else
            Trace1HairlineValue.text = "-----"
        End if
    Case sTrace2TagID
        If HairlineDataValid then
            Trace2HairlineValue.text =
HairlineValue
        Else
            Trace2HairlineValue.text = "-----"
        End if
    Case sTrace3TagID
        If HairlineDataValid then
            Trace3HairlineValue.text =
HairlineValue
        Else
            Trace3HairlineValue.text = "-----"
        End if
    Case sTrace4TagID
        If HairlineDataValid then
            Trace4HairlineValue.text =
HairlineValue
        Else
            Trace4HairlineValue.text = "-----"
        End if
End Select

End Sub
```

OnScrollLeft

Syntax

OnScrollLeft (oReserved as object)

Description

The OnScrollLeft event fires when the trace is scrolled to the left.

Comments

The OnScrollLeft event returns an object pointer for internal use only.

OnScrollRight

Syntax

OnScrollRight (oReserved as object)

Description

The OnScrollRight event fires when the trace is scrolled to the right.

Comments

The OnScrollRight event returns an object pointer for internal use only.

OnZoomIn

Syntax

OnZoomIn

Description

The OnZoomIn event fires when the trace is zoomed in.

Comments

The OnZoomIn event does not return any parameters.

5. GProTrend Object

5.4. GProTrend Events

Example

```
Sub OnZoomIn()  
  
    Dim years As Integer  
    Dim months As Integer  
    Dim days As Integer  
    Dim hours As Integer  
    Dim minutes As Integer  
    Dim seconds As Single  
    Dim intSeconds As Integer  
  
    'Need to get the current time span and display it  
  
    ProTrend.DateToTimeSpan _  
        ProTrend.VisibleTimeSpan, years, months, days, _  
        hours, minutes, seconds  
    intSeconds = cint(seconds)  
  
    txtVisibleTimeSpan.text = "Visible Time Span: " & _  
        years & ":" & months & ":" & days & ":" & _  
        hours & ":" & minutes & ":" & intSeconds  
    txtVisibleEarliestTime.text = "Earliest Time: " & _  
        & ProTrend.VisibleEarliestTime  
    txtVisibleLatestTime.text = "Latest Time: " & _  
        ProTrend.VisibleLatestTime  
  
End Sub
```

OnZoomOut

Syntax

OnZoomOut

Description

The OnZoomOut event fires when the trace is zoomed out.

Comments

The OnZoomOut event does not return any parameters.

Example

```
Sub OnZoomOut()  
  
    Dim years As Integer  
    Dim months As Integer  
    Dim days As Integer  
    Dim hours As Integer  
    Dim minutes As Integer  
    Dim seconds As Single  
    Dim intSeconds As Integer  
  
    'Need to get the current time span and display it  
  
    ProTrend.DateToTimeSpan _  
        ProTrend.VisibleTimeSpan, years, months, days, _  
        hours, minutes, seconds  
    intSeconds = Cint(seconds)  
  
    txtVisibleTimeSpan.text = "Visible Time Span: " & _  
        years & ":" & months & ":" & days & ":" & _  
        hours & ":" & minutes & ":" & intSeconds  
    txtVisibleEarliestTime.text = "Earliest Time: " & _  
        ProTrend.VisibleEarliestTime  
    txtVisibleLatestTime.text = "Latest Time: " & _  
        ProTrend.VisibleLatestTime  
End Sub
```

5. GProTrend Object

5.4. GProTrend Events

6. GProDataSources Object

6.1 Introduction

Purpose

The GProDataSources object is an OLE collection that contains GProDataSource objects. It exposes the IEnumVARIANT interface, which identifies it as an OLE collection. (The IEnumVARIANT interface is documented in Microsoft's online development documentation.) The following sections describe the methods and properties of the GProDataSources object, which are standard, or recommended, for OLE collections.

Default DataSourcesIf no specific Data Sources are configured, when the first trace is added to the trend a Default Update Data Source ("LocalTPN") and a Default History Data Source ("DefaultPHD") is created. For most applications, no other Data Sources are needed.

The Default History Data Source is the PHD Server named in the GPRO_PHD_HOST environment variable. If it is NULL, then the PHD_HOST environment variable is used.

The Default Update Data Source is determined by the GPRO_TREND_RTSSOURCE environment variable. If its value is "PHD" then the trend uses the same PHD Server as the History default. Otherwise, it is treated as the (TPS alias) name of a TPN Server. If it is NULL, the local private TPN Server is used.

Accessing the GProDataSources Object

Before using the GProDataSources object in a GUS graphic, a reference to it must be created. Any object name may be used. In the following example (and subsequent references), the GProDataSources object is assigned to an object called "oDataSources".

```
Dim oDataSources as object
```

```
set oDataSources = ProTrend.DataSources
```

The GProDataSources object is created as a result of instantiating its GProTrend object. It is then returned through the GProTrend's data sources method.

6.2 GProDataSources Properties

_NewEnum

Description

The `_NewEnum` property is requested by OLE collection clients. The following code example shows how to step through a collection using the For/Each idiom:

```
For Each FoundDataSource In MyTrendControl.DataSources
    Print FoundDataSource.Name
Next FoundDataSource
```

Example

'Cycle through all the datasources and look at the `hResult` property

```
Sub OnPeriodicUpdate()

    Dim oDS As Object
    For Each oDS in ProTrend.DataSources
        If oDS.Name = "LocalTPN" Then
            If oDS.hResult = 0 Then
                indLocalTPNStatus.fillcolor = TDC_Green
            Else
                indLocalTPNStatus.fillcolor = TDC_Red
            End If
        End If

        If oDS.Name = "DefaultPHD" Then
            If oDS.hResult = 0 Then
                indDefaultPHDStatus.fillcolor =
TDC_Green
            Else
                indDefaultPHDStatus.fillcolor = TDC_Red
            End If
        End If
    Next

End Sub
```

Count

Syntax

oDataSources.Count

Description

The Count property contains a long value equal to the number of GProDataSource objects in this collection.

Comments

- The Count property does not take any parameters.
- It is persistable. Persisting the Trend object also persists its GProDataSources OLE collection.
- It has a default value of 0.
- It has a default value of 0.
- It is read-only.

Example

```
TxtNumberOfDataSources.text = ProTrend.DataSources.count
```

6.3 GProDataSources Methods

Item

Syntax

oDataSources.Item(Name) where: *Name* is the name of the data source object (e.g. "MyLCN").

Description

The Item method returns a GProDataSource object, which is used to identify the data source object.

The Item method supports the “Item” idiom for this OLE collection. Using this idiom, the following two lines of code are equivalent:

```
Set FoundSource = ProTrend.DataSources("My LCN")
```

```
Set FoundSource = ProTrend.DataSources.Item ("My LCN")
```

Comments

The Item method takes the following parameter:

Parameter	Description
Name	The Name parameter is of the VARIANT type. When the Name is a string, the Item method returns the data source object with the same name. The match is case-sensitive.

Example

```
If ProTrend.DataSources.Item("LocalTPN").hResult <> 0 then  
    MsgBox ProTrend.DataSources.Item("LocalTPN").hResult
```

Add

Syntax

oDataSources.Add Name

Description

The Add method returns a GProDataSource object, which is used to do the following:

- Create a new GProDataSource object.
- Add the new GProDataSource object to the GProDataSources collection.
- Return the new GProDataSource object's interface.

Comments

The Add method takes the following parameter:

Parameter	Description
Name	A string which contains the name that is given to the new GProDataSource object within the collection, and which is accessible from the data source's Name property (described in GProDataSource Properties section).

Example

```
'Add a data source

dim oSource as object
oSource = ProTrend.DataSources.Add ("newDataSource")

'Now all the data source properties and methods can be
'referenced

ProTrend.DataSources.Item("NewDataSource").Host = "164.178.1.2"
```

Remove

Syntax

oDataSources.Remove Name

Description

DataSources.Remove removes the data source whether or not it is active.

If the data source was active, it is deactivated before removal and any traces configured to use that data source stops receiving updates.

Comments

The Remove method takes the following parameter:

Parameter	Description
Name	The Name parameter is of the VARIANT type. When the Name is a string, the Remove method attempts to delete the data source object with the same name. The match is case-sensitive.

Example

'Remove the data source added in the previous example

```
ProTrend.DataSources.Remove "NewDataSource"
```

Trend**Syntax**

oDataSources.Trend

Description

The Trend method returns a GProTrend object, which identifies the trend object that owns the GProDataSources object.

Comments

The Trend method does not take any parameters.

7. GProDataSource Object

7.1 Introduction

Purpose

The GProDataSource object relates to a data source. The GProDataSource object abstracts the idea of a TPN Server, PHD Server or any other data source from which process data can be retrieved. In most such data sources, each retrieval operation retrieves a group of items together in order to make the most of bandwidth. The GProDataSource object also incorporates this concept so that the tags that belong to the GProDataSource are internally managed into collection groups so that they may be retrieved optimally. A GProDataSource object may therefore be thought of as encapsulating both the data source and the groups of tags that is retrieved from the data source. The involved tags are the active tags that are configured to the data source through their DataSource properties.

It is not necessary use the programmatic interface to associate traces to a GproDataSource. When a Tag is added to the trend, it is automatically assigned to the default Update or History data source configured for this GUSPro console.

7.2 GProDataSource Properties

The following table provides a synopsis of each of the properties of GProDataSource objects.

NOTE: Before changing the properties of a Data Source, that source must be deactivated.

Property	Type	Scripting Access	Usage	Default Value
Type	long	Read/ Write	All types	0 = kInvalid
Name	BSTR	Read Only	All types	provided by container
HCIServerAliasName	BSTR	Read/ Write	Type = kTPN, If specified, overrides Host & Server	Blank
Host	BSTR	Read/ Write	KTPN & kPHDHistory	Blank; For PHD, environment variable "PHD_HOST" For TPN, assumes local node's network name.
Server	BSTR	Read/ Write	kTPN	Blank. For TPN uses the local TPN server
Username	BSTR	Read/ Write	kPHDHistory	Blank
Password	BSTR	Read/ Write	kPHDHistory	Blank
LastCollectionTime	DATE	Read	All types	OLE date epoch
Timeout	double	Read/ Write	All types	5
State	long	Read Only	All types	kInactive
NativeTagPrefix	BSTR	Read/ Write	Any type – but designed for PHD	Blank

Further detail is provided in the following subsections.

long Type

The *Type* property contains one of the enumerated constants in the following table:

Constant	Numerical Value
KTPN	0
kPHDHistory	2
KTest	4

The following subsections describe each of the enumerated constants.

KTPN

This data source type identifies an HCI / OPC data server.

If HCIServerAliasName, Host and Server are all blank, this is the PrivateTPN server on the local node.

kPHDHistory

This data source uses a PHD server, using PHD's own non-COM interface.

KTest

This data source type generates sine wave data for testing and demonstration purpose.

kInvalid

This data source type is used, as a placeholder, when there is no valid data source. When this data source type is asked for data, it generates an error. This enumerated constant is the default for this *Type* property, indicating that no valid data source type is selected.

BSTR Name

This property is the name of the data source as chosen by the trend client program.

NOTE: References to the DataSource Name are case sensitive.

BSTR Host

This property bears the name of the host computer from which the data source retrieves the data.

BSTR Username

For data sources, that utilize a username/password combination, this property bears the username.

BSTR Password

For data sources, that utilize a username/password combination, this property bears the (unencrypted) password.

BSTR Server

This property bears the server from which the data source is to collect data. This is distinct from the *Host* property since any given computer may host more than one server process.

BSTR HCIServerAliasName

This property bears the user-friendly HCI component name as described, for the in the `HCI::GetComponentInfo ()` method in the *HCI Client Developer's Guide*. When this property is written to, `HCI::GetComponentInfo ()` is used to determine the corresponding Host (described in BSTR Host section) and Server (described in BSTR Server section) properties, and those properties are updated accordingly.

DATE LastCollectionTime

This read-only property contains the date, and time, at which the last data was collected. Depending on the data source server, this may be the timestamp returned in the data packet from the data source server. For data sources that don't support this, it bears the time provided by the timer, which is the time at which the timer must trigger in order not to drift. This is described in the third paragraph of section 2.4.2.1.3 above.

double Timeout

This is the timeout configured for the data source in seconds. If the data source receives a collection request, which it hasn't fulfilled, for longer than this number of seconds, a timeout condition occurs, with results described in section 2.4.4.1.13.3 below.

long State

This property can be one of the enumerated constants in the following table:

Constant	Numerical Value
kNormal	0
kInactive	1

Constant	Numerical Value
kTimedout	2
kOverrun	3

A separate *kError*, or *kFailed*, constant is not necessary because, if the data source is inactive, an error *hResult* shows that the data source became inactive because of an error. The following subsections describe each of the enumerated constants.

kNormal

When the *State* property contains *kNormal*, this indicates that the data source is active and there is no request overrun. When there is an overrun, the *State* property contains *kOverrun*, described below.

kInactive

When this property contains *kInactive* it can mean any of the following:

1. The data source has been deactivated, by invoking the data source's Deactivate () method.
2. The data source has become inactive because of a fail-over, for some other reason than a time-out.
3. The data source has not yet been activated, by calling the Activate () method.

kInactive is the default value of the *State* property.

kTimedOut

When the *State* property contains *kTimedOut*, it means that the data source timed out. Successive timer requests continues to queue up until the data collection thread recovers and services the requests (in which case *State* goes back to *kNormal*), or until the data source becomes inactive.

kOverrun

When the *State* property contains *kOverrun*, it is equivalent to *kNormal*, except that a timer request has been made before the previous timer request can be completed. If the data source is both overrun, and timed out, the state is *kTimedOut*.

BSTR NativeTagPrefix

This property contains a prefix for converting tags IDs, from a different data source type, to the current data source. This is useful when, for example, a particular site's convention is to name all PHD tags by prefixing their associated TPN tag names with standard prefix. Then, converting from a TPN tag ID, to a PHD tag ID, consists of simply prefixing the TPN tag ID with the standard prefix. This feature exists to allow tag ID conversions that cannot be practically supported otherwise.

7.3 GProDataSource Methods

Activate()

Invoking this method causes the DataSource object to do the following

1. If the *State* property is not *kInactive*, the DataSource inactivates, as described in section 2.4.4.2.2 below, except that its tags are not failed-over.
2. Spawn a data collection thread
3. Log on to the data source server.
4. If it has active tags, collect the high and low ranges and alarm trip points for each of its active tags.
5. If it has active tags, configure its collection groups to collect the data for each tag that references it as its data source

Each data source has its own data collection thread so that, must the data source server hang, the GPTC still continues to update the tags belonging to other data sources (provided they are operating sufficiently), and also retain the ability to act upon the latency of the hung data source.

Logging on to the data source entails different specific behavior, depending on which type of data source is involved. With the PHD data source, logging on entails forming a connection and supplying the username and password to the PHD server. For TPN data sources, logging consists of contacting the TPN server and acquiring its relevant interfaces.

Logging on is always undertaken by the data collection thread, regardless of which data source it is, to avoid any assumption as to the actual server's threading model. If logging in is done in response to writing a property, or through an explicit method call, that would imply that consequently acquired interfaces would belong to the main thread, while the interfaces would be accessed by the data collection thread. Since this can cause some servers to fail, the assumption is avoided entirely and the data collection thread logs in to the server as one of the first things it does after it is spawned.

When all this is successful, the *Status* property transitions to *kNormal*.

Whenever there is an active data collection thread, that thread receives messages from the timer as described in section 2.4.2.1.4 above.

Deactivate()

Invoking this method causes the data source to inactivate itself. This is done by sending a message, to the data source object's data collection thread, with the instruction to self-destruct. When the thread self-destructs, it performs all the necessary steps of logging out of its server and releasing interfaces as appropriate.

7. GProDataSource Object

7.3. GProDataSource Methods

8. GProTag Object

8.1 Introduction

Purpose

The GProTag object represents an individual tag that can be collected by a data source. This is distinct from a trace. Any given GProTrace represents a visual trace in the trend display area, whereas a GProTag object represents a tag ID and other properties that establish data collection parameters from a data source, specific to the data source.

This section describes the GProTag properties and methods.

Accessing the GProTag Object

Before using the GProTag object in a GUS graphic, a reference to it must be created. Any object name may be used. In the following example (and subsequent references), the GProTag object is assigned to an object called “oTag”.

```
Dim oTag as object  
  
set oTag = oTagPair.CreatePrimaryTag 'Create  
  
set oTag = oTagPair.PrimaryTag 'Access
```

8.2 GProTag Properties

TagID

Syntax

oTag.TagID

Description

The TagID property returns string with the external tag ID as it is known to the data source server. For example, if the data source type is kTPN, then a valid tag ID may appear as “AM001.PV”. If the type is kPHDHistory, then the tag ID is the PHD tag.

8. GProTag Object

8.2. GProTag Properties

Comments

- The TagID property does not take any parameters.
- It is persistable.
- It has a default of "".

Example

```
oTag.TagID = "REG1001.PV"
```



TIP

To trend a Boolean or Enumeration parameter, include the .Internal attribute in the TagID. For example, the following trend mode changes as integer state values:

```
Otag.TagID = "REG1001.Mode.Internal"
```

Type

Syntax

oTag.Type

Description

The Type property returns a GProTagType that identifies the type of tag. The GProTagType object can be set to the enumerated constant described in the following table.

Constant	Numerical Value	Description
kPrimary	0	Indicates that the tag is the primary tag within its tag pair.

Comments

- The Type property does not take any parameters.
- It is persistable.
- It does not have a default.
- It is read-only.

Example

```
txtTagType.text = oTag.Type
```

InternalID

Syntax

oTag.InternalID

Description

The InternalID property returns a string containing the internal tag ID as it is known to the data source server. For the PHD data source, this is the four-byte tag number by which the tag is known to the PHD Server.

This property is available as a hexadecimal representation of the binary form of the internal ID. There is no assumption as to its size, in order to accommodate future data source types for which the internal ID may be of an arbitrary length.

The contents of this property are not valid until the tag's data source fires its OnGotData event to indicate successful data retrieval. This is because the internal IDs are gathered during the data source log on process.

Comments

- The InternalID property does not take any parameters.
- It is not persistable.
- It has a default of "".
- It is read-only.

8. GProTag Object

8.2. GProTag Properties

Example

```
txtTagInternalID.text = oTag.InternalID
```

ValueDataSource

Syntax

oTag.ValueDataSource

Description

The ValueDataSource property returns the GProDataSource object from which the GProTag gets the data for the following:

- It's trending value.
- The trending value's tag description.

If the ValueDataSource property is NULL, then activating this tag results in the following behavior:

- If the tag is in a history tag pair, the data source is set to whichever data source is configured in the trend's DefaultHistoryDataSource property. If the DefaultHistoryDataSource property is NULL, then the ValueDataSource property is set to whichever data source has the name "PHD". If no such data source exists, a default PHD data source is created, and then the tag's data source is set to it.
- If the tag is in an update tag pair, the data source is set to whichever data source is configured in the trend's DefaultUpdateDataSource property. If the DefaultUpdateDataSource property is NULL, then the ValueDataSource property is set to whichever data source is named LocalTPN. If this data source doesn't exist, a default TPN data source is created.

Comments

- The ValueDataSource property does not take any parameters.
- It is persistable.
- It has a default of NULL.

Example

```
'Get the name of the value data source
```

```
txtTagValueDataSource.text = oTag.ValueDataSource.name
```

RangingDataSource

Syntax

oTag.RangingDataSource

Description

The RangingDataSource property returns the GProDataSource object from which GProTag gets the lower and upper range data for scaling.

If this property is NULL, then when this tag is activated, the ranging data source is set to whichever data source is configured in the trend's DefaultUpdateDataSource property. If the trend's DefaultUpdateDataSource property is NULL, then the RangingDataSource property is set to whichever data source is named LocalTPN. If this data source doesn't exist, a default TPN data source is created.

Comments

- The RangingDataSource property does not take any parameters.
- It is persistable.
- It has a default of NULL.

Example

```
'Get the name of the value data source
```

```
txtTagRangingDataSource.text = oTag.RangingDataSource.name
```

ReadFromCache

Syntax

oTag.ReadFromCache

Description

The ReadFromCache property returns Boolean which determines whether the data source reads from the data source server's cache, or request the data directly from the field device. When this property is True, the TPN data sources retrieves the tag's value using the OPC_DS_CACHE option (described in the *OLE For Process Control Data Access Custom Interface Standard*). If this property is False, the OPC_DS_DEVICE option is used.

Comments

- The ReadFromCache property does not take any parameters.
- It is persistable.
- It has a default of False.

Example

```
oTag.ReadFromCache = True
```

RequestedCacheUpdatePeriod

Syntax

oTag.RequestedCacheUpdatePeriod

Description

The RequestedCacheUpdatePeriod property returns a double value that indicates the desired accuracy of cached data in seconds. This property is intended only to control the behavior of the interface. How the server deals with the update rate and how often it actually polls the hardware internally is an implementation detail of the data source. A value of 0 indicates that the server must use the fastest practical rate.

Comments

- The RequestedCacheUpdatePeriod property does not take any parameters.
- It is persistable.
- It has a default of 0.

Example

```
oTag.RequestedCacheUpdatePeriod = 5
```

RetrievedCacheUpdatePeriod

Syntax

oTag.RetrievedCacheUpdatePeriod

Description

The RetrievedCacheUpdatePeriod property returns a double value that contains the actual period (in seconds) at which the tag's value is updated in the data source server's cache. This may be different from the refresh period requested. This property appears on the tag so that the data source may build each tag into an appropriate collection group.

This property does not become valid until the data source collects data for this tag.

Comments

- The RetrievedCacheUpdatePeriod property does not take any parameters.
- It is not persistable.
- It has a default of 0.
- It is read-only.

Example

```
TxtCacheUpdatePeriod.text = oTag.RetrievedCacheUpdatePeriod
```

RangeHigh

Syntax

oTag.RangeHigh

Description

The RangeHigh property returns a VARIANT that determines the maximum measurement value that is retrieved for the tag. Writing this property updates the trace's RetrievedRangeHigh, and sets the trace's RangeHighFixed property to 0.

The value must be specified as a Variant of type real or type String. If the variant type is real, then RangeHigh or RangeLow is set to an integer value. The RetrievedRangeHigh/Low values are not updated if the RangeHigh/Low is set to a negative value. But if a single type (real) variant is used, the RetrievedRangeHigh/Low values are updated even if the RangeHigh or RangeLow are set to negative values.



TIP

If a constant value is assigned to the real variant, it must include a "." decimal to guarantee an accurate numeric assignment.

If the variant type is a string, then the RangeHigh string is treated as an external tag ID. The data source collects the range high value during the following operations:

- The log on process.
- When the tag is active and becomes configured to an active data source.
- Whenever the RangeHigh property is written to an active tag that is configured to an active data source.

Then the trace's RetrievedRangeHigh is updated with resulting numeric value.

If the variant type is empty or null, then the data source makes assumptions as to how the range high value must be determined, and writes that numeric value to the trace's RetrievedRangeHigh property.

If the data source is unsuccessful in assigning the trace's RetrievedRangeHigh property, then the trace's RetrievedRangeHigh property is set to the first Value collected, and the trace's RangeHighFixed property is set to the appropriate failure HRESULT.

Comments

- The RangeHigh property does not take any parameters.
- It is persistable.
- It has a default of empty.

Example

```
oTag.RangeHigh = 100
```

RangeLow

Syntax

oTag.RangeLow

Description

The RangeLow property returns a VARIANT that determines the minimum measurement value that is retrieved for the tag. Writing this property updates the trace's RetrievedRangeLow, and sets the trace's RangeLowFixed property to 0.

If the variant type is numeric, then the RangeLow value is treated as a constant, converted to a double (if is a different kind of numeric value) and written directly to the trace's RetrievedRangeLow whenever this tag becomes the active update tag.

If the variant type is a string, then the RangeLow string is treated as an external tag ID. The data source collects the range low value during the following operations:

- The log on process.
- When the tag is active and becomes configured to an active data source.
- Whenever this property is written to an active tag that is configured to an active data source.

Then the trace's RetrievedRangeLow is updated with the resulting numeric value whenever this tag becomes the active update tag.

8. GProTag Object

8.2. GProTag Properties

If the variant type is empty or null, then the data source makes assumptions as to how the minimum limit must be determined, and writes that numeric value to the trace's RetrievedRangeLow property whenever this tag becomes the active update tag. These assumptions are documented for each type of data source in Assigning RetrievedRangeHigh and RetrievedRangeLow Properties section.

If the data source is unsuccessful in assigning the trace's RetrievedRangeLow property, then the trace's RetrievedRangeLow property is set to the first Value collected, and the trace's RangeLowFixed property is set to the appropriate failure HRESULT.

Comments

- The RangeLow property does not take any parameters.
- It is persistable.
- It has a default of empty.

Example

```
oTag.RangeLow = 0
```

DescriptionTagID

Syntax

```
oTag.DescriptionTagID
```

Description

The DescriptionTagID property returns a string that contains the tag ID that is used for reading the trace's RetrievedDescription property from the data source. If this is an empty string, then the configured data source infers which tag ID ought to be used depending on the type of the data source.

Comments

- The DescriptionTagID property does not take any parameters.
- It is persistable.
- It has a default of "".

Example

```
TxtDescriptionTagID.text = oTag.DescriptionTagID
```

Description

Syntax

oTag.Description

Description

The Description property returns a string that contains descriptive text as written by the user program.

Comments

- The Description property does not take any parameters.
- It is persistable.
- It has a default of "".

Example

```
oTag.Description = "Update tag for Flow1"
```

8. GProTag Object

8.2. GProTag Properties

State

Syntax

oTag.State

Description

The State property returns the GProTagState that specifies the tag's state. The GProTagState can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kTagNormal	0	Indicates one of the following: - If the tag's data source is active, the data source collects data for the tag. - If the tag's data source is inactive, then activating the data source causes the data source to begin collecting data for the tag. When the tag is active, its Value property is used to update the tag's tag pair trace.
kTagInactive	1	Indicates that the data source does not collect data for this tag.

Comments

- The State property does not take any parameters.
- It is not persistable.
- It has a default of 1 (kTagInactive).

Example

```
TxtTagState.text = oTag.State
```

8.3 GProTag Methods

TagPair

Syntax

oTag.TagPair

Description

The TagPair method returns the GProTagPair object that identifies the tag pair to which this tag belongs.

Comments

The TagPair method does not take any parameters.

Example

'Get the tags TagPair name

`txtTagPairName.text = oTag.TagPair.Name`

Activate

Syntax

oTag.Activate

Description

If the tag's data source is active, calling the Activate method causes the data source object to do the following:

1. Update its list of tags for which to collect data so that it begins collecting data for the tags.
2. Collect the RangeHigh and other such parameters of the tag.

If the tag's data source is not active, then the data source reacts as follows:

1. Still update its list of tags, but does not collect any data until it becomes active.
2. Not collect the RangeHigh, and other such parameters; but it does as soon as it becomes active.

8. GProTag Object

8.3. GProTag Methods

Comments

The Activate method does not take any parameters.

Example

```
oTag.Activate
```

Deactivate

Syntax

```
oTag.Deactivate
```

Description

Calling the Deactivate method causes the tag to become inactive. The tag's data source stops collecting data for that tag.

Comments

The Deactivate method does not take any parameters.

Example

```
oTag.Deactivate
```

9. GProTagPair Object

9.1 Introduction

Purpose

The GProTagPair object represents a single GProTag object, which is the primary tag. This section describes the GProTagPair properties and methods.

Accessing the GProTagPair Object

Before using the GProTagPair object in a GUS graphic, a reference to it must be created. Any object name may be used. In the following example (and subsequent references), the GProTagPair object is assigned to an object called “oTagPair”.

```
Dim oTagPair as object  
set oTagPair = oTrace.UpdateTagPair
```

where oTrace is a GProTrace object.

The GProTagPair object's instantiation is implicit in the instantiation of its containing GProTrace object.

9.2 GProTagPair Properties

PrimaryTag

Syntax

oTagPair.PrimaryTag

Description

The PrimaryTag property returns a GProTag object that contains the primary tag object of the tag pair. If the primary tag doesn't exist, this property contains NULL.

9. GProTagPair Object

9.2. GProTagPair Properties

Comments

- The PrimaryTag property does not take any parameters.
- It is persistable.
- It has a default of NULL.
- It is read-only.

Example

```
oTagPair.PrimaryTag.activate
```

Type

Syntax

```
oTagPair.Type
```

Description

The Type property returns a GProTagPairType that identifies the type of tag pair. The GProTagPairType can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kHistory	0	Indicates that the tag pair is the history tag pair.
kUpdate	1	Indicates that the tag pair is the update tag pair.

Comments

- The Type property does not take any parameters.
- It is persistable.
- It does not have a default.
- It is read-only.

Example

```
TxtTagPairType.text = oTagPair.Type
```

AggregateType

Syntax

oTagPair.AggregateType

Description

The AggregateType property returns a GProAggregateType that indicates the type of historical aggregate that is to be requested from the data source. The GProAggregateType can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kRaw	0	In this mode, the data source retrieves actual raw data for the requested aggregate interval, without interpolation. It is equivalent to the "raw" aggregate type of PHD.
kSnapshot	1	In this mode, the data source does not retrieve an aggregate, but retrieved values may be interpolated ones. It is equivalent to the PHD snapshot aggregate type.
		ATTENTION Requesting data beyond that which PHD has collected results in false extrapolated data being returned from PHD. To avoid this when in snapshot mode, make sure the update rate specified for the trend is at least twice as much as the update period on PHD.
kAverage	3	In this mode, the data source retrieves the average data over the aggregate interval. It is equivalent to the PHD PHR_Avg constant.
kStDev	6	In this mode, the data source retrieves the standard deviation of data over the aggregate interval.

9. GProTagPair Object

9.2. GProTagPair Properties

Constant	Numerical Value	Description
kMinimum	8	In this mode, the data source retrieves the minimum value in the resample interval. It is equivalent to the PHD PHR_Min constant and to the MinimumAtInterval reduction type (documented in the <i>Uniformance Visual PHD Developer's Guide</i>).
kMaximum	10	In this mode, the data source retrieves the maximum value in the aggregate interval. It is equivalent to the PHD PHR_Max constant and to the MaximumAtInterval reduction type (documented in the <i>Uniformance Visual PHD Developer's Guide</i>).
kStart	11	In this mode, the data source retrieves the value at the beginning of the aggregate interval. The timestamp at the beginning of the interval is used and is plotted on the trend display. It is equivalent to the PHD PHR_First constant.
kEnd	12	In this mode, the data source retrieves the value at the end of the aggregate interval. The timestamp at the end of the interval is used and is plotted on the trend display. It is equivalent to the PHD PHR_Last constant.
kDelta	13	In this mode, the data source retrieves the difference between the first and last value in the resample interval.
kRange	18	In this mode, the data source retrieves the difference between the minimum and maximum value over the aggregate interval.

Exclusion of Interpolated Data by a PHD Data Source

For every aggregate type, except kSnapshot, the PHD data source collects raw values before requesting any aggregate data. The raw data is used to ensure that this data source never requests an aggregate that contains a time interval for which data has not yet been collected.

When the timer triggers this data source it follows the steps below for each of its tags.

- Check for the latest raw data.
- Calculate the latest complete timer period that contains the latest raw data. The interval is complete when none of it is in the future relative to the time requested by the timer.
- Request the data for that period and update the trend display. If there are two or more aggregate periods, within the requested period, then a point is plotted for each such aggregate period.

Comments

- The AggregateType property does not take any parameters.
- It is persistable.
- It has a default of 0 (kRaw).

Example

```
oTagPair.AggregateType = 0 'kRaw
```

AggregateOffset

Syntax

oTagPair.AggregateOffset

Description

The AggregateOffset property returns a GProAggregateOffset that determines the relationship between aggregate intervals and timestamps. In historical data, timestamps have a limited resolution and the AggregateOffset provides a means of determining which interval is implied by a given timestamp.

9. GProTagPair Object

9.2. GProTagPair Properties

The GProAggregateOffset can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kBefore	0	Indicates that the intervals are those that precede the timestamp.
kAfter	1	Indicates that the intervals are those that follow the timestamp.
kAround	2	Indicates that the intervals are those for which the timestamp is the midpoint.

Comments

- The AggregateOffset property does not take any parameters.
- It is persistable.
- It has a default of 0 (kBefore).

Example

```
oTagPair.AggregateOffset = 0 'kBefore
```

AggregateSampleInterval

Syntax

```
oTagPair.AggregateSampleInterval
```

Description

The AggregateSampleInterval property returns a date that defines the size of each interval, for which the aggregate is applied. If this property is set to zero, representing the Epoch, then the UpdatePeriod property of the GProTrend object is applied.

For times less than 24 hours, GPB scripts can automatically convert a string of the form “hh:mm:ss” and assign it to this property. In other circumstances, it is necessary to use the TimeSpanToDate method of the trend object.

Comments

- The AggregateSampleInterval property does not take any parameters.
- It is persistable.
- It has a default of Epoch. In this case the UpdatePeriod is used.

Example

```
oTagPair.AggregateSampleInterval = "0:20:0" '20 minutes  
oTagPair.AggregateSampleInterval = _  
    ProTrend.TimeSpanToDate(0,0,0,0,20,0) '20 minutes
```

AggregateInterval

Syntax

oTagPair.AggregateInterval

Description

The AggregateInterval property returns a date that defines the size of the requested aggregate interval. If this property is zero, representing the Epoch, then the UpdatePeriod property of the GProTrend object is applied.

For times less than 24 hours, GPB scripts can automatically convert a string of the form “hh:mm:ss” and assign it to this property. In other circumstances, it is necessary to use the TimeSpanToDate method of the trend object.

This property is useful when the aggregate interval must be longer than the update period. For example, if the trend is updating every five seconds, it is possible to plot a new one-hour average with each trace update. If the update period is five seconds, the sample interval is 30 minutes, and the aggregate interval is one hour, then every five seconds, two thirty second averages are plotted for the past hour.

Comments

- The AggregateInterval property does not take any parameters.
- It is persistable.
- It has a default of Epoch. In this case the UpdatePeriod is used.

Example

```
oTagPair.AggregateInterval = "0:20:0" '20 minutes  
oTagPair.AggregateInterval = _  
    ProTrend.TimeSpanToDate(0,0,0,0,20,0) '20 minutes
```

9.3 GProTagPair Methods

ActiveTag

Syntax

oTagPair.ActiveTag

Description

The ActiveTag method returns a GProTag object that identifies the active tag object of the tag pair. For example, if the primary tag is the one that is active, then the ActiveTag method returns the primary tag object.

Comments

The ActiveTag method does not take any parameters.

Example

```
TxtActiveTag.text = oTagPair.ActiveTag.TagID
```

Trace

Syntax

oTagPair.Trace

Description

The Trace method returns a GProTrace object that identifies the trace to which this tag pair belongs.

Comments

The Trace method does not take any parameters.

Example

```
TxtTagPairTrace.text = oTagPair.Trace.Name
```

ActivatePrimaryTag

Syntax

```
oTagPair.ActivatePrimaryTag
```

Description

The ActivatePrimaryTag method calls Activate on the primary tag, if the primary tag is not active.

Comments

The ActivatePrimaryTag method does not take any parameters.

RemovePrimaryTag

Syntax

```
oTagPair.RemovePrimaryTag
```

Description

The RemovePrimaryTag method attempts to remove the primary tag as follows:

- If the primary tag's State property is kTagNormal, the RemovePrimaryTag method attempts to failover to the failover tag. If this fails, the method raises the HRESULT that is raised from the FailOver method.
- If the primary tag's State property is not kTagNormal, it removes the tag.

When a primary tag is removed (or not yet created), the PrimaryTag property contains NULL.

Comments

The RemovePrimaryTag method does not take any parameters.

CreatePrimaryTag

Syntax

oTagPair.CreatePrimaryTag

Description

The CreatePrimaryTag method attempts to create a tag object and assign it to the PrimaryTag property.

Comments

The CreatePrimaryTag method does not take any parameters.

Example

```
OtagPair.CreatePrimaryTag
```

10. GProTraces Object

10.1 Introduction

Purpose

The GProTraces object is an OLE collection that contains GProTrace objects. It exposes the IEnumVARIANT interface, which identifies it as an OLE collection. (The IEnumVARIANT interface is documented in Microsoft's online documentation.) The following sections describe the methods and properties of the GProTraces object, which are standard, or recommended, for OLE collections.

Accessing the GProTraces Object

Before using the GProTraces object in a GUS graphic, a reference to it must be created. Any object name may be used. In the following example (and subsequent references), the GProTraces object is assigned to an object called "oTraces".

```
Dim oTraces as object  
set oTraces = ProTrend.Traces 'Access
```

The GProTraces object's instantiation is implicit in the instantiation of its containing GProTrend object.

10.2 GProTraces Properties

_NewEnum

Description

The _NewEnum property is requested by OLE collection clients. The following code example shows how to step through a collection using the For/Each idiom:

```
For Each FoundTrace In MyTrend.Traces  
    Print FoundTrace.Name  
Next FoundTrace
```

10. GProTraces Object

10.2. GProTraces Properties

Example

```
Dim oTrace as object
For Each oTrace in ProTrend.Traces
    If oTrace.Name = "Flow1" then
        MsgBox "Found"
    End if
next
```

Count

Syntax

oTraces.Count

Description

The Count property returns a long value equal to the number of GProTrace objects in this collection.

Comments

- The Count property does not take any parameters.
- It is persistable. Persisting the Trend object also persists its GProTraces OLE collection.
- It has a default value of 0.
- It is read-only.

Example

```
TxtNumberOfTraces.text = ProTrend.Traces.count
```

10.3 GProTraces Methods

Item

Syntax

oTraces.Item(Name)

Description

The Item method returns a GProTrace object that is used to identify the trace object.

The Item method supports the “Item” idiom for this OLE collection. Using this idiom the following two lines of code are equivalent:

```
Set FoundTrace = MyTrend.Traces("My Trace")
```

```
Set FoundTrace = DataSource.Trace.Item ("My Trace")
```

Comments

The Item method takes the following parameter:

Parameter	Description
Name	The Name parameter is of the VARIANT type. When the Name is a string, the Item method returns the trace object with the same name. The match is case-insensitive.

Example

```
If ProTrend.Traces.Item("Flow1").hResult = 0 then  
    MsgBox "No error in Flow1"  
End if
```

Add

Syntax

oTraces.Add Name

Description

The Add method returns a GProTrace object, which is used to do the following:

- Create a new GProTrace object.
- Add the new GProTrace object to the GProTraces collection.
- Return the new GProTrace object's interface.

Comments

The Add method takes the following parameter:

Parameter	Description
Name	A string which contains the name that is given to the new GProTrace object within the collection, and which is accessible from the trace's Name property (described in Name section). If the Name parameter is the null string, then the new trace object's Name property defaults to TraceN, where N represents an integer that makes the name unique within the GProTraces OLE collection.

Example

```
Dim oTrace2 as object  
Set oTrace2 = ProTrend.Traces.Add("Flow2")
```

Remove

Syntax

oTraces.Remove Name

Description

Given the name (or index within the GProTraces collection) of a GProTrace object, the Remove method removes it from the collection. Then:

- Its GProTagPair objects are destroyed with all of their respective tag objects.
- The visible trace is no longer be visible in the trend display area.

Comments

The Remove method takes the following parameter:

Parameter	Description
Name	The Name parameter is of the VARIANT type. When the Name is a string, the Remove method attempts to delete the GProTrace object with the same name. The match is case-insensitive.

Example

```
ProTrend.Traces.Remove "Flow2"
```

GetEldest

Syntax

oTraces.GetEldest

Description

The GetEldest method returns a string that contains the name of the oldest trace with a Deletable property set to True. This can be used to remove the eldest trace in order to add a new trace.

Comments

The GetEldest method does not take any parameters.

Example

```
Dim sNameEldest as string
sNameEldest = ProTrend.Traces.GetEldest
'now remove this trace
oTraces.Remove sNameEldest
```

Trend

Syntax

oTraces.Trend

Description

The Trend method returns a GProTrend object that identifies the trend object that owns the GProTraces object.

Comments

The Trend method does not take any parameters.

Example

'The following example will display a message box containing
'the string ProTrend

```
msgbox ProTrend.Traces.Trend.Name
```

11. GProTrace Object

11.1 Introduction

Purpose

The GProTrace object represents an individual trace that can appear on the trend display. Trace objects receive updates from their ActiveTag objects. When such updates are received, they are plotted in the trend area. This section describes the GProTrace properties and methods.

Accessing the GProTrace Object

Before using the GProTrace object in a GUS graphic, a reference to it must be created. Any object name may be used. In the following example (and subsequent references), the GProTrace object is assigned to an object called "oTrace".

```
Dim oTrace as object  
set oTrace = oTraces.Add("My Trace") 'Create  
set oTrace = oTraces("My Trace") 'Access
```

11.2 GProTrace Properties

Name

Syntax

oTrace.Name

Description

The Name property returns a string with the name of trace by which it can be identified within the GProTraces collection. This name can be used to retrieve the GProTrace interface, using the GProTraces object.

11. GProTrace Object

11.2. GProTrace Properties

Comments

- The Name property does not take any parameters.
- It is persistable.
- It has a default of "TraceN".
- It is read-only.

Example

```
TxtTraceName.text = oTrace.Name
```

Description

Syntax

```
oTrace.Description
```

Description

The Description property returns a string with a description of the trace.

Comments

- The Description property does not take any parameters.
- It is persistable.
- It has a default of "".

Example

```
oTrace.Description = "Flow1 Trace"
```

Color

Syntax

oTrace.Color

Description

The Color property returns a long value for the trace color as it appears on the trend display. Writing this property causes the color of the trace to be immediately changed on the trend display.

Comments

- The Color property does not take any parameters.
- It is persistable.
- The default trace color is collected from the GProTrend object's ColorPallet property, as described in ColorPallet section.

Example

```
oTrace.Color = 50331647 'white
```

Deletable

Syntax

oTrace.Deletable

Description

The Deletable property returns a Boolean which determines whether a tag is identified for removal, by using the GetEldest method described in GetEldest section.

Comments

- The Deletable property does not take any parameters.
- It is persistable.
- It has a default of False.

11. GProTrace Object

11.2. GProTrace Properties

Example

```
oTrace.Deletable = false
```

CanonicalDataType

Syntax

```
oTrace.CanonicalDataType
```

Description

The CanonicalDataType property returns a short value that contains the variant type of the data being requested for the active update tag. Data sources always request the data as a double; however the data may have a different type as maintained by the data source server. This property contains the data type of the value as stored in the data source server.

Comments

- The CanonicalDataType property does not take any parameters.
- It is not persistable.
- It has a default of 0 (VT_EMPTY).
- It is read-only.

Example

```
txtCanonicalDataType.text = oTrace.CanonicalDataType
```

Value

Syntax

oTrace.Value

Description

The Value property returns a double value that contains the last value for the tag that was collected by the data source. Data sources always request the data as a double or convert the data to a double if it is not. For example, if the data source server maintains the data as an enumeration (that is, a string), the data source requests the data as a double so that the data source server provides it as the corresponding ordinal value in double form.



TIP

This Value does not identify NaN values or error status. Scripts must also check the OPCQuality property.

Comments

- The Value property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtValue.Text = oTrace.value
```

TimeStamp

Syntax

oTrace.TimeStamp

Description

The TimeStamp property returns the timestamp of the last value that was collected for the active update tag. This is maintained on the trace object because the data source server may maintain a distinct timestamp for each value even though the data source may request all of its tags as a single transaction.

Comments

- The TimeStamp property does not take any parameters.
- It is not persistable.
- It has a default of Epoch.
- It is read-only.

Example

```
TxtTraceTimeStamp.text = oTrace.TimeStamp
```

hResult

Syntax

oTrace.hResult

Description

The hResult property returns a long value that holds the HRESULT relating to the last value that was collected by the update data source. This may contain a failure HRESULT, even if data collection succeeded, if the update data source server returns a failure code with the individual tag's value.

If no data has been collected for the tag since its data source was last activated, hResult contains -2147221497 (OLE_E_BLANK).

The value of this property remains valid for the duration of the data source object's OnGotData event handler code.

Comments

- The hResult property does not take any parameters.
- It is not persistable.
- It has a default of -2147221497 (OLE_E_BLANK).
- It is read-only.

Example

```
If oTrace.hResult <> 0 then msgbox "Error in trace"
```

OPCQuality

Syntax

oTrace.OPCQuality

Description

The OPCQuality property returns a short integer value that contains the OPC quality associated with the last value collected for this tag by its data source. (OPC quality is described in the *OLE For Process Control Data Access Custom Interface Standard*.)

Non-OPC data sources attempt to generate meaningful OPC quality values that match the semantics of their own condition codes.

Comments

- The OPCQuality property does not take any parameters.
- It is not persistable.
- It has a default of 0 (OPC_QUALITY_BAD).
- The value for Good data is 192 (OPC_QUALITY_OK).
- It is read-only.

Example

```
TxtOPCQuality.text = oTrace.OPCQuality
```

PercentConfidence

Syntax

oTrace.PercentConfidence

Description

The PercentConfidence property returns a short integer value that contains the percent confidence of the last value collected for this tag by its data source. This confidence exists for historical data sources that may interpolate based on missing or uncertain values.

The value of this property remains valid for the duration of the trend object's OnUpdateCollected event handler code.

Comments

- The PercentConfidence property does not take any parameters.
- It is not persistable.
- It has a default of 0.
- It is read-only.

Example

```
TxtPercentConfidence.text = oTrace.PercentConfidence
```

ConfidenceThreshold

Syntax

oTrace.ConfidenceThreshold

Description

The ConfidenceThreshold property returns a short integer value that contains the confidence threshold. Each time data is fetched for a tag, if the PercentConfidence is numerically less than the ConfidenceThreshold, the trend display is not updated with the data. The trace line shows a gap and the latest value is not changed.

Comments

- The ConfidenceThreshold property does not take any parameters.
- It is persistable.
- It has a default of 50.

Example

```
oTrace.ConfidenceThreshold = 95
```

CollectionUnit

Syntax

oTrace.CollectionUnit

Description

The CollectionUnit property returns a string that contains the unit in which the update source server collects data for the trace's active update tag. This property is not valid until an OnGotData event is fired from the data source, because the data collection thread acquires it.



TIP

The data collection thread is not the same thread that triggers OnDataChange events in GUS.

Data Source Specific Details

For a TPN data source, whenever an active tag becomes configured to an already active data source, the data source's data collection thread reforms its OPC collection groups accordingly, and assign the CollectionUnit property for the trace's active update tag. This also occurs whenever an inactive tag becomes active, or with all the active tags configured to a data source, whenever the data source becomes active. This is done by appending .EUDESC to the point name and collecting the value of the resulting point.parameter.

11. GProTrace Object

11.2. GProTrace Properties

For a PHD data source, whenever an active tag becomes configured to an active data source, the data source's data collection thread reforms its data collection group, and assign the CollectionUnit property for the tag from the units member of the PHD SHORT_TAGDEF structure. This also occurs whenever an inactive tag becomes active, or with every active tag configured to a data source, whenever the data source becomes active.



ATTENTION

If PHD is configured to use converted units in history and the update data source is a different server the trend may display inconsistent data.

Comments

- The CollectionUnit property does not take any parameters.
- It is not persistable.
- It has a default of "".
- It is read-only.

Example

```
TxtCollectionUnit.text = oTrace.CollectionUnit
```

hResultCollectionUnit

Syntax

oTrace.hResultCollectionUnit

Description

The hResultCollectionUnit property returns a string that holds the HRESULT that results from attempting to read the collection unit from the data source server.

Comments

- The hResultCollectionUnit property does not take any parameters.
- It is not persistable.
- It has a default of -2147221497 (OLE_E_BLANK).
- It is read-only.

Example

```
If oTrace.hResultCollectionUnit.text <> 0 then
    MsgBox "Error in collection unit"
End if
```

RequestUnit

Syntax

oTrace.RequestUnit

Description

The RequestUnit property returns a string that holds the units in which the data is requested from the data source server. Some data source types, such as kPHDHistory, can convert data from the collection unit into the request unit.

The GProTrace object uses this property to display units. If this property is left blank, the GProTrace object uses the CollectionUnit property instead.

Comments

- The RequestUnit property does not take any parameters.
- It is persistable.
- It has a default of "".

Example

```
oTrace.RequestUnit = "m3/h"
```

HairlineDateTime

Syntax

oTrace.HairlineDateTime

Description

The HairlineDateTime property returns the date and time at the current hairline cursor position in the trend display. (This is the last date-time for which the OnHairlineCursor event was fired.) This value is updated each time the user moves the hairline cursor or the trace moves beneath the hairline cursor.

Comments

- The HairlineDateTime property does not take any parameters.
- It is not persistable.
- It has a default of Epoch.
- It is read-only.

Example

```
TxtHairlineDateTime.text = oTrace.HairlineDateTime
```

HairlineValue

Syntax

oTrace.HairlineValue

Description

The HairlineValue property returns a double value that contains the tag value at the current hairline cursor position in the trend display. (This is the last value for which the OnHairlineCursor event was fired.) This value is updated each time the user moves the hairline cursor or the trace moves beneath the hairline cursor.

Comments

- The HairlineValue property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtHairlineValue.text = oTrace.HairlineValue
```

HairlineDataValid

Syntax

oTrace.HairlineDataValid

Description

The HairlineDataValid property returns a Boolean indicating if the value at the hairline is valid.

Comments

- The HairlineDataValid property does not take any parameters.
- It is not persistable.
- It has a default of False.
- It is read-only.

Example

```
If oTrace.HairlineDataValid = false then  
    MsgBox "Hairline data invalid"  
End if
```

ScalingMode

Syntax

oTrace.ScalingMode

Description

The ScalingMode property returns a GProScalingMode that defines how the y-axis scale range values are determined. The GProScalingMode object can be set to one of the enumerated constants described in the following table.

Constant	Numerical Value	Description
kTag	0	The trace is scaled according to the RetrievedRangeLow and RetrievedRangeHigh property values.
kExplicit	1	The trace is scaled according to the trace's own ManualRangeLow and ManualRangeHigh property values.
kAuto	2	The trace is automatically scaled to fit the visible portion of the trace as precisely as possible into the axes.

Comments

- The ScalingMode property does not take any parameters.
- It is persistable.
- It has a default of 0 (kTag), except when the AxesMode is kShared, in which case the default is 2 (kAuto).

Example

```
oTrace.ScalingMode = 0 'kTag
```

ManualRangeHigh

Syntax

oTrace.ManualRangeHigh

Description

The ManualRangeHigh property returns a double value that contains the maximum measurement value that is anticipated for the tag. This number is used for scaling when the ScalingMode property is set to kExplicit.

Comments

- The ManualRangeHigh property does not take any parameters.
- It is persistable.
- It has a default of 0.0.

Example

```
oTrace.ScalingMode = 1 'kExplicit  
oTrace.ManualRangeHigh = 100  
oTrace.ManualRangeLow = 0
```

ManualRangeLow

Syntax

oTrace.ManualRangeLow

Description

The ManualRangeLow property returns a double value that contains the minimum measurement value that is anticipated for the tag. This number is used for scaling when the ScalingMode property is set to kExplicit.

Comments

- The ManualRangeLow property does not take any parameters.
- It is persistable.
- It has a default of 0.0.

Example

See ManualRangeHigh example in ManualRangeHigh section.

RangeHighFixed

Syntax

oTrace.RangeHighFixed

Description

The RangeHighFixed property returns a long value which contains an HRESULT connected with attempting to interpret and retrieve a RetrievedRangeHigh based on the active tag's ManualRangeHigh. User code may also set this property to S_FALSE to intentionally invoke automatic scaling. When this property contains either a failure HRESULT or S_FALSE, then maximum limit floating is set.

When maximum limit floating is set, each time the Value property is updated and the Value property exceeds the RetrievedRangeHigh, the following occurs:

- The RetrievedRangeHigh property is changed to the new value.
- The trend display is rescaled to reflect the new maximum limit.

Comments

- The RangeHighFixed property does not take any parameters.
- It is persistable.
- It has a default of 1 (S_FALSE).

Example

`oTrace.RangeHighFixed = 100`

RangeLowFixed

Syntax

oTrace.RangeLowFixed

Description

The RangeLowFixed property returns a long value which contains an HRESULT connected with attempting to interpret and retrieve a RetrievedRangeLow based on a non-numeric RangeLow. User code may also set this property to S_FALSE to intentionally invoke automatic scaling. When this property contains either a failure HRESULT or S_FALSE, then minimum limit floating is set.

When minimum limit floating is set, each time the Value property is updated and the Value property is less than the RetrievedRangeLow, the following occurs:

- The RetrievedRangeLow property is changed to the new value.
- The trend display is rescaled to reflect the new minimum limit.

Comments

- The RangeLowFixed property does not take any parameters.
- It is persistable.
- It has a default of 1 (S_FALSE).

Example

```
oTrace.RangeLowFixed = 0
```

RetrievedRangeHigh

Syntax

oTrace.RetrievedRangeHigh

Description

The RetrievedRangeHigh property returns a double value that contains the maximum measurement value that is anticipated for the tag. This value can be used for determining the scaling for the trend on the trend display.

If a value is above this limit, the tag is not visible on the trend display, but the hairline cursor displays the value.

Comments

- The RetrievedRangeHigh property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtRangeHigh.text = oTrace.RetrievedRangeHigh
```

RetrievedRangeLow

Syntax

oTrace.RetrievedRangeLow

Description

The RetrievedRangeLow property returns a double value that contains the minimum measurement value that is anticipated for the tag. This value can be used for determining the scaling for the trend on the trend display.

If a value is below this limit, the tag is not visible on the trend display, but the hairline cursor displays the value.

Comments

- The RetrievedRangeLow property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtRangeLow.text = oTrace.RetrievedRangeLow
```

Assigning RetrievedRangeHigh and RetrievedRangeLow Properties

The RetrievedRange values used for kTag scaling are set according to the rules described below.

1) If the RangeHigh / RangeLow property of the Tag is set (not blank)

RetrievedRangeHigh is set to the value of the parameter specified as the tag's RangeHigh.

RetrievedRangeLow is set to the value of the parameter specified as the tag's RangeLow.

2) Else if the Data Source is a TPN server

The data source attempts to infer the appropriate tag name for the trace's RetrievedRangeHigh and RetrievedRangeLow based on its parameter. The following table describes the rules by which this is attempted.

Tag ID Parameter	RetrievedRangeLow set from	RetrievedRangeHigh set from
.PV .PVTV	Tag's point name with .PVEULO appended	Tag's point name with .PVEUHI appended
.SP	Tag's point name with .SPEULO appended If the .SPEULO parameter does not exist, then use the .PVEULO parameter.	Tag's point name with .SPEUHI appended If the .SPEUHI parameter does not exist, then use the .PVEUHI parameter.

11. GProTrace Object

11.2. GProTrace Properties

Tag ID Parameter	RetrievedRangeLow set from	RetrievedRangeHigh set from
.OP	0 (constant)	100 (constant)

If the tag does not match any of the options in the table or if the point does not have the requested.PVEUHI parameter, continue with step 4.

3) Else if the Data Source is PHD

The data source queries the PHD Server for the tag's HI_EXTREME value. Whenever a tag is added to an already active data source, the data source's data collection thread reforms the PHLIST array, and assigns the RetrievedRangeHigh property if the RangeHigh is blank, and assigns the RetrievedRangeLow property if the RangeLow property is blank.

PHLIST is a structure of the PHD API, while phd_sethost and phd_login are functions of the PHD API. They are described in the Uniformance Process History Database System Manual.

4) Otherwise

The trace's RangeHighFixed property is set to a failure HRESULT. RetrievedRangeHigh floats, initially set to 0, increases to match any collected value that exceeds its current value.

Likewise the trace's RangeLowFixed property is set to a failure HRESULT. RetrievedRangeLow floats, initially set to 0, decreases to match any collected value that is less than its current value.

DynamicRangeHigh

Syntax

oTrace.DynamicRangeHigh

Description

The DynamicRangeHigh property returns a Boolean that indicates whether the high range is dynamic (True) or static (False). When the high range is dynamic, then RetrievedRangeHigh is collected every time the value is collected. If high range is static, then it is collected only under the following conditions:

- When the tag is active and its configured data source becomes active.
- When the tag is active and becomes configured to an active data source.

- When the tag is configured to an active data source and becomes active.
- Whenever this property is written on an active tag that is configured to an active data source.

Regardless, the RetrievedRangeHigh property is updated with the resulting numeric value, if the RetrievedRangeHigh was collected successfully, whenever a tag becomes the active update tag.

Comments

- The DynamicRangeHigh property does not take any parameters.
- It is persistable.
- It has a default of False.

Example

```
oTrace.DynamicRangeHigh = false
```

DynamicRangeLow

Syntax

oTrace.DynamicRangeLow

Description

The DynamicRangeLow property returns a Boolean that indicates whether the low range is dynamic (True) or static (False). When the low range is dynamic, then the tag's RetrievedRangeLow is collected every time the value is collected. If the low range is static, then it is collected only under the following conditions:

- When the tag is active and its configured data source becomes active.
- When the tag is active and becomes configured to an active data source.
- When the tag is configured to an active data source and becomes active.
- Whenever this property is written on an active tag that is configured to an active data source.

Regardless, the RetrievedRangeLow property is updated with the resulting numeric value, if the RetrievedRangeLow was collected successfully, whenever a tag becomes the active update tag.

11. GProTrace Object

11.2. GProTrace Properties

Comments

- The DynamicRangeLow property does not take any parameters.
- It is persistable.
- It has a default of False.

Example

```
oTrace.DynamicRangeLow = false
```

RetrievedDescription

Syntax

oTrace.RetrievedDescription

Description

The RetrievedDescription property returns a string that contains the actual tag description as read from the active update tag.

Data Source Specific Details

For a TPN or PHD data source, whenever an active tag becomes configured to an active data source, the data source's data collection thread reforms its data collection groups accordingly, and assigns the trace's RetrievedDescription property for the tag. This also occurs whenever an inactive tag becomes active, or with all the active tags configured to a data source, whenever the data source becomes active. If the DescriptionTagID property is *not* the null string, then that tag ID is used to retrieve the description. If the DescriptionTagID property is the null string, then one of the following occurs:

- For a TPN data source, the description is set by appending .PTDESC to the point name.
- For a PHD data source, the description is set to the dscr member of the PHD SHORT_TAGDEF structure.

Comments

- The RetrievedDescription property does not take any parameters.
- It is not persistable.
- It has a default of "".
- It is read-only.

Example

```
TxtTraceDescription.text = oTrace.RetrievedDescription
```

hResultDescription

Syntax

oTrace.hResultDescription

Description

The hResultDescription property returns a long value that contains an HRESULT connected with attempting to retrieve the RetrievedDescription property, based on the DescriptionTagID property of the active update tag.

Comments

- The hResultDescription property does not take any parameters.
- It is not persistable.
- It has a default of -2147221497 (OLE_E_BLANK).
- It is read-only.

Example

```
If oTrace.hResultDescription <> 0 then  
    MsgBox "Error in description"  
End if
```

VisibleRangeHighLimit

Syntax

oTrace.VisibleRangeHighLimit

Description

The VisibleRangeHighLimit property returns a double value which contains the maximum value that can be displayed on the visible part of the trace.

Comments

- The VisibleRangeHighLimit property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtRangeHigh.text = oTrace.VisibleRangeHighLimit
```

VisibleRangeLowLimit

Syntax

oTrace.VisibleRangeLowLimit

Description

The VisibleRangeLowLimit property returns a double value that contains the minimum value that can be displayed on the visible part of the trace.

Comments

- The VisibleRangeLowLimit property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtRangeLow.text = oTrace.VisibleRangeLowLimit
```

CollectedRangeHigh

Syntax

```
oTrace.CollectedRangeHigh
```

Description

The CollectedRangeHigh property returns a double value that contains the maximum value represented by the trend, including both visible and invisible time intervals.

Comments

- The CollectedRangeHigh property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtColRangeHigh.text = oTrace.CollectedRangeHigh
```

CollectedRangeLow

Syntax

```
oTrace.CollectedRangeLow
```

Description

The CollectedRangeLow property returns a double value that contains the minimum value represented by the trend, including both visible and invisible time intervals.

11. GProTrace Object

11.2. GProTrace Properties

Comments

- The CollectedRangeLow property does not take any parameters.
- It is not persistable.
- It has a default of 0.0.
- It is read-only.

Example

```
TxtColRangeLow.text = oTrace.CollectedRangeLow
```

LatestTimeCollected

Syntax

oTrace.LatestTimeCollected

Description

The LatestTimeCollected property returns a date that contains the date-time of the latest retrieved value that is represented in the trace, whether visible or not. This property is particular to each trace because different traces may experience different data source latencies or separate failures retrieving updates or history.

Comments

- The LatestTimeCollected property does not take any parameters.
- It is not persistable.
- It has a default of Epoch.
- It is read-only.

Example

```
TxtLatestTime.text = oTrace.LatestTimeCollected
```

EarliestTimeCollected

Syntax

oTrace.EarliestTimeCollected

Description

The EarliestTimeCollected property returns a date that contains the date-time of the earliest retrieved value that is represented in the trace, whether visible or not. This property is particular to each trace because different traces may experience different data source latencies or separate failures retrieving updates or history.

Comments

- The EarliestTimeCollected property does not take any parameters.
- It is not persistable.
- It has a default of Epoch.
- It is read-only.

Example

```
TxtEarliestTime.text = oTrace.EarliestTimeCollected
```

Visible

Syntax

oTrace.Visible

Description

The Visible property returns a Boolean that determines whether the trace is visible on the trend display (True) or not (False).

Comments

- The Visible property does not take any parameters.
- It is persistable.
- It has a default of True.

11. GProTrace Object

11.2. GProTrace Properties

Example

```
oTrace.visible = false
```

TraceWidth

Syntax

```
oTrace.TraceWidth
```

Description

The Visible property returns a Long that determines the width (in pixels) of the trace.

Comments

- The TraceWidth property does not take any parameters.
- It is persistable.
- The default is determined by the TraceWidth property of the GproTrend object..

Example

```
oTrace.visible = 3
```

LineStyle

Syntax

```
oTrace.LineStyle
```

Description

The Visible property returns a Long that determines the type of line used to draw the trace on the trend display. The support values are:

- o 0 = Solid line
- o 1 = Dashes
- o 2 = Dots
- o 3 = Dash/Dot
- o 4 = Dash/Dot/Dot
- o 5 = Clear / transparent

Comments

- The Line property does not take any parameters.
- It is persistable.
- It has a default of 0 (solid line).

Example

```
oTrace.LineStyle = 2 ' dotted line
```

11.3 GProTrace Methods

HistoryTagPair

Syntax

```
oTrace.HistoryTagPair
```

Description

The HistoryTagPair method returns the GProTagPair object that contains the history GProTag objects for the trace.

Comments

The HistoryTagPair method does not take any parameters.

Example

```
TxtName.text = oTrace.HistoryTagPair.PrimaryTag.Name
```

UpdateTagPair

Syntax

oTrace.UpdateTagPair

Description

The UpdateTagPair method returns the GProTagPair object that contains the update GProTag objects for the trace.

Comments

The UpdateTagPair method does not take any parameters.

Example

```
TxtName.text = oTrace.UpdateTagPair.PrimaryTag.Name
```

Trend

Syntax

oTrace.Trend

Description

The Trend method returns the GProTrend trend object that owns the GProTrace object.

Comments

The Trend method does not take any parameters.

Example

```
'Will display a message box with the text ProTrend
```

```
Msgbox ProTrend.SelectedTrace.Trend.Name
```

Refresh

Syntax

oTrace.Refresh

Description

The Refresh method re-fetches from the data source all history and update points.

Comments

Any update points that were plotted up until the point that the refresh event was called is replaced with a fetch of history data.

Example

```
Sub OnLButtonClick()  
    oTrace.Refresh  
End Sub
```

11. GProTrace Object
11.3. GProTrace Methods

12. ProTrend Scripting Examples

12.1 ProTrend Object Scripting Examples

In these examples ProTrend is the actual Trend Object that has been inserted in the GUS display.

Adding a Trace to a Trend

```
Global oTrace As Object
Global oTraceUTagPair As Object
Global oTraceHTagPair As Object

Sub OnLButtonUp()

    'Add oTrace

    If Not (oTrace Is Nothing) Then
        ProTrend.Traces.Remove oTrace.name
    End If

    Set oTrace = ProTrend.Traces.Add("REG1001")
    oTrace.UpdateTagPair.CreatePrimaryTag
    Set oTraceUTagPair = oTrace.UpdateTagPair
    oTrace.HistoryTagPair.CreatePrimaryTag
    Set oTraceHTagPair = oTrace.HistoryTagPair

    'Create the Update Tag Pair

    oTraceUTagPair.PrimaryTag.TagID = "REG1001.PV"
    oTrace.UpdateTagPair.PrimaryTag.activate

    'Create the History Tag Pair

    oTraceHTagPair.PrimaryTag.TagID = "REG1001"
    oTrace.HistoryTagPair.PrimaryTag.activate

    'Set some properties on the trace
```

12. ProTrend Scripting Examples

12.1. ProTrend Object Scripting Examples

```
oTrace.Color = TDC_RED
ProTrend.AxesMode = 2 'kHarmonized
oTrace.ScalingMode = 0 'kTag
```

End Sub

Removing a Trace

```
Global oTrace as Object
```

```
Sub OnLButtonUp()
```

```
    If Not oTrace Is Nothing Then
        ProTrend.Traces.Remove oTrace.name
        Set oTrace = Nothing
    End If
```

End Sub

Setting Trend Properties on Startup

```
Sub OnDisplayStartup()
```

```
    ProTrend.BackgroundColor = TDC_BLACK
    ProTrend.VisibleTimeSpan = "00:20:00" '20 minutes
    ProTrend.LegendPosition = 0 'kOff
    ProTrend.UpdatePeriod = 1 '1 Second
    ProTrend.HairlineColor = 50331647 'White
    ProTrend.GridColor = 45857723 'Gray
    ProTrend.BackgroundColor = 48094685 'Gray
    ProTrend.BorderColor = 48094685 'Gray
    ProTrend.LegendColor = 50331647 'White
```

End Sub

12.2 ProTrend Tool Interface Examples

Adding a Tag to the ProTrend Tool

```
Sub OnLButtonUp()  
  
    Dim oGPTT as object  
    Set oGPTT = CreateObject("GProTrendTool.GUSProTrendTool")  
    oGPTT.AddTag "FIC100"  
    Set oGPTT = nothing  
  
End Sub
```

Loading A Saved Trend in the ProTrend Tool

```
Sub OnLButtonUp()  
  
    Dim oGPTT as object  
    Set oGPTT = CreateObject("GProTrendTool.GUSProTrendTool")  
  
    'In this example Trend1 is the saved name of the trend  
    'and Common is the Trend Tool folder it is located in  
  
    oGPTT.LoadTagSet "Trend1", "Common"  
    Set oGPTT = nothing  
  
End Sub
```

12. ProTrend Scripting Examples

12.2. ProTrend Tool Interface Examples

13. Appendix

13.1 Color Registry Settings

The following registry settings are used by the ProTrend Object and by the ProMenus, and are located in HKEY_LOCAL_MACHINE\Software\BasicScript Program Settings\GUSPro\Colours\.



ATTENTION

Changes to the registry must be made in HKEY_LOCAL_MACHINE. These changes are propagated to HKEY_CURRENT_USER when users log on. This is done with the InitializeGUSProSettings program that has a shortcut in the startup directory. More details on this program are located in the ProMenu User Guide.

Registry Key	Default Color		
	Value	Color Description	RGB()
ErrorFill	33554432	Black	0, 0, 0
Text	33554432	Black	0, 0, 0
TitleBack	45857723	Gray	187, 187, 187
DisplayBack	48094685	Light Gray	221, 221, 221
ButtonNormal	46186688	Gray	192, 192, 192
ButtonSelected	48292064	Gray	224, 224, 224
ButtonText	33554432	Black	0, 0, 0
EmerAlarm	33554687	Red	255, 0, 0
ErrorBack	33554432	Black	0, 0, 0
ErrorText	50331647	White	255, 255, 255
HighAlarm	33619967	Yellow	255, 255, 0
LowAlarm	50331392	Cyan	0, 255, 255
MainBacking	46976204	Light Gray	204, 204, 204

13. Appendix

13.1. Color Registry Settings

Registry Key	Default Color		
	Value	Color Description	RGB()
MainText	33554432	Black	0, 0, 0
ModeHighlight	44739242	Medium Gray	170, 170, 170
OPTrace	45741567	Light Yellow	255, 245, 185
PVTrace	33587236	Dark Green	36, 128, 0
SecondaryBacking	44739242	Medium Gray	170, 170, 170
SecondaryText	50331647	White	255, 255, 255
Selected	39821195	Dull Green	139, 159, 95
SPTTrace	46102272	Medium Blue	0, 119, 191
TrendBack	44739242	Medium Gray	170, 170, 170
DisplayBackground	48094685	Light Gray	221, 221, 221
Trace1	39288832	Dark Blue Green	0, 128, 87
Trace2	41963264	Dark Blue	0, 79, 128
Trace3	41943104	Purple	64, 0, 128
Trace4	39059584	Dark Pink	128, 0, 84
HighlightBack	50331647	White	255, 255, 255
TitleText	33554432	Black	0, 0, 0

13.2 Group Trend Registry Settings

The following registry settings are used by the displays created using the Group Trend template and are located in HKEY_LOCAL_MACHINE\Software\BasicScript Program Settings\GUSPro\GroupTrend.

Registry Key	Default Value	Comment
TagPrefixSP	Empty	This string is appended on the front of every PHD tag for SP and PVTV values
TagPrefixOP	Empty	This string is appended on the front of every PHD tag for OP values
TagPrefixPV	Empty	This string is appended on the front of every PHD tag for PV values
TagSuffixSP	.SP	This string is appended at the end of every PHD tag for SP values
TagSuffixPVTV	.PVTV	This string is appended at the end of every PHD tag for PVTV values
TagSuffixOP	.OP	This string is appended at the end of every PHD tag for OP values
TagSuffixPV	.PV	This string is appended at the end of every PHD tag for PV values
OPLabel	OP:	This string is displayed in front of the OP value in the tag detail section
PVLabel	PV:	This string is displayed in front of the PV value in the tag detail section

13. Appendix

13.2. Group Trend Registry Settings

Registry Key	Default Value	Comment
SPLabel	SP:	This string is displayed in front of the SP value in the tag detail section
Time1	00:10:00	These are the time span settings allowed in the time pull down box. The first one is the default.
Time2	00:20:00	
Time3	00:30:00	
Time4	01:00:00	
Time5	02:00:00	
Time6	04:00:00	
Time7	08:00:00	
Time8	16:00:00	
Update1	4	These are the update rate settings allowed in the update rate pull down box. The first one is the default.
Update2	10	
Update3	15	
Update4	30	
Update5	60	
Update6	120	
Update7	600	
Update8	3600	

13.3 Overview Trend Template Registry Settings

The following registry settings are used by the displays created using the Overview Trend template, and are located in HKEY_LOCAL_MACHINE\Software\BasicScript Program Settings\GUSPro\ 48PointTrend.

Registry Key	Default Value	Comment
Time1	02:00:00	These are the time span settings allowed in the time pull down box. The first one is be the default.
Time2	00:10:00	
Time3	00:20:00	
Time4	01:00:00	
Time5	04:00:00	
Time6	08:00:00	
Time7	16:00:00	
Time8	24:00:00	
Update1	30	These are the update rate settings allowed in the update rate pull down box. The first one is be the default.
Update2	45	
Update3	60	
Update4	120	
Update5	180	
Update6	240	
Update7	300	
Update8	360	

13. Appendix

13.4. Supported Variant Types

13.4 Supported Variant Types

The ProTrend Object supports the variant types in the following table.

VB Type	C/C++/MFC Type	C/C++ VARTYPE	VariantTypeConstant	Numeric Constant
String	BSTR	VT_BSTR	vbVString	8
Int	short	VT_I2	vbVInteger	2
Long	long	VT_I4	vbVLong	3
Single	float	VT_R4	vbVSingle	4
Double	double	VT_R8	vbVDouble	5
DATE	DATE	VT_DATE	vbVDate	7

The VariantTypeConstant is an enumeration in the VBRUN type library. Other variant types are not supported because they are not well supported by the C/C++ run-time and the MFC.

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